

BPG-01 · BEST PRACTICE GUIDES

Building a project controls function from zero

The order you stand things up in, and why the order is the whole decision

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.
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— Summary

How to establish a project controls function where none exists, in the order that works: coding structure before tools, cut-off before dashboards, and a defined control-account level before anyone is asked for a variance. The guide gives the dependency chain, a method for sizing the function against the smallest variance it has to detect, a worked reporting calendar derived backwards from the governance meeting, and an explicit list of what to defer and how to say so out loud.

— About this document

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Building a project controls function from zero

In one paragraph. How to establish a project controls function where none exists, in the order that works: coding structure before tools, cut-off before dashboards, and a defined control-account level before anyone is asked for a variance. The guide gives the dependency chain, a method for sizing the function against the smallest variance it has to detect, a worked reporting calendar derived backwards from the governance meeting, and an explicit list of what to defer and how to say so out loud.

Who this is for. Project controls managers and heads of project management office asked to build a function on a new programme or in an organisation that has never had one; project directors deciding what to fund first; finance business partners who will have to live with whatever coding scheme gets chosen.

— 1. The only real decision is sequence

Almost every element of a controls function is uncontroversial. Nobody argues that a project should be without a work breakdown structure, a cost baseline, a progress measurement method or a monthly report. The arguments are about **order** — and order is where functions are won or lost, because most of these elements depend on each other in one direction only.

A cost report cannot be produced before there is an agreed cut-off date. An agreed cut-off is worthless if cost arrives coded to a structure nobody can roll up. A rollup needs a coding structure decided before the first purchase order was raised, because a purchase order carries its code into every transaction it later generates. And the coding structure cannot be designed until somebody has decided what the reportable units of scope are. That chain runs one way; reversing a link is possible, but it is paid for in rework, and [BPG-03](#) puts a number on the most common reversal.

The failure this guide is written against is not incompetence. It is the reasonable instinct to start with what the sponsor asked for — usually a dashboard — and backfill the structure later. The dashboard gets built. It reports whatever the accounts payable team processed that week, against a budget nobody has time-phased, at a level of detail chosen by whoever configured the tool. Then it is very hard to remove, because it exists and people look at it.

— 2. What "zero" looks like from the inside

Real starting positions are rarely blank. They are usually one of three states, and each needs a different opening move.

Genuinely nothing. A new programme, no systems configured, no historical postings. The easiest and rarest case: everything below applies in order.

A finance system already posting. Cost is recorded against a general ledger and perhaps a rudimentary project code, but there is no scope structure, no baseline and no progress measurement. Transactions already exist and carry codes, so the opening move is to count them — that number

decides whether you design a structure that maps onto the existing one or accept a retrofit (see [BPG-03 – Cost breakdown structure and the code of accounts](#)).

A controls function that has stopped working. Reports are produced, nobody uses them, and the schedule and the cost report disagree. This looks like the hardest case and is often the quickest to fix, because the data usually exists — what is missing is a cut-off, a rules-of-credit register or an agreed control-account level. Diagnose before you rebuild; [BPG-19 – Project controls assurance and health checks](#) covers that diagnosis.

State which of the three you are in, in writing, in the first week. Pretending a rebuild is a start-up is how a rebuild gets the wrong budget.

— 3. The dependency chain

The order below is not a preference. Each item is genuinely blocked by the one above it.

3.1 Reportable scope. Decide what the project is made of, expressed as deliverables rather than as departments. This is the work breakdown structure (WBS), and it parents everything else because it determines what a variance can be about. [BPG-02 – The work breakdown structure](#) owns the method.

3.2 The coding structure. Decide how cost will be classified — by scope element, by cost type, by resource class — and fix the code format. This must happen before the first commitment is raised, because every commitment stamps a code onto a stream of downstream transactions. [BPG-03](#) owns it.

3.3 The control-account level. Decide where scope, budget, cost and schedule meet, and therefore where a named person is accountable for a variance. It cannot be settled before 3.1 and 3.2, because the control account is defined by their intersection with the organisational breakdown structure. §7 gives the arithmetic test.

3.4 The cut-off calendar. Decide the date on which the month ends for controls purposes, and the working-day sequence from that date to the governance meeting. §9 works it backwards from the meeting.

3.5 The rules of credit. Decide, per work package, how physical progress will be measured and what evidence earns each increment — before anyone is asked for a percentage, because the first percentage anybody gives you sets the precedent. [BPG-06 – Progress measurement and rules of credit](#) owns this.

3.6 The baseline. Freeze the scope, the time-phased budget and the schedule under a change-control gate; [BPG-04 – Baseline and baseline change control](#) owns it. A baseline set before 3.1 to 3.5 is a number, not a baseline: there is nothing holding it in place.

3.7 The first report. Produce it, then produce the second on the same day of the cycle in the same shape, finished or not. Cadence is established by repetition, not by design.

3.8 Tooling and visualisation. Only now — because only now do you know the structure, the cadence, the level and the measurement rules, which is to say the requirement. Buying earlier is buying somebody else's requirement.

— 4. The minimum viable controls function

The smallest set of artefacts that constitutes a real function — not a good one, a real one — is shorter than most implementation plans suggest.

Artefact	What it must contain to count
Work breakdown structure and dictionary	Every element defined in a sentence, with an owner and an inclusion/exclusion note
Code of accounts	Segment definitions, the valid-combination rule, and who may create a new code
Control-account register	Account, WBS element, responsible manager, budget, measurement technique
Time-phased cost baseline	Budget spread across the schedule, at control-account level
Cut-off calendar	Cut-off date and the working-day sequence to issue
Rules-of-credit register	Technique and evidence per work package
Change register	Every trend, its status and its cost impact, from day one
Monthly report	One page of numbers, one page of narrative, control-account detail beneath

Everything on that list is a document or a register; none needs a tool beyond a spreadsheet. That is the point — the function is the discipline, and software is an efficiency layer over it. [TPL-01 – Project controls execution plan](#) names all of them, their owners and their frequencies in one controlled document, and that is the artefact a sponsor should approve.

— 5. Cut-off before dashboards

The cut-off date is the least glamorous item on the list and the one that most changes everything downstream, because it is what makes a period a period.

Without a fixed cut-off, cost enters the report according to when invoices happened to be processed. The period cost figure becomes a measure of the accounts payable team's throughput rather than of the project's consumption — and, worse, the cost figure and the progress figure then describe different windows of time, progress measured to the day the planner asked the field and cost to the day the ledger was extracted, so every index computed from them compares two different months.

A cut-off is real only when three things hold: the date is fixed and published in advance; every data source reports to that date and no other; and the gap between cut-off and complete data is filled by accrual rather than by waiting. That last condition makes the cut-off enforceable, which is why [BPG-07 – Accruals and cut-off discipline](#) is a prerequisite for a credible report rather than an accounting refinement.

A dashboard adds nothing the underlying number does not already contain; it changes how quickly the number is read. Over a sound cut-off and a stable coding structure it is genuinely valuable. Built first, it industrialises whatever error is in the data and gives it a colour.

— 6. What to defer, and how to defer it honestly

Deferral is a decision and should be recorded as one. The defaults below are the Institute's recommended practice for a new function; the grounds matter more than the list, because a different risk profile will legitimately reorder it.

Defer quantitative schedule risk analysis until the schedule passes a quality review. Simulating a network with open ends and unjustified constraints produces a distribution around a number that was never a forecast. See [BPG-05 – Schedule quality – a practical review](#) and [BPG-17](#).

Defer earned value indices until the rules of credit are written and signed. A cost performance index derived from subjective percentages teaches the organisation that the index is negotiable, and that lesson is expensive to unlearn.

Defer resource-loaded scheduling unless the project is genuinely resource-constrained and the data will be maintained. An unmaintained resource-loaded schedule is worse than an unloaded one, because it looks like it means something. **Defer integration** between the cost system and the scheduling tool until both are stable alone, because integration multiplies the consequences of an error in either.

Do not defer the change register. It costs half a day to open and it is the only artefact that cannot be reconstructed retrospectively — records can rebuild a document, but not the reasoning behind a decision taken in month two.

State each deferral with a trigger: "quantitative schedule risk analysis will be introduced once the schedule quality review returns no open ends on the critical path and fewer than five unjustified constraints." A deferral with a trigger is a plan; one without is a gap somebody else will find.

— 7. Sizing the function against the work it has to detect

The most consequential sizing decision is the number of control accounts, because it sets both the monthly effort of running the function and the smallest problem the function can see. Too few accounts and a material overrun hides inside a large one, diluted below the reporting threshold until it is too late to act. Too many and measurement overhead consumes the analysis time that justified the function.

The Institute's recommended practice is to derive the floor from the detection requirement, not from a rule of thumb:

$$\text{Minimum control accounts} = \frac{(\text{Budget at completion} \times \text{variance reporting threshold})}{\div \text{the smallest variance you must act on}}$$

The logic is direct. If a variance is reported only when it exceeds a percentage of its account's budget, the account has to be small enough that the smallest problem you care about clears that percentage. Both inputs are explicit organisational decisions and both belong in the controls execution plan. Set the floor from the calculation, then check the ceiling against capacity: accounts multiplied by monthly effort per account must leave room for analysis. §9.2 runs both directions.

One qualification: the formula uses the average account size, so a portfolio with a wide spread of account values needs the test applied to the largest accounts individually.

— 8. How this goes wrong

The tool is procured first, and it defines the structure. A configuration workshop asks what the cost breakdown should look like three weeks before anyone has decided what the deliverables are. The resulting structure is whatever the implementation consultant used last time, and it survives the life of the project, because by the time anyone objects there are transactions posted against it.

Cost coding is left to procurement. Purchase orders are raised against whichever ledger account gets them approved fastest, and six months later the same physical scope sits under three different codes. [BPG-03](#) sets out the consequence.

The cut-off moves to suit the invoice run. Somebody notices a large invoice will land on the 2nd and holds the cut-off open for it. That breaks the comparability of the period, and it establishes the precedent that cut-off is negotiable whenever the number is unhelpful.

The baseline is set before the scope is decomposed. A total is agreed at sanction and called a baseline. With no time-phasing and no control-account allocation, the first variance question — is this account over, or just early? — has no answer, and the function spends its opening quarter reverse-engineering the thing it was supposed to be measuring against.

Everything is measured, nothing is owned. Two hundred work packages, each with a percentage, none with a named control-account manager. Progress collected by the controls team belongs to the controls team, so the numbers get argued with rather than acted on. Accountability comes from who signs.

The function reports to the wrong meeting. The pack goes to a monthly review with no authority to approve a change, so decisions defer to a body that meets quarterly and the variance is two months old on arrival. Establish which meeting owns the change-control gate before designing the calendar.

Level of effort creeps into the baseline. Supervision, project management and the controls function itself are budgeted as packages that earn by the calendar, cannot show a schedule variance, and dilute every index they are aggregated into. Segregate them from the outset; retrofitting the separation means restating the trend.

— 9. Worked example

Illustrative figures. All values are in generic currency units. No jurisdiction, sector or real project is implied.

9.1 The reporting calendar, derived backwards

Assumptions. A monthly cycle; a five-day working week, holidays ignored. The steering committee, which may approve a baseline change, meets on the tenth working day. Readers need three working days with the pack. Cut-off is the last calendar day of the preceding month. Working backwards:

Working day	Activity
WD 10	Steering committee
WD 8-9	Reading time
WD 7	Pack issued
WD 6	Project manager review and sign-off
WD 5	Narrative drafted with control-account managers
WD 4	Earned value computed; variances analysed
WD 3	Accruals determined and posted; schedule progressed to the status date
WD 2	Cost extracted from the ledger; goods-received-not-invoiced report pulled
WD 1	Field progress returns and timesheets submitted against the cut-off

Seven working days run from cut-off to issue, of which two are pure data collection and one is judgement. That is the whole argument for sequence: **only four working days in the month are available for analysis**, and they exist only because the three before them were protected.

Now move the committee to the seventh working day, as sponsors regularly request. Three working days disappear and the pack must issue on WD 4, compressing the chain to data in, extract and accrue, compute and draft, review and issue. The casualty is the accrual judgement, because it is the only step with no external deadline attached to it. [BPG-07](#) sets out what that compression costs in reported performance; the point here is that the reporting calendar is a controls design decision and should be negotiated as one rather than inherited.

9.2 Sizing the control-account population

Assumptions. Budget at completion 24,000,000 currency units. A variance is reported when it exceeds 10 % of its account's budget. The smallest variance the project director requires to be visible is 150,000. Monthly effort per control account — data check, reconciliation, status conversation, commentary — is 40 minutes. The function has 1.5 full-time equivalents at 150 productive hours a month.

Floor, from the detection test.

$$\begin{aligned}
 \text{Minimum control accounts} &= (24,000,000 \times 0.10) \div 150,000 \\
 &= 2,400,000 \div 150,000 \\
 &= 16 \text{ control accounts}
 \end{aligned}$$

At exactly 16 accounts the average account is $24,000,000 \div 16 = 1,500,000$, and a 150,000 variance is $150,000 \div 1,500,000 = 10.0\%$ — sitting precisely on the threshold, which is not a margin. Take 20 accounts instead: the average account is $24,000,000 \div 20 = 1,200,000$, and the same variance is $150,000 \div 1,200,000 = 12.5\%$, clearing the threshold with room for an account that runs larger than average.

Ceiling, from capacity.

$$\begin{aligned}
 \text{Monthly review effort} &= 20 \text{ accounts} \times 40 \text{ minutes} = 800 \text{ minutes} = 13.3 \text{ hours} \\
 \text{Available capacity} &= 1.5 \text{ FTE} \times 150 \text{ hours} = 225 \text{ hours} \\
 \text{Share consumed} &= 13.3 \div 225 = 5.9\%
 \end{aligned}$$

Roughly six per cent of the function's month goes on the control-account review pass, before any analysis, forecasting or change assessment. That is affordable.

The counterfactual. Structured as 60 accounts — attractive, because the average account falls to 400,000 and a 150,000 variance becomes 37.5 % of its account, impossible to miss — the review pass becomes $60 \times 40 = 2,400$ minutes = 40.0 hours, or $40.0 \div 225 = 17.8$ % of capacity. The visibility is genuinely better. Whether it is worth three times the monthly effort is a resourcing decision for the project director, and it should be put to them in those terms rather than settled by whoever configures the tool.

What the answer depends on. The 40-minute figure is the sensitive input, and it varies with data quality and with whether control-account managers write their own commentary. Measure it after two cycles and redo the ceiling calculation. The detection test itself depends on the threshold and the materiality figure being set by someone with authority to set them — if they have not been, that is the finding, not the arithmetic.

— 10. Checklist

Take this into the meeting where the function is being scoped. Every line is answerable yes or no by someone in the room; a "we'll come back to that" is itself a finding.

Before anything is procured

- [] Which of the three starting positions in §2 are we in, and who has written that down?
- [] Is there a deliverable-oriented WBS, or only a list of departments and phases?
- [] Has the code of accounts been designed, with a named owner who may authorise a new code?
- [] How many cost transactions have already posted under a provisional code?
- [] Has anyone been asked to approve a tool before the four questions above were answered?

Structure

- [] Is the control-account count derived from the §7 detection test, with both inputs written down?
- [] Does every control account have a named manager who has agreed in person to own its variance?
- [] Is level-of-effort work segregated into its own accounts, with its share of the baseline recorded?
- [] Does every WBS element have a dictionary entry — a sentence, an owner, inclusions and exclusions?

Cadence

- [] Is the cut-off fixed, published, and identical for cost, progress and schedule?
- [] Has the working-day sequence from cut-off to issue been drawn, with the analysis days identified?
- [] Which meeting may approve a baseline change, and does the calendar reach it?
- [] What happens when a large invoice lands after cut-off? (The correct answer is: it is accrued.)

Measurement and deferrals

- [] Is there a signed rules-of-credit register covering every package that will be asked for a percentage?
- [] Is the evidence for each credit increment named — which document, signed by whom?
- [] Is every deferred capability listed with a trigger condition in the controls execution plan?
- [] Is the change register open today, even though there is nothing in it yet?
- [] Has the sponsor been told in writing what they are not getting in the first two cycles, and why?

— Related

- [BPG-02 – The work breakdown structure](#) — the first link in the chain, and what everything else is defined against.
- [BPG-03 – Cost breakdown structure and the code of accounts](#) — the coding decision that must precede the first commitment, and what reversing it costs.
- [BPG-07 – Accruals and cut-off discipline](#) — why the cut-off in §5 is enforceable only if the accrual process behind it is.
- [BPG-14 – Monthly reporting that gets read](#) — what the cadence in §3.7 should produce.
- [TPL-01 – Project controls execution plan](#) — the controlled document naming every §4 artefact, its owner and its frequency.
- [TPL-15 – Project controls health check](#) — for diagnosing the third starting position in §2.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 3 (Budgeting and Forecasting) for the time-phased budget, Domain 5 (Cost Management and Cost Control) for the control account and the commitment-accrual-actual cycle, and Domain 8 (Project Management Lifecycle) for scope decomposition and work authorisation.

The control-account sizing method in §7 is PCI recommended practice, derived arithmetically in §9.2 rather than borrowed; the reporting threshold and materiality inputs are organisational decisions, not published values. No external standard is reproduced here, and no threshold in this guide is attributed to any named publication.

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BPG-02 · BEST PRACTICE GUIDES

The work breakdown structure

Decomposition rules, the 100 % rule, and why a WBS built to mirror the org chart cannot answer a cost question

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
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— Summary

How to decompose scope so that the resulting structure can carry a budget, a schedule and an owner: the 100 % rule and what it actually forbids, how far to decompose and the test that tells you when to stop, the difference between deliverable, phase and discipline structures, what belongs in a WBS dictionary entry, and how the WBS crosses the organisational breakdown structure to form control accounts. Includes a two-level decomposition with the 100 % rule verified at each level and a demonstration of why an organisation-shaped WBS cannot answer the question it will be asked most.

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The work breakdown structure

In one paragraph. How to decompose scope so that the resulting structure can carry a budget, a schedule and an owner: the 100 % rule and what it actually forbids, how far to decompose and the test that tells you when to stop, the difference between deliverable, phase and discipline structures, what belongs in a WBS dictionary entry, and how the WBS crosses the organisational breakdown structure to form control accounts. Includes a two-level decomposition with the 100 % rule verified at each level and a demonstration of why an organisation-shaped WBS cannot answer the question it will be asked most.

Who this is for. Planners and cost engineers building or inheriting a WBS; project managers who have to approve one; anyone who has been handed a structure and asked why the numbers will not add up.

— 1. What the structure has to do before it is elegant

A work breakdown structure (WBS) is judged by whether it can answer questions, not by whether it looks tidy. Four questions are asked of it constantly, and a structure that cannot answer all four will be worked around within a quarter:

- What is the total cost of this thing we are building?
- Who is accountable for that number?
- Is this element late, over, or both?
- If scope is added, where does it attach?

Every rule in this guide exists to keep one of those four answerable. The 100 % rule keeps the totals true. Deliverable orientation keeps the first question answerable in one place rather than four. Decomposition depth keeps the third answerable early enough to matter. The dictionary keeps the fourth from being settled by whoever raises the purchase order.

The WBS is also the parent artefact of the whole controls function — the cost breakdown structure, the schedule, the progress measurement register and the change register are all defined against it. That is why [BPG-01 – Building a project controls function from zero](#) puts it first in the dependency chain, and why changing it later is expensive in a way that changing almost anything else is not.

— 2. Decomposition: what it is and what it is not

Decomposition is the division of the total scope of work into progressively smaller components, each of which is a piece of the thing being delivered rather than a piece of the effort to deliver it. The distinction sounds pedantic until you try to cost something.

"Foundations" is a deliverable. "Civil engineering" is a discipline. "Detailed design phase" is a period of time. All three appear in real structures, and only the first can carry a completion state that

anybody can verify. A deliverable is either there or it is not; a discipline is never finished until the project is; a phase ends on a date that can be moved.

The practical rule is that each element should be describable as a **noun with a completion condition**. "Inlet works structure — concrete poured, cured and surveyed to tolerance" is decomposable, costable, schedulable and verifiable. "Engineering support" is none of those things.

This does not mean discipline and phase never appear. They appear as codes, not as branches — see §7 and [BPG-03 – Cost breakdown structure and the code of accounts](#). A transaction can be tagged with its discipline, its phase, its cost type and its resource class simultaneously, and none of those tags needs to be a level of the WBS. Confusing a tag with a branch is the single most common structural error in project controls, and §8 shows what it costs.

— 3. The 100 % rule

The 100 % rule states that the children of any element together represent all of that element's scope — no more, and no less. It is checked at every level, not once at the bottom.

What it forbids is more interesting than what it requires. It forbids **omission**: scope that exists but has no home, which is how commissioning spares, temporary works and site clean-up disappear from budgets. It forbids **overlap**: two elements that both plausibly contain the same work, which is how the same activity gets costed twice in the estimate and then charged to whichever code is open. And it forbids **invention**: elements that describe work not in the project's scope, which is how a speculative future phase ends up carrying budget.

The rule applies to scope. Because budget follows scope, the arithmetic check is that each parent's budget equals the sum of its children's budgets — but the check is a symptom test, not the rule itself. A structure can balance perfectly and still omit scope, if the omitted scope was never estimated.

Three practices make the rule enforceable rather than aspirational:

Check upward as well as downward. Ask of each child, "what parent does this belong to?" — omissions show up on the way up more reliably than on the way down.

Give rework a home in advance. Decide, and write down, that rework is charged to the element it repairs rather than to a general remediation code. Without that decision, rework becomes a floating cost that appears in whichever account has headroom.

Name the exclusions. Every dictionary entry states what the element does not include. Exclusions are how the 100 % rule survives contact with an interface, because two adjacent elements that each disclaim the same interface scope reveal the gap immediately.

— 4. Deliverable, phase and discipline structures

Three organising principles compete at level 2, and the choice has consequences that persist for the life of the project.

Deliverable-oriented. Level 2 elements are the physical or logical things being produced. This is the Institute's recommended default because it makes the most-asked question — what did this cost? — answerable at a node rather than by a query. Its weakness is that it obscures phase: the design cost of everything is scattered across every branch.

Phase-oriented. Level 2 is design, procurement, construction, commissioning. This suits projects where the phases are contractually distinct, funded separately, or executed by different organisations, and it makes stage-gate reporting natural. Its weakness is the mirror of the deliverable structure's: the total cost of any one deliverable is spread across four branches.

Discipline-oriented. Level 2 is civil, mechanical, electrical, instrumentation. This is almost always a mistake dressed as convenience, because a discipline is not a deliverable and never completes. It survives because it matches how the estimate was built and how the resource pool is organised.

The hybrid that usually wins puts deliverables at level 2 and phase at level 3 within a deliverable — so the design of the inlet works sits under the inlet works, not under a global design branch. This keeps the deliverable total intact while preserving stage visibility, at the cost of repeating the phase pattern across branches. Where phase is genuinely the primary funding and accountability boundary, invert it — but do so deliberately, and record why in the WBS dictionary, because the next planner will assume it was an accident.

— 5. How far to decompose

Decomposition stops when the lowest element — the work package — satisfies four conditions simultaneously: it can be estimated with acceptable confidence; it can be scheduled as one or a small number of activities; it has a single accountable owner; and its completion can be verified against something other than an opinion.

That leaves a wide range, so use the **reporting-cycle test**: no work package should be able to run for more than two reporting periods without producing an objective completion event. On a monthly cycle, that means a package longer than about two months needs interim milestones with defined evidence — which is a progress measurement decision, and belongs with the rules of credit in [BPG-06 – Progress measurement and rules of credit](#).

The counterweight is population. §9.3 shows the arithmetic: the number of work packages multiplied by the effort of statusing each one is a fixed monthly tax, and a structure that produces four hundred packages on a monthly cycle needs someone to status twenty of them every working day. That is achievable with a mature field reporting process and impossible without one.

Where the scope is not yet definable, do not decompose it artificially. Hold it as a **planning package** — a real element with a budget and a rough duration but no work-package detail — and convert it later. That conversion is rolling wave, not re-baselining, and [BPG-04 – Baseline and baseline change control](#) explains why the distinction matters to the variance.

— 6. The WBS dictionary

The dictionary is what turns a numbered list into a controlled artefact. Without it, the meaning of an element lives in the head of whoever created it, and a WBS whose meaning is undocumented will be interpreted differently by the estimator, the planner and the person coding the invoice.

A usable entry is short and contains, at minimum: the code and title; a one-sentence scope statement in the form of a noun with a completion condition; explicit inclusions and, more importantly, exclusions; the responsible organisational unit; the budget and its estimate basis reference; the measurement technique that will be used to credit progress; and the acceptance evidence that closes the element.

Two of those fields do most of the work. **Exclusions** prevent the 100 % rule failing at interfaces. **Acceptance evidence** prevents the argument at the end, because it fixes in advance what "done" is going to look like. An entry without those two is a title with extra words.

TPL-02 – [Work breakdown structure and WBS dictionary](#) provides the instrument, with the fields defined and a worked fragment.

— 7. WBS × OBS = the control account

The WBS says what. The organisational breakdown structure (OBS) says who. Their intersection is the **control account** — the point where scope, budget, schedule and cost come together under a named manager, and the level at which performance is actually measured.

The crossing matters because a single WBS element is often delivered by more than one organisation: a structure built partly by a subcontractor and partly by direct labour, or a package where equipment comes through a central procurement function and installation through the site team. Splitting the element by organisation produces two control accounts whose budgets sum to the element's budget — the 100 % rule again, applied through the crossing rather than down the branch. §9.2 works this through.

Two consequences follow. First, the number of control accounts is not the number of WBS elements; it is the number of populated intersections, and it is a design decision with an arithmetic test set out in [BPG-01](#) §7. Second, a WBS element with no identifiable owning organisation is a warning: either the scope has not been assigned, or the element is not a deliverable.

— 8. How this goes wrong

The WBS mirrors the org chart. Level 2 becomes Engineering, Procurement, Construction, Commissioning — or worse, the names of the departments. Every deliverable's cost is then spread across four branches, and the only person who can produce a deliverable total is whoever remembers which cost codes belong together. §9.4 puts a number on it. This structure is chosen because it matches the way the organisation is funded and the way the estimate was assembled, and it fails the moment someone asks what a thing cost.

Overlapping elements that both look right. "Piping" and "Mechanical installation" both plausibly contain pipe supports. Nobody notices at estimate stage because the estimator only used one of them. The site team charges to whichever is open, and by month four the two accounts cannot be compared to anything.

The structure is changed after transactions exist. An element is split, renamed or moved because the original decomposition proved inconvenient. The historical postings stay where they were, so the trend breaks at the change and every comparison across it needs a manual bridge. If the structure must change, [BPG-03](#) sets out what the remapping actually costs.

Level of effort branches. "Project management", "Site establishment" and "Project controls" appear as level 2 elements. They are real cost, they must be budgeted, and they are not deliverables — so they never complete, cannot show a schedule variance, and dilute every index they are rolled into. Budget them, segregate them, and keep their share of the baseline visible.

Decomposition to the activity. The WBS is taken down to individual schedule activities, producing several thousand elements. The structure is now the schedule, the two artefacts can no longer

be maintained independently, and every schedule change becomes a scope change. The WBS stops at the work package; the activity network sits beneath it.

The dictionary is written after the estimate. Elements are named during estimating and defined afterwards, so the definitions are reverse-engineered from what the estimate happened to contain. The exclusions are then whatever nobody priced, which is exactly the scope that will be argued about.

A code is reused. An element is deleted and its number given to something else. Every historical report now refers to a different thing under the same code. Codes are permanent, even when the scope they described was cancelled — retire them, never reissue them.

— 9. Worked example

Illustrative figures. Generic currency units. No real project, organisation or sector is implied. A water treatment plant upgrade is used because it has clearly separable physical deliverables.

9.1 Two levels of decomposition, both checked

Project budget at completion: **12,400,000**.

Level 2 — deliverable-oriented.

Code	Element	Budget
1.1	Project management and controls (level of effort)	900,000
1.2	Design and engineering	1,650,000
1.3	Civil and structural works	4,200,000
1.4	Mechanical and process equipment	3,850,000
1.5	Electrical, control and commissioning	1,800,000
Total		12,400,000

Check: $900,000 + 1,650,000 + 4,200,000 + 3,850,000 + 1,800,000 = 12,400,000$. The 100 % rule holds at level 2. Note that 1.1 is segregated level of effort, not a deliverable — it is budgeted here so it cannot leak into the deliverable accounts, and its share of the baseline is $900,000 \div 12,400,000 = 7.26\%$, a figure worth stating explicitly because it is the proportion of the baseline that cannot show a schedule variance.

Level 3 — decomposition of 1.3 Civil and structural works (4,200,000).

Code	Element	Budget
1.3.1	Site preparation and earthworks	620,000
1.3.2	Inlet works structure	1,140,000
1.3.3	Aeration basin structures	1,580,000
1.3.4	Access roads and hardstanding	490,000
1.3.5	Site drainage and containment	370,000
Total		4,200,000

Check: $620,000 + 1,140,000 + 1,580,000 + 490,000 + 370,000 = 4,200,000$. The rule holds independently at level 3; it is not inherited from the level-2 check.

9.2 Crossing the OBS to form control accounts

Element 1.3.3 (Aeration basin structures, 1,580,000) is delivered by two organisational units: a subcontracted civils package, and a direct-works team responsible for the liner installation and hydrostatic testing.

Control account	WBS	Organisation	Budget
CA-1.3.3-A	1.3.3	Civils subcontract	1,310,000
CA-1.3.3-B	1.3.3	Direct works	270,000
Total			1,580,000

Check: $1,310,000 + 270,000 = 1,580,000$. The 100 % rule survives the crossing, which is the property that makes control-account performance roll up into WBS-element performance without adjustment.

9.3 Choosing the depth

Assumption. Statusing a work package at month end — collecting the progress claim, checking it against the evidence named in the dictionary, and resolving queries — takes an average of 12 minutes. The month has 20 working days.

At an average work package of 100,000: $12,400,000 \div 100,000 = \mathbf{124 \text{ packages}}$. Statusing effort = $124 \times 12 = 1,488$ minutes = **24.8 hours** a month, or $124 \div 20 = \mathbf{6.2 \text{ packages a working day}}$.

At an average work package of 30,000: $12,400,000 \div 30,000 = \mathbf{413 \text{ packages}}$ (rounded down). Statusing effort = $413 \times 12 = 4,956$ minutes = **82.6 hours** a month, or $413 \div 20 = \mathbf{20.65 \text{ packages a working day}}$.

The finer structure is not wrong. It is a commitment to roughly 82.6 hours of statusing every month — more than half a full-time equivalent doing nothing else — and that commitment should be made knowingly, with the field reporting process that supports it, rather than discovered in month three. The sensitive input is the 12 minutes: measure it after two cycles and redo the arithmetic.

9.4 What the org-chart WBS costs you

Suppose the same project had been decomposed at level 2 by department. The aeration basins' true cost would then be spread as follows:

Where the cost sits	Amount
Engineering — basin design	185,000
Procurement — liner and embedded items	240,000
Construction — civils and direct works	1,310,000
Commissioning — hydrostatic testing and handover	95,000
True deliverable cost	1,830,000

Check: $185,000 + 240,000 + 1,310,000 + 95,000 = 1,830,000$.

In the deliverable structure, 1,830,000 is a node — visible, owned, comparable to its estimate, and capable of showing a variance. In the department structure it is a query that has to be assembled from four branches by someone who knows which cost codes belong to the basins. The number is not lost; it is unownable. Nobody has a variance to explain, because nobody has the total.

The consequence is the point. When the basins overrun by, say, 140,000 — which is $140,000 \div 1,830,000 = 7.7\%$ of the deliverable — the department structure shows four small percentages against four large departmental budgets, none of which trips a 10 % reporting threshold. The overrun is real, it is material, and the structure has made it invisible.

— 10. Checklist

Take this into the WBS review. It is written to be answered by the person who built the structure, in front of the people who will have to use it.

Structure

- [] Can every level 2 element be stated as a noun with a completion condition?
- [] Does any level 2 element name a department, a discipline or a phase? If so, why, and is it recorded?
- [] Is level-of-effort work segregated, and what percentage of the baseline does it represent?
- [] Has any code ever been reused for different scope?

The 100 % rule

- [] Does every parent's budget equal the sum of its children, at every level, checked independently?
- [] Working upward, does every element have an unambiguous parent?
- [] Where two elements could both contain the same work, which one does, and is it written down?
- [] Where does rework charge — to the element it repairs, or to a general code?
- [] Are commissioning spares, temporary works, site clean-up and interface scope each in exactly one place?

Depth

- [] Can any work package run more than two reporting periods without an objective completion event?
- [] How many work packages are there, and what is the monthly statusing effort at your measured rate?
- [] Is undefined future scope held as a planning package rather than decomposed speculatively?
- [] Has the WBS been taken down to activity level? (If yes, it has absorbed the schedule.)

Dictionary and ownership

- [] Does every element have a dictionary entry with inclusions and exclusions?
- [] Does every entry name the acceptance evidence that will close the element?
- [] Does every element cross to at least one organisational unit, and does every crossing have a named manager?
- [] Do the control-account budgets at each crossing sum to the WBS element's budget?

— Related

- [BPG-01 – Building a project controls function from zero](#) — where the WBS sits in the dependency chain, and the control-account sizing test.
- [BPG-03 – Cost breakdown structure and the code of accounts](#) — the second axis: how cost type and resource class are carried as codes rather than as branches.
- [BPG-04 – Baselining and baseline change control](#) — what happens to the structure when scope is added, and why rolling wave is not a re-baseline.
- [BPG-06 – Progress measurement and rules of credit](#) — how the work packages defined here are credited with progress.
- [TPL-02 – Work breakdown structure and WBS dictionary](#) — the instrument, with every dictionary field defined.
- [TPL-03 – Cost breakdown structure and code of accounts](#) — the companion template for the coding axis.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 8 (Project Management Lifecycle) for decomposition and the 100 % rule, Domain 5 (Cost Management and Cost Control) for control accounts and work packages, and Domain 1 (Foundations of Accounting for Project Controls) for the mapping between scope structure and cost coding.

The 100 % rule is a long-established principle of scope decomposition, described here in the Institute's own words. The reporting-cycle depth test in §5 and the statusing-effort arithmetic in §9.3 are PCI recommended practice; the 12-minute input is an assumption of the example, not a benchmark. No external standard, table or diagram is reproduced.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

BPG-03 · BEST PRACTICE GUIDES

Cost breakdown structure and the code of accounts

The coding decision that has to be made before the first commitment is raised

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
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— Summary

How to design the cost breakdown structure and the code of accounts that every project transaction will carry: the difference between the scope axis and the cost axis, cost types and resource classes, segment design and the valid-combination matrix, how the structure maps to the general ledger without either side distorting the other, and why the decision cannot wait. Includes a worked comparison of designing the structure up front against retrofitting it eight months in, with every assumption named.

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Cost breakdown structure and the code of accounts

change your mind later.

In one paragraph. How to design the cost breakdown structure and the code of accounts that every project transaction will carry: the difference between the scope axis and the cost axis, cost types and resource classes, segment design and the valid-combination matrix, how the structure maps to the general ledger without either side distorting the other, and why the decision cannot wait. Includes a worked comparison of designing the structure up front against retrofitting it eight months in, with every assumption named.

Who this is for. Cost engineers and cost managers designing or inheriting a coding structure; finance business partners who own the general ledger the project has to reconcile to; project managers about to approve the first purchase orders on a new project.

— 1. Two questions, two axes

Every project cost report is answering one of two questions, and they need different structures.

What did the aeration basins cost? is a **scope** question. It is answered by the work breakdown structure (WBS), and [BPG-02 – The work breakdown structure](#) covers how to build one.

What did we spend on subcontract labour, across the whole project? is a **cost type** question. No amount of scope decomposition answers it, because subcontract labour appears in every branch. It is answered by the **cost breakdown structure (CBS)** — the decomposition of cost by element or type rather than by deliverable.

The two axes are independent, and a single transaction carries a position on both. An invoice for concrete supplied to the inlet works is coded to WBS element 1.3.2 and to cost type "permanent materials". Crossing the axes gives a matrix in which every cell is meaningful: the material cost of the inlet works, the total material cost of the project, and the total cost of the inlet works are all sums along a row, a column or both.

The **code of accounts** is the notation carrying both axes, plus whatever else the project needs to distinguish, on every transaction. Once a transaction is stamped with it, the stamp is what everything downstream sees.

— 2. Cost types

The cost type axis classifies cost by what was bought, and its top level is remarkably stable across sectors. A workable default:

Cost type	What sits in it	Why it is separated
Direct labour	Own-workforce hours charged to project scope	Hours are measurable independently of money; productivity analysis needs them
Subcontract	Work bought as a package, priced by the subcontractor	Commitment behaviour and accrual timing differ from labour
Permanent materials	Materials incorporated into the deliverable	Quantity-controllable; recoverable through valuation
Consumables	Materials consumed in the process	Not quantity-controllable against the deliverable
Plant and equipment	Owned or hired plant, and its running cost	Hire versus own is a distinct commercial decision
Temporary works	Scaffolding, formwork, site facilities	Real cost, no permanent deliverable to attach it to
Freight, duties and logistics	Movement of goods to site	Often the largest surprise on international scope
Professional services	Design, survey, inspection, specialist advice	Bought differently and accrued differently
Indirects and site overhead	Supervision, site establishment, welfare	Must be visible; must not be spread invisibly
Escalation and currency provision	Explicitly held allowances	Must never be mixed into base cost

Two design rules matter more than the list itself. **Direct labour is always split out, and always carries hours as well as money**, because hours are the earliest and most reliable productivity signal on labour-driven work and they are lost the moment labour is bought as a lump sum. And **allowances are never mixed into base cost**, because an escalation allowance buried inside a materials line cannot be released, tracked or reported, and it will be spent on something else.

Contingency and management reserve are held separately again — they are not cost types but reserves, and [BPG-10 – Contingency and management reserve](#) covers their treatment.

— 3. Resource classes

The resource class axis sits beneath cost type and answers what kind of resource. Within direct labour: welder, pipefitter, electrician, supervisor. Within plant: crane, excavator, generator.

It is optional and should be added only where a rate-and-quantity analysis will genuinely be performed, because it multiplies both the code space and the discipline needed to keep it clean. The test: if nobody will ever ask "how many welder-hours at what rate?", do not create a welder resource class. An unused segment does not stay empty — it fills with whatever appeared first in the drop-down list, and that is worse than nothing, because somebody will eventually report from it. Where resource classes are used, define them as rigorously as cost types: a definition, a unit of measure, and what does not belong in them.

— 4. Designing the code

A code of accounts is a concatenation of segments, each of which answers one question. A workable structure for a mid-sized project:

Segment	Length	Answers	Example
Project	2	Which project or contract	W4
WBS element	5	Which scope element	13302
Cost type	3	What was bought	SUB
Resource class	2	What kind of resource	CV
Transaction state	1	Commitment, accrual or actual	A

Four rules govern segment design.

One question per segment. A segment that sometimes means discipline and sometimes means phase is not a segment; it is two segments sharing a field, and every report built on it will need an exception list.

Fixed length, no intelligence beyond position. Variable-length segments break every string operation downstream. Embedding meaning in a digit — "the third character is 7 for offshore scope" — creates a rule that lives nowhere except in the head of the person who invented it.

Room to grow, but not much. A five-character WBS segment supports far more elements than any project needs; a three-character one runs out during execution, and running out means either overloading an existing code or extending the field, both of which break history.

A valid-combination matrix. The theoretical code space is the product of the segments, and most of it is nonsense. With 120 codeable WBS elements, 18 cost types and 6 resource classes, the space is $120 \times 18 \times 6 = 12,960$ combinations. Perhaps 1,400 of those are legitimate on any given project. The matrix that says which are permitted is the difference between a coding structure and a coding suggestion — without it, a requisition can be coded to "temporary works" against a design element, and nothing will stop it.

Name the owner of the code list, and make creation of a new code an authorised act. A code created ad hoc to get a purchase order through approval is the mechanism by which every structure decays.

— 5. Mapping to the general ledger

The project code of accounts and the general ledger chart of accounts are different instruments with different owners and different purposes, and forcing either to be the other produces a structure that serves neither.

The general ledger exists to produce financial statements. Its classifications follow financial reporting: what is capitalised and what is expensed, which categories must be disclosed separately, which period. It is owned by finance, subject to audit, and it changes rarely. The project code of accounts exists to produce management control. Its classifications follow forecasting and accountability: which scope, whose responsibility, what kind of resource, which state of commitment. It is

owned by the project and may legitimately be more granular than the ledger by an order of magnitude.

The correct relationship is a **defined many-to-one mapping** from project codes to ledger accounts, maintained as a controlled document, with two properties: every project code maps to exactly one ledger account, and the mapping is reconciled at every period close. Many project codes mapping to one ledger account is normal and healthy. One project code mapping to two ledger accounts is a defect — it means the code is carrying a distinction the ledger needs but the code does not express, and the split will be made by hand, differently, every month.

Two cautions. The treatment of what may be capitalised, how borrowing costs are handled, and when revenue may be recognised varies by reporting framework and by jurisdiction, and none of it should be assumed from one country's practice; the project's coding structure should carry enough information for finance to apply whatever framework governs, rather than pre-judging it. And where the project and the ledger disagree at close, the reconciliation is the deliverable — an unexplained difference between the cost report and the ledger is a finding, not a rounding.

— 6. Why the decision cannot wait

A purchase order is not a document; it is the head of a stream. Once raised, it generates goods receipts, invoices, accruals, retentions, variations and payment records, and every one of them inherits the code the purchase order carried. Change the code afterwards and you are not editing a field — you are re-coding a stream, in a system that usually requires a formal amendment to do it.

That is why the coding structure sits early in the dependency chain in [BPG-01 – Building a project controls function from zero](#), before tooling and before any reporting. The cost of the decision is roughly constant whenever it is made; the cost of reversing it grows with the transactions already posted and the commitments already open, and \$9 puts numbers on both sides.

Where the structure genuinely cannot be finalised early, there is a legitimate middle path: fix the segment layout and the cost-type list, which are stable, and populate the WBS segment's lower levels as scope is defined. What must not be deferred is the shape, because a shape change is what forces the re-code.

— 7. Direct, indirect and the allocation question

Some cost cannot be attributed to a single scope element without a rule: site supervision, site establishment, shared plant, quality assurance. There are three honest treatments and one dishonest one.

Charge it to a dedicated indirect element. Simple and transparent, and it keeps the deliverable accounts clean — but no deliverable then carries its full cost, so benchmarking against a fully-loaded rate needs a further step.

Allocate it on a defined driver. Spread supervision across deliverables in proportion to direct labour hours, or shared plant in proportion to plant-hours booked. Fully-loaded deliverable costs, at the price of a rule that must be documented, applied consistently and understood by every reader.

Do both, in different views — the indirect element for control, allocation only in a clearly labelled fully-loaded view for estimating. This is the Institute's recommended practice where the project can maintain two views without confusing them.

The dishonest treatment is to allocate without documenting the driver, or to change the driver mid-project. Both make period-on-period comparison meaningless, and both are usually discovered when somebody asks why a deliverable's unit rate moved without anything happening to it.

— 8. How this goes wrong

Coding is decided by whoever raises the first requisition. They select the code that gets it approved. Everyone after copies them, because it worked. Six months later the structure is whatever the first requisition happened to need.

The ledger is used as the project code of accounts. The project inherits finance's chart, which has no scope dimension, so cost can be reported by type and by period but never by deliverable. The scope dimension gets rebuilt in a spreadsheet, maintained by one person, reconciled to nothing.

A segment is overloaded. The cost type segment starts carrying discipline for a few elements because there was no discipline segment. Reports built on cost type now need an exception list, and that list is not in the report — it is in the analyst's head.

Escalation is buried in base rates. The estimate applied escalation inside the unit rates rather than as a separate allowance. Two years in, nobody can say how much of the spend was escalation and how much was quantity growth, so the forecast cannot distinguish a price problem from a scope problem.

Codes are created to solve approval problems. A cost that does not fit anywhere gets a new code rather than a decision. The list grows monotonically, half of it is used once, and the hierarchy stops rolling up cleanly.

Hours are lost. Labour is coded to money only, so the productivity signal — earned hours against burned hours — is unavailable, and the project loses its earliest indicator of a labour overrun.

The mapping to the ledger is undocumented. It lives as a formula in a workbook; the person who wrote it moves on; the monthly reconciliation becomes an exercise in re-deriving somebody else's intent, and the difference is written off as timing.

— 9. Worked example

Illustrative figures. All effort is in person-hours; monetary values are in generic currency units. No real project, organisation or system is implied. The figures below are the shape of the decision, not a benchmark — the inputs must be measured on your own project before the answer means anything.

9.1 The situation

A project is eight months into execution. The coding structure was deferred at start-up: a provisional two-segment code (project and a rough scope grouping) has been used, with no cost type and no resource class. A proper structure is now needed, because the client has asked for cost by cost type and the forecast cannot separate a price problem from a quantity problem.

Volumes at the point of decision.

Item	Volume
Cost transactions posted (invoice lines, timesheet lines, journals)	4,200
Open commitments requiring a coding amendment	190
Monthly cost reports already issued	12

9.2 The retrofit

Assumptions. 70 % of transactions map mechanically by a lookup rule on the old segment; the remaining 30 % need a human decision. A manual decision takes 4 minutes including the check. A purchase order coding amendment takes 25 minutes including supplier notification and internal approval. Restating an issued report so the trend is continuous takes 3 hours each. Two reconciliation cycles at the new code level are needed before the ledger and the cost report agree, at 16 hours each.

Task	Calculation	Hours
Build and test the crosswalk	Fixed effort	40.0
Manual re-coding	$4,200 \times 0.30 = 1,260 \text{ items} \times 4 \text{ min} = 5,040 \text{ min}$	84.0
Commitment amendments	$190 \times 25 \text{ min} = 4,750 \text{ min}$	79.2
Restating issued reports	$12 \times 3 \text{ h}$	36.0
Reconciliation to the ledger	$2 \times 16 \text{ h}$	32.0
Subtotal		271.2
Contingency at 20 %	271.2×0.20	54.2
Total		325.4

Checks: $4,200 \times 0.30 = 1,260$; $1,260 \times 4 = 5,040$ minutes; $5,040 \div 60 = 84.0$ hours. $190 \times 25 = 4,750$ minutes; $4,750 \div 60 = 79.17$, rounded to 79.2 hours. Subtotal $40.0 + 84.0 + 79.2 + 36.0 + 32.0 = 271.2$. Contingency $271.2 \times 0.20 = 54.24$, rounded to 54.2. Total $271.2 + 54.2 = 325.4$ hours.

Elapsed time. Assigning one person at 32 productive hours a week to the exercise: $325.4 \div 32 = 10.2$ weeks. That is roughly two and a half reporting cycles during which the comparative trend is unavailable, because the old codes and the new codes describe different things.

9.3 The up-front alternative

Assumptions. A two-day design workshop with three people — cost engineering, finance and procurement — at 7 productive hours a day. Configuration of the code list and the valid-combination matrix in the cost system, plus the mapping document to the ledger: 24 hours. Test postings and sign-off: 8 hours.

Task	Calculation	Hours
Design workshop	3 people × 2 days × 7 h	42.0
Configuration and mapping document	Fixed effort	24.0
Test postings and sign-off	Fixed effort	8.0
Total		74.0

Check: $42.0 + 24.0 + 8.0 = 74.0$ hours.

9.4 The comparison

```
Ratio = retrofit effort ÷ up-front effort
      = 325.4 ÷ 74.0
      = 4.4
```

The retrofit costs about **4.4 times** the up-front design in direct effort. Expressed as an addition rather than a ratio, the avoidable cost is $325.4 - 74.0 = \mathbf{251.4 \text{ hours}}$ — because the up-front work has to be done either way; deferring it does not remove it, it only adds the re-coding to it.

What this does not include, and it matters more than what it does. The 10.2 weeks of broken comparative reporting has no line in the table, and on most projects it is the larger cost: for two and a half cycles the project cannot answer "is this getting better or worse?", which is the only question the cost report exists to answer. Nor does the table include the decisions taken during those cycles on a cost report that cannot separate price from quantity.

Sensitivity. The dominant inputs are the transaction volume and the manual proportion. At 8,000 transactions rather than 4,200, with everything else unchanged, the manual re-coding becomes $8,000 \times 0.30 = 2,400$ items $\times 4$ minutes = 9,600 minutes = 160.0 hours, and the subtotal becomes $40.0 + 160.0 + 79.2 + 36.0 + 32.0 = 347.2$ hours, giving a total of $347.2 \times 1.20 = \mathbf{416.6 \text{ hours}}$ and a ratio of $416.6 \div 74.0 = \mathbf{5.6}$. The lesson is directional and robust: the retrofit cost scales with elapsed time, and the up-front cost does not.

10. Checklist

Take this into the coding design workshop, or into the review of a structure you have inherited.

Design

- [] Does every segment answer exactly one question, and is that question written down?
- [] Are all segments fixed-length, with no meaning embedded in individual characters?
- [] Is direct labour separated, and does it carry hours as well as money?
- [] Are escalation, currency provision and any other allowance held outside base cost?
- [] Does a resource class exist that nobody will ever report from? (If so, delete it now.)
- [] Is there a valid-combination matrix, and does the system enforce it?

Ownership

- [] Who may authorise a new code, and how long does it take? (If it is faster to invent one, they will.)

- [] Is the code list a controlled document with a version and an owner?
- [] Are retired codes retired permanently, never reissued?

The ledger

- [] Does every project code map to exactly one general ledger account?
- [] Is the mapping a controlled document rather than a workbook formula?
- [] Is it reconciled every period, with differences explained rather than written off as timing?
- [] Has finance confirmed the structure carries enough information for their reporting framework, without the project pre-judging the treatment?

Indirects

- [] Is every indirect cost either charged to a named indirect element or allocated on a documented driver?
- [] If costs are allocated, is the driver written down, and has it changed during the project?
- [] Is there a clearly labelled distinction between the control view and any fully-loaded view?

If you are considering deferring the decision

- [] How many transactions are currently posting each month?
- [] How many commitments are open, and what does an amendment cost in your system?
- [] Run §9's arithmetic with your own numbers, and take the answer to the person asking for the delay.

— Related

- [BPG-01 – Building a project controls function from zero](#) — where the coding decision sits in the sequence, and what else it blocks.
- [BPG-02 – The work breakdown structure](#) — the scope axis this structure crosses.
- [BPG-07 – Accruals and cut-off discipline](#) — the transaction-state segment in practice, and why commitment, accrual and actual must be distinguishable.
- [BPG-11 – Change orders and variations](#) — how new scope acquires a code without breaking the structure.
- [TPL-03 – Cost breakdown structure and code of accounts](#) — the instrument, with segment definitions and a valid-combination matrix.
- [TPL-02 – Work breakdown structure and WBS dictionary](#) — the companion template for the scope axis.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 5 (Cost Management and Cost Control) for the cost breakdown structure and the commitment-accrual-actual states, Domain 1 (Foundations of Accounting for Project Controls) for cost coding and the relationship to the general ledger, and Domain 11 (Business Process Cycles) for the procure-to-pay controls that the coding structure depends on.

The retrofit arithmetic in §9 is constructed for this guide from stated assumptions. It is not survey data and not a benchmark; the inputs must be measured on your own project. Where financial reporting frameworks govern capitalisation, revenue recognition or disclosure, the treatment varies by framework and by jurisdiction, and none is described here as universal. No external standard, chart of accounts or table is reproduced.

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BPG-04 · BEST PRACTICE GUIDES

Baselining and baseline change control

What a baseline actually freezes, what the gate has to test,
and why a variance you baseline away is still there

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

What a project baseline is composed of and what it genuinely freezes, what a change-control gate must test before it approves anything, how re-baselining differs from rolling wave, and why moving a baseline to absorb a variance destroys the only signal the project had. Includes a worked comparison of the correct treatment of an approved scope addition against a re-baseline to forecast, showing what each does to the reported variance and to the funding question underneath it.

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Baselining and baseline change control

visible.

In one paragraph. What a project baseline is composed of and what it genuinely freezes, what a change-control gate must test before it approves anything, how re-baselining differs from rolling wave, and why moving a baseline to absorb a variance destroys the only signal the project had. Includes a worked comparison of the correct treatment of an approved scope addition against a re-baseline to forecast, showing what each does to the reported variance and to the funding question underneath it.

Who this is for. Project controls managers and control account managers who maintain a baseline; project managers and sponsors who sit on the change board; anyone being asked to approve a re-baseline and wondering what they are actually approving.

— 1. A baseline is a claim, not a number

A baseline is the project's formal statement that this scope, delivered this way, will take this long and cost this much. Its value comes entirely from being fixed: a number that moves whenever performance disappoints measures nothing, because every comparison against it is a comparison with itself.

That is the whole argument, and it is worth stating before the mechanics, because most baseline failures are not procedural. They happen because somebody found the variance uncomfortable and the baseline was the softest thing in the room.

A useful test for whether a project has a baseline at all: can you produce, in under an hour, the exact version of the scope, schedule and cost plan that the current variance is measured against, together with the list of changes approved into it since sanction? If not, the project has a plan and a forecast but nothing in between.

— 2. What the baseline is composed of

Three components, approved together, each with its own version.

The scope baseline — the work breakdown structure (WBS), the dictionary, and the scope statement it implements. This is the reference for every subsequent question about whether something is a change or was always included. [BPG-02 – The work breakdown structure](#) covers its construction.

The schedule baseline — the network, durations, logic, calendars and constraints as approved, with the schedule basis document that records what the durations assume. A baselined schedule with no basis document is a set of dates whose reasoning has been discarded.

The cost baseline — the budget, time-phased across the schedule at control-account level. This is the planned value curve, and its time-phasing is what makes a cost variance distinguishable from a timing difference.

Two totals sit around these. **Budget at completion** is the sum of the time-phased budgets plus distributed contingency where the organisation's convention places it inside the baseline. Management reserve normally sits outside the performance measurement baseline and is not available to a control account without a formal transfer. Conventions differ legitimately between organisations, so the rule is to state yours in the controls execution plan and apply it consistently rather than to assume; the treatment of both reserves is owned by [BPG-10 – Contingency and management reserve](#).

— 3. What a baseline actually freezes

This is where most disputes originate, because the baseline freezes less than people assume and more than they would like.

It freezes the measurement reference. Every variance reported from now on is measured against this version. That is its primary function.

It freezes the scope commitment. Work not in the scope baseline is, by definition, a change — which is why an ambiguous WBS dictionary entry is a commercial exposure and not just untidiness.

It freezes the time-phasing at control-account level. Not at activity level: activities can and will move within a control account as the schedule is updated, and that movement is progress management, not baseline change. The distinction between "the schedule has been updated" and "the schedule baseline has been changed" is one every planner must be able to state on demand.

It does not freeze the forecast. The forecast is expected to move every period; that is what makes it a forecast. Confusing the two produces the request to "update the baseline to match the forecast", which is §7's subject.

It does not freeze the method. How a work package earns progress can be corrected if the rule was wrong, though the correction must be applied to the whole package and disclosed, because it will move the reported earned value without anything physical happening.

It does not freeze reality. A baseline that has become impossible should be said to be impossible. The correct response is a change request with a defensible recovery plan, not a quiet re-phasing.

— 4. The change-control gate

A change-control gate is not an approval meeting; it is a test. A change request that passes it should have answered six questions in writing, and a board that approves without them is rubber-stamping.

1. **What exactly changes in the scope baseline?** Named WBS elements, with dictionary amendments drafted. "Additional works to the inlet structure" is not an answer.
2. **What is the cost impact, in full?** Direct cost, indirect and time-related cost, escalation on the new scope, and the effect on any allowance. A cost impact stated without its time-related component is usually understated.

3. **What is the schedule impact?** Not "four weeks" but which activities, which logic, and what happens to the critical path and to float elsewhere. A change consuming somebody else's float is still consuming a project asset.
4. **Where does the money come from?** Contingency drawdown, management reserve transfer, client variation, or an unfunded increase in budget at completion. "To be confirmed" means the change is not ready.
5. **What is the impact of not approving it?** The counterfactual is the only way a board can distinguish a genuine requirement from a preference.
6. **What is the effect on reported performance?** Specifically: does approval change any historical variance? The correct answer is almost always no, and if the proposal says yes, §7 applies.

Authority thresholds should be set so that the great majority of changes are approved at the lowest competent level, with a documented escalation. A gate that sends everything to the sponsor is a gate that will be bypassed. [TPL-04 – Baseline change request](#) provides the instrument.

The trend register is the front end of the gate. A trend is a change that has been identified but not yet approved — a design development, a quantity growth, an early warning from a subcontractor. It is logged with an estimated value the moment it is identified, and it appears in the forecast before it appears in the baseline. Projects that log trends only when they become change requests report a forecast that jumps in steps, always upward, always late. [BPG-11 – Change orders and variations](#) covers the commercial instrument; the trend register is the internal one, and it should run ahead of it.

— 5. Rolling wave is not re-baselining

Rolling wave planning holds future scope as **planning packages** — real elements with budget and approximate duration, but no work-package detail — and converts them to detailed work packages as the scope becomes definable.

The conversion moves budget within the baseline. The budget at completion is unchanged, the total time-phased budget is unchanged, and no historical variance is affected. It is a planning act, not a change, and it does not go through the change gate — though it should be recorded, because a conversion that changes the phasing of a planning package's budget will move the planned value curve and therefore the schedule performance index.

Two disciplines keep this honest. The conversion must be **budget-neutral**: if the detailed estimate of the newly defined work exceeds the planning package's budget, that excess is a change and goes through the gate — it does not get absorbed by quietly enlarging the package. And planning packages must have a **conversion deadline**: undefined scope held indefinitely is not rolling wave, it is undistributed budget with no plan, and it hides both scope risk and estimate risk until very late.

— 6. Re-baselining

There are legitimate reasons to re-baseline. A major approved scope change that reshapes the project; a contractual reset; a suspension and restart; an original baseline so far from reality that variance reporting against it has stopped conveying information. In each case the honest test is whether the existing baseline can still support a decision, not whether it is uncomfortable.

A re-baseline done properly has five properties. The old baseline is **archived and remains retrievable**, not overwritten. The **reason is documented** and approved at the level that approved the original. The **cumulative variance to date is stated explicitly** in the approval — the number that is about to disappear from the report is written into the record. The **funding position is resolved**: if the new baseline is larger, the source of the additional funds is named. And **reporting continues against both** for at least one cycle, so readers can see the step.

Where none of those five is present, what is happening is not a re-baseline.

— 7. Keeping the variance visible

The most consequential discipline in this guide is the refusal to let a baseline absorb a performance problem.

The request is always reasonable-sounding. The baseline is out of date; the variance is historical; the team has a credible recovery plan; reporting a large negative variance every month is demoralising and distracts from the work. All of that can be true, and the answer is still no — because the variance is the project's only quantitative record that performance has not matched the plan, and removing it removes the basis for every remaining forecast.

Three consequences follow from baselining a variance away, and §9 demonstrates each.

The forecast loses its evidence. A cost performance index of 0.88 is the empirical basis for expecting the remaining work to cost more than budgeted. Reset it to 1.00 and the forecast has to be justified from optimism instead of from data. [BPG-09 – Estimate at completion](#) depends on this history existing.

The funding gap survives. Moving the baseline changes what the report compares against; it does not change what the project will cost or what has been sanctioned. The difference between the new baseline and the authorised funding is a real gap that now belongs to nobody.

The trend is destroyed. Performance trend — is this getting worse, stable, or recovering? — is the single most useful thing a controls function produces, and it requires an unbroken series. A reset breaks it, and no amount of subsequent reporting reconstructs it.

The alternative is neither difficult nor dramatic: report the variance, report the cause, report the recovery plan and the evidence for it, and let the trend show whether the recovery is happening. A board that sees the same negative variance narrowing over four periods is being told something true. A board that sees it vanish is being told nothing.

— 8. How this goes wrong

The baseline is set before the scope is decomposed. A sanction total is declared the baseline. With no time-phasing, the first variance question — over, or early? — is unanswerable, and the project quietly rebuilds the baseline retrospectively from whatever was spent.

Approved changes are not incorporated. Change requests are approved but never applied to the baseline. The baseline and the authorised scope drift apart, and the reported variance becomes the sum of a real performance problem and an administrative backlog, with no way to separate them.

Historical earned value is restated when scope is added. New scope is added to the baseline and backdated into the earlier periods, so past performance improves without anything having happened. Added scope enters from the approval date forward.

The schedule is re-baselined and the cost baseline is not. The two now describe different plans. Every schedule performance index computed from them is meaningless, and nobody notices for a quarter because the numbers still look like numbers.

Contingency is drawn down without a change record. Money moves from the allowance into a control account to cover an overrun. The control account now shows no variance, the allowance is depleted, and there is no record connecting the two. This is baselining away a variance with extra steps.

"Re-baseline" means "delete the history". The old baseline is overwritten in the tool. Nobody can now state what the project originally committed to, which makes the closeout lessons exercise in [BPG-20](#) impossible to run.

Planning packages never convert. Undefined budget sits in planning packages until late execution, by which time the estimate is worse and the schedule is fixed. The rolling wave discipline exists to prevent exactly this, and it works only with conversion deadlines.

— 9. Worked example

Illustrative figures. Generic currency units. Period: month 8 of a monthly reporting cycle. Values are cumulative to the data date unless stated. Rounding: indices to three decimal places, currency to the nearest 1,000 where a rounded figure is presented. No real project is implied.

9.1 The position at month 8

Measure	Value
Budget at completion (BAC)	6,000,000
Planned value (PV)	2,400,000
Earned value (EV)	2,040,000
Actual cost (AC)	2,310,000

Cost variance (CV), schedule variance (SV), cost performance index (CPI) and schedule performance index (SPI) follow directly:

$$\begin{aligned}
 CV &= EV - AC = 2,040,000 - 2,310,000 = -270,000 \\
 SV &= EV - PV = 2,040,000 - 2,400,000 = -360,000 \\
 CPI &= EV \div AC = 2,040,000 \div 2,310,000 = 0.883 \\
 SPI &= EV \div PV = 2,040,000 \div 2,400,000 = 0.850
 \end{aligned}$$

Forecasting on the assumption that performance to date persists — one method among several, and [BPG-09 – Estimate at completion](#) owns the choice — gives an estimate at completion (EAC) and a variance at completion (VAC):

$$\begin{aligned} \text{EAC} &= \text{BAC} \div \text{CPI} = 6,000,000 \div 0.883116 = 6,794,000 \quad (\text{to the nearest } 1,000) \\ \text{VAC} &= \text{BAC} - \text{EAC} = 6,000,000 - 6,794,000 = -794,000 \end{aligned}$$

The unrounded index $2,040,000 \div 2,310,000 = 0.883116$ is used in the division; using the rounded 0.883 would give 6,795,000, a difference of 1,000 that illustrates why indices are carried unrounded through a calculation and rounded only at presentation.

9.2 An approved scope addition, treated correctly

A change request for additional scope valued at **480,000** is approved at the month-8 board, funded by a client variation. The correct treatment adds the new scope's time-phased budget to the baseline from the approval date forward, and leaves every historical figure alone.

$$\begin{aligned} \text{New BAC} &= 6,000,000 + 480,000 = 6,480,000 \\ \text{EV, AC and the historical CPI are unchanged: CPI remains } 0.883 \\ \text{EAC} &= \text{AC} + (\text{BAC} - \text{EV}) \div \text{CPI} \\ &= 2,310,000 + (6,480,000 - 2,040,000) \div 0.883116 \\ &= 2,310,000 + 4,440,000 \div 0.883116 \\ &= 2,310,000 + 5,028,000 \\ &= 7,338,000 \quad (\text{to the nearest } 1,000) \\ \text{VAC} &= 6,480,000 - 7,338,000 = -858,000 \end{aligned}$$

The forecast overrun has grown from 794,000 to 858,000 — because the added scope is assumed to be delivered at the same efficiency as the work so far, which is an assumption the board should be asked to accept or reject explicitly. The performance signal is intact: CPI is still 0.883, the trend is continuous, and the board can see both that scope was added and that the project is running over.

9.3 The same month, re-baselined to forecast

Now suppose the request is instead to re-baseline the account to the current forecast, resetting earned value to actual cost at the data date so that the account "starts clean".

Measure	Correct treatment (§9.2)	Re-baselined to forecast
BAC	6,480,000	7,338,000
CPI to date	0.883	1.000
CV to date	-270,000	0
VAC	-858,000	0
Forecast cost of the project	7,338,000	7,338,000
Authorised funding	6,480,000	6,480,000
Unresolved funding gap	858,000, reported	858,000, not reported

Both columns describe the same project, costing the same money. Nothing physical differs. What differs is that in the right-hand column the 858,000 gap between what the project will cost and what has been authorised no longer appears anywhere in the performance report — and the 0.883 that was the evidence for expecting it has been replaced by a 1.000 that is evidence of nothing.

Check: $7,338,000 - 6,480,000 = 858,000$, which equals the VAC in the correct treatment. The gap is the same number under both treatments, which is the point.

9.4 Rolling wave, for contrast

The same project holds a planning package of **1,200,000** for later scope. In month 9 that scope becomes definable and is decomposed into five work packages:

$$210,000 + 260,000 + 190,000 + 300,000 + 240,000 = 1,200,000$$

Budget at completion is unchanged at 6,480,000. No historical variance moves. No change request is required, because no scope and no budget has changed — only the level of detail at which it is planned. Had the detailed estimate come to 1,340,000, the excess of $1,340,000 - 1,200,000 = 140,000$ would be a change, and would go through the gate in §4 with its funding source named.

— 10. Checklist

Take this into the change board, or into the meeting where a re-baseline is being proposed.

Before approving any change

- [] Which named WBS elements change, and are the dictionary amendments drafted?
- [] Does the cost impact include time-related cost, escalation on the new scope, and any allowance effect?
- [] Which activities and which logic change, and what happens to the critical path and to others' float?
- [] Where does the money come from — contingency, reserve, client variation, or an unfunded increase?
- [] What happens if this is not approved?
- [] Does approval alter any historical variance? (If yes, ask why before anything else.)

Baseline hygiene

- [] Can the current baseline version and its change history be produced in under an hour?
- [] Have all approved changes actually been incorporated, and how many are outstanding?
- [] Do the schedule baseline and the cost baseline describe the same plan?
- [] Is there a schedule basis document, and was it updated at the last change?
- [] Is management reserve outside the performance measurement baseline, and is the convention written down?

Rolling wave

- [] Does every planning package have a conversion deadline?
- [] Have conversions been budget-neutral, and where they were not, did the excess go through the gate?
- [] What proportion of the budget at completion is still undefined?

If a re-baseline is proposed

- [] Is the old baseline archived and retrievable, or is it about to be overwritten?

- [] Is the cumulative variance being written off stated as a number in the approval?
 - [] Is the funding gap between the new baseline and the authorised amount identified and owned?
 - [] Will both baselines be reported for at least one cycle?
 - [] If the answer to any of the above is no: what is being proposed is not a re-baseline.
-

— Related

- [BPG-02 – The work breakdown structure](#) — the scope baseline this depends on, and why dictionary exclusions are a change-control asset.
- [BPG-06 – Progress measurement and rules of credit](#) — the earning rules that make the baseline measurable.
- [BPG-08 – Earned value in practice](#) — how the variances in §9 are computed and read.
- [BPG-10 – Contingency and management reserve](#) — the reserve conventions referenced in §2, and drawdown discipline.
- [BPG-11 – Change orders and variations](#) — the commercial instrument that sits alongside the internal gate.
- [TPL-04 – Baseline change request](#) — the instrument implementing the six-question gate in §4.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 3 (Budgeting and Forecasting) for the time-phased cost baseline and reserves, Domain 5 (Cost Management and Cost Control) for change control and cost impact assessment, Domain 8 (Project Management Lifecycle) for the scope baseline and integrated change control, and Domain 10 (Project Scheduling) for the schedule baseline and its basis document.

The six-question gate in §4 and the five properties of a legitimate re-baseline in §6 are PCI recommended practice. Conventions on whether contingency sits inside or outside the performance measurement baseline differ legitimately between organisations and between published bodies of practice; this guide names the difference rather than asserting one convention as universal. No external standard is reproduced.

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BPG-05 · BEST PRACTICE GUIDES

Schedule quality — a practical review

The checks that tell you whether a schedule can be trusted to recalculate, and how to report the answer

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

How to review a schedule you did not build and decide whether its dates are results or assertions: logic completeness and open ends, constraints, lags and leads, calendars, negative and excessive float, out-of-sequence progress and update integrity, duration distribution, and a walk of the critical path. Every check is described in the Institute's own words, with the difference between a rule and a convention stated explicitly. Includes a full review of a worked schedule and the verdict it supports.

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Schedule quality — a practical review

In one paragraph. How to review a schedule you did not build and decide whether its dates are results or assertions: logic completeness and open ends, constraints, lags and leads, calendars, negative and excessive float, out-of-sequence progress and update integrity, duration distribution, and a walk of the critical path. Every check is described in the Institute's own words, with the difference between a rule and a convention stated explicitly. Includes a full review of a worked schedule and the verdict it supports.

Who this is for. Planners and schedulers inheriting a schedule; project controls managers who have to sign a baseline; anyone asked to accept a contractor's programme, or to say whether a completion date can be relied on.

— 1. What the review is actually testing

A critical path method schedule earns its authority from one property: when something changes, it recalculates, and the new dates are consequences of the change rather than opinions about it. A schedule quality review tests whether that property holds.

It is not a test of whether the plan is good. A schedule can pass every check here and still describe a sequence that will not work, durations that are heroic and a completion date nobody believes — those are questions for the schedule basis, the estimators and the delivery team. Conversely, a schedule can fail several checks and still have roughly the right end date, arrived at by judgement.

The distinction determines what the review can conclude. A schedule that fails these checks is not necessarily wrong; it is **not defensible**, which is a different and more actionable finding — its dates cannot be shown to follow from anything. That matters when the schedule is used to support a delay claim, justify an acceleration cost, set a contractual completion date or feed a quantitative risk analysis, all of which assume a network that recalculates honestly.

A word about thresholds. Several checks below are naturally expressed as a percentage: how many open ends is too many, how many constraints are acceptable. There is no universal answer, and this guide does not pretend otherwise. Where a number appears, it is either a starting point the Institute recommends agreeing and recording before the review, or it is an observation from the worked example. No threshold here is attributed to any published standard, and none should be quoted as though a governing body had set it. The professional discipline is to agree your thresholds with the schedule's owner in advance, record them, and apply them consistently — an argument about the threshold after the result is known is not a review.

— 2. Before you open the file

Three preconditions determine whether the review will mean anything.

The data date. Every metric below is relative to it. Confirm it, and confirm that it matches the period the progress report covers. A schedule progressed to a different date than the cost report is a finding before any check is run.

The schedule basis. The controlled document recording what the baseline assumes — calendars and their justification, productivity and crew assumptions behind key durations, sequencing rationale, exclusions, third-party dependencies, approvals. Without it, every subsequent question ("why is this duration 40 days?") has no answer, and the review can only observe structure, never judgement. Its absence is itself the most significant finding a review can make.

The calendars in use. List them before anything else. Calendar effects are invisible on a printed bar chart and change elapsed durations without changing the duration field, so a review that has not enumerated the calendars is reviewing something it cannot see.

— 3. Logic completeness and open ends

Every activity except the project's genuine start and finish should have at least one predecessor and at least one successor. An activity missing either is an **open end**: it floats free of the network, so when reality changes around it, its dates do not move. The forward and backward passes quietly stop being true for that activity and, more damagingly, for anything downstream of it.

Two refinements matter in practice. First, **state the base**. Open ends as a percentage of all activities and open ends as a percentage of incomplete activities are different numbers, and the second is usually the more useful, because a completed activity's missing successor no longer affects the forecast. Second, **exclude the legitimate ends explicitly** rather than silently — a schedule with three "project finish" milestones has a structural problem the count would otherwise hide.

Logic density — total relationships divided by total activities — is a coarse companion measure. A value close to 1.0 suggests a chain with little cross-linking, which usually means real dependencies are missing. A very high value can mean redundant logic that makes the network hard to reason about. Density is a prompt to look, never a verdict on its own.

Relationship types. Finish-to-start is the type that most reliably expresses a physical dependency. Start-to-start and finish-to-finish pairs are legitimate and often necessary for overlapping work, but a start-to-start without a matching finish-to-finish leaves the successor's finish unconstrained, which is an open end in disguise. Start-to-finish relationships are rare enough that each one should be individually justified. Report the mix, and interrogate the non-finish-to-start share rather than assuming it is wrong.

— 4. Constraints

A date constraint overrides the network. That is its purpose and its danger: a constrained activity stops responding to its predecessors, so the schedule reports a date it has been told rather than a date it has computed.

Some constraints are legitimate — a contractual milestone, a permit date, a fixed possession window, a client-supplied item with a known delivery date. Each should be individually justified in the schedule basis, naming the external fact that fixes it.

Distinguish **soft** constraints, which restrict a date without overriding logic (start no earlier than, finish no later than), from **hard** ones, which pin a date outright (must start on, must finish on).

Hard constraints are the ones that generate silent nonsense: an activity pinned to a date its predecessors cannot support does not report a conflict, it reports negative float somewhere else, or nothing at all.

The review question is not "how many constraints?" but "how many constrained activities lie on the longest path?" — because those are the ones whose dates are assertions. A schedule can carry a hundred constraints on peripheral work and still recalculate honestly through its critical path; one hard constraint on the critical path means the critical path is partly declared.

— 5. Lags, leads and calendars

Lags inject elapsed time that no activity owns. A finish-to-start relationship with a fifteen-day lag adds fifteen days that have no resource, no progress measurement, no owner and no scrutiny. Some lags are physically real — concrete curing, paint drying, a mandated notice period — and those should be justified in physical terms. Where a lag represents work, it should be an activity.

The review approach is threshold-based and simple: expose **every** lag above an agreed length, require a written physical justification for each, and convert anything unjustified into a resourced activity. Then sum the lag on the critical path, because that is the number that shows how much of the project's duration is unowned. §9 does this, and the result is usually the most persuasive line in the report.

Leads — negative lags — allow a successor to start before its predecessor finishes. They are almost always a modelling shortcut for an overlap that should be expressed as start-to-start with an appropriate lag, and they behave unpredictably when progress is applied out of sequence. Treat every lead as a finding requiring justification.

Calendars change elapsed duration without changing the duration field: moving an activity from a five-day to a seven-day calendar shortens its elapsed time with no visible change anywhere. Conventions also differ on which calendar a lag follows, so the same numeric lag can mean different elapsed times in different parts of one schedule. Enumerate every calendar; confirm every assignment is deliberate rather than inherited; fix the lag-calendar convention once, in writing; and confirm that seasonal non-work days on weather-sensitive work sit in a calendar rather than buried in durations.

— 6. Float

Negative float means a late date earlier than an early date — the schedule is telling you it cannot meet a constraint or a deadline as modelled. It is never a metric to be normalised. Either the constraint is wrong, the logic is wrong, or the project is already late and nobody has said so. Report the worst value and the count, and identify which of the three explanations applies.

Excessive float is the more interesting signal, because it usually indicates missing logic rather than genuine slack: an activity with no successor that needs it shows enormous total float and means nothing. Cluster the high-float activities and look at what they share — frequently a whole area or subcontract package that was never linked in. What counts as excessive depends on project duration, so express the threshold relative to remaining duration and agree it before the review.

— 7. Update integrity

These checks apply to a progressed schedule and are the fastest way to find out whether the update is trustworthy.

Actuals beyond the data date. An activity showing an actual start or finish later than the data date is a data error, not a judgement. There is no acceptable count other than zero. **Remaining duration on complete activities** and **progress on activities with no actual start** are the same class of error: both indicate the update was typed rather than derived.

Out-of-sequence progress — a successor started before its finish-to-start predecessor finished — means either the logic was wrong or the update is. Either way the current critical path is suspect, because the tool's handling of out-of-sequence work depends on a setting most readers of the schedule have never seen. Report the count, and check the setting.

Stale updates. Activities whose remaining duration has not moved for several periods while showing progress are work that is being progressed on paper and not in fact.

— 8. Durations and the critical path

Duration distribution. Long remaining durations hide detail: a 60-day activity reports one percentage and can absorb months of slippage before its finish date moves. Very short ones in large numbers inflate maintenance without adding control. Express the check against the reporting cycle — an activity whose remaining duration exceeds two reporting periods cannot be meaningfully statused — and see [BPG-06 – Progress measurement and rules of credit](#) for the ones that must stay long.

The critical path walk is the check that no metric replaces, and the only one that requires judgement rather than a query. Print the longest path and read it end to end, asking of each link: does this activity physically have to finish before that one starts? A credible critical path tells a story a site manager would recognise. An incredible one jumps between unrelated areas, passes through administrative activities, or runs through a chain of lags and constraints rather than work.

Three specific tests. Does the path run through **activities or through lags and constraints**? What proportion of its duration is lag (§9 computes this)? And does it pass through **level-of-effort or management activities** — "project management", "monthly reporting" — which is a sign the network is being held together by activities that describe overhead rather than production.

— 9. How this goes wrong

The review becomes a metric report. Twelve percentages are computed, tabulated and sent; nobody walks the critical path, so the one finding that would have changed a decision is missed. Metrics locate the questions; they do not answer them.

Thresholds are argued after the fact. The reviewer reports 8 % constraints and the owner says the acceptable level is 10 %. Agree thresholds in writing beforehand and the conversation is about the schedule rather than about the reviewer.

A failed check is treated as a verdict. A schedule with open ends is reported as "wrong". The finding is that its dates cannot be shown to follow from its logic — defensibility, not accuracy.

The schedule is fixed to pass the checks. Open ends are closed by linking activities to the project finish milestone; constraints are converted to long lags; high float is removed by adding logic that does not exist. Every metric improves and the schedule becomes less true. This is the most damaging failure in this guide, because it is invisible in the next review.

Calendars are never examined. The review covers logic, constraints and float, and never lists the calendars. An activity on a seven-day calendar in an otherwise five-day schedule quietly shortens the critical path, and no metric in the report would show it.

Out-of-sequence progress is normalised. A recurring count of out-of-sequence activities is accepted as "how the site works". It may be — in which case the logic is wrong and should be corrected, because the network is now describing a sequence that is not being followed.

The review has no owner and no follow-up. Findings are issued, the next baseline is accepted anyway, and the same findings recur. A review without a named person accountable for closing each finding is an audit trail for someone else's later argument.

— 10. Worked example

Illustrative figures. A single progressed schedule at one data date. Percentages are stated with their base. Working-day durations on a five-day calendar unless otherwise stated. No real project is implied, and no threshold below is attributed to any published standard — each is a starting point agreed with the schedule's owner before the review.

10.1 The population

Item	Count
Total activities	1,480
— of which milestones	96
Complete	612
In progress	174
Not started	694
Total relationships	3,120

Check: $612 + 174 + 694 = 1,480$. Incomplete activities = $174 + 694 = 868$. Activities carrying progress = $612 + 174 = 786$.

10.2 Logic

Logic density = relationships ÷ activities = $3,120 \div 1,480 = 2.11$

Relationship mix: finish-to-start 2,410; start-to-start 428; finish-to-finish 233; start-to-finish 49. Check: $2,410 + 428 + 233 + 49 = 3,120$.

Non-finish-to-start share = $(3,120 - 2,410) \div 3,120 = 710 \div 3,120 = 22.8 \%$

Open ends: 63 activities lack a predecessor or a successor. Against all activities that is $63 \div 1,480 = 4.3 \%$. Two of the 63 are the project's genuine start and finish milestones, so the meaningful figure excludes them on both sides of the ratio: $61 \div 1,478 = 4.1 \%$. The 49 start-to-finish relationships are each a separate question, since that type is rare enough to require justification one by one.

10.3 Constraints, lags and leads

Check	Finding
Activities carrying a date constraint	118 ($118 \div 1,480 = 8.0 \%$)
— of which hard constraints	22
Relationships carrying a lag	274 ($274 \div 3,120 = 8.8 \%$)
— of which exceed 10 working days	41, totalling 690 days
Relationships carrying a lead (negative lag)	37
Calendars in use	4 (five-day, six-day, seven-day, shutdown)

Mean length of the 41 long lags = $690 \div 41 = 16.8$ working days. Each requires a physical justification; the 37 leads each require a justification or conversion to an explicit overlap.

10.4 Float and update integrity

Check	Finding
Activities with negative total float	84; worst value -27 days
Activities with total float above 60 working days	212 ($212 \div 1,480 = 14.3 \%$)
Incomplete activities with remaining duration above 44 working days	133 ($133 \div 868 = 15.3 \%$)
Out-of-sequence progress	46 ($46 \div 786 = 5.9 \%$ of activities carrying progress)
Actual dates later than the data date	9

The 44-working-day threshold is two reporting periods on a monthly cycle with a five-day calendar, which is where the check comes from rather than from any published figure.

10.5 The critical path

The longest path returned by the tool contains **41 activities** and spans **318 working days** from the data date to project completion. Reading it end to end:

- **11 of the 41** activities carry a date constraint.
- **5 relationships** on the path carry lags, totalling **62 working days**.

Lag share of the critical path = $62 \div 318 = 19.5 \%$

Nearly a fifth of the remaining critical path is elapsed time that no activity owns, that no resource is assigned to, that no progress report mentions and that nobody is accountable for delivering.

That single number does more to explain why the completion date has not moved in four months than every other metric in this review combined.

10.6 The verdict

The schedule is **not currently defensible as a forecast**, for three reasons that are all remediable: 9 actual dates beyond the data date are a data error with no acceptable count other than zero; 37 leads make out-of-sequence behaviour unpredictable; and 62 working days of unjustified lag on the critical path mean the completion date is partly asserted rather than computed.

It may still be approximately right about the end date. That is not the same claim, and a review that conflates the two helps nobody. The report should say what it can support: these dates cannot at present be shown to follow from this network, and here are the six things that would change that.

— 11. Checklist

Take this into the schedule review. Agree every threshold with the schedule's owner before you run a single query, and record what you agreed.

Preconditions

- [] What is the data date, and does it match the cost report's cut-off?
- [] Is there a schedule basis document, and when was it last updated?
- [] How many calendars are in use, and is every assignment deliberate?
- [] Which calendar do lags follow, and is that written down anywhere?

Logic

- [] Open ends: how many, against which base, and how many are on the longest path?
- [] Logic density, and does a low value point to missing cross-links?
- [] What is the non-finish-to-start share, and is every start-to-finish relationship justified?
- [] Does every start-to-start have a matching finish-to-finish?

Steering instruments

- [] How many constrained activities, how many hard, and how many on the longest path?
- [] Is every constraint justified against a named external fact in the schedule basis?
- [] Every lag above the agreed length: listed, with a physical justification each?
- [] Every lead: justified, or converted to an explicit overlap?

Float and updates

- [] Negative float: count, worst value, and which of the three explanations applies?
- [] High-float clusters: do they share an area, a package or a subcontractor?
- [] Actual dates beyond the data date: the count must be zero.
- [] Out-of-sequence progress: count, and what is the tool's out-of-sequence setting?
- [] Any activity progressed for several periods with no movement in remaining duration?

The path

- [] Has someone read the longest path end to end and can they narrate it?

- [] What proportion of the critical path's duration is lag?
- [] Does the path run through level-of-effort or administrative activities?
- [] Would a site manager recognise this as the sequence that governs the finish date?

Reporting

- [] Is each finding written as a defensibility statement rather than a verdict on accuracy?
- [] Does each finding have a named owner and a date?
- [] Has anyone checked that last review's findings were closed by correction rather than by cosmetics?

— Related

- [BPG-04 – Baselining and baseline change control](#) — the review runs before a schedule baseline is accepted, and at every re-baseline.
- [BPG-12 – Claims and extension of time](#) — why a network that cannot recalculate cannot support a delay analysis.
- [BPG-17 – Quantitative schedule risk analysis](#) — the analysis that must not be run on a schedule that fails these checks.
- [BPG-19 – Project controls assurance and health checks](#) — where the schedule review sits within a wider assurance cycle.
- [TPL-14 – Schedule quality review checklist](#) — the instrument, with the threshold-agreement fields.
- [TPL-11 – Quantitative schedule risk analysis input sheet](#) — what a schedule must satisfy before it becomes a risk model.

— Sources and standards

Drawn from the Institute's Body of Knowledge, Domain 10 (Project Scheduling): network analysis and the forward and backward passes, leads and lags, calendars, schedule health checks and the schedule basis document.

Several published bodies of practice define schedule assessment metrics, and organisations frequently adopt one as a house standard. This guide deliberately describes the underlying checks in the Institute's own words and reproduces no third party's metric definitions, thresholds or scoring. Where a number appears here it is either an assumption of the worked example or a threshold the Institute recommends agreeing locally, and it is labelled as such. No threshold in this document should be quoted as the requirement of any named standard.

— Status and version

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BPG-06 · BEST PRACTICE GUIDES

Progress measurement and rules of credit

Choosing a technique determines how easy the number is to argue with

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

How physical progress is measured and credited: the rules-of-credit register as a controlled artefact, the six measurement techniques and what each one costs in objectivity, how to set milestone weights so they reflect budget rather than duration, what evidence has to exist before a credit is taken, and how packages roll up. Includes a worked incremental-milestone measurement of a piping package, the same package under mis-set weights, and the arithmetic of level-of-effort dilution.

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Progress measurement and rules of credit

is to argue with.

In one paragraph. How physical progress is measured and credited: the rules-of-credit register as a controlled artefact, the six measurement techniques and what each one costs in objectivity, how to set milestone weights so they reflect budget rather than duration, what evidence has to exist before a credit is taken, and how packages roll up. Includes a worked incremental-milestone measurement of a piping package, the same package under mis-set weights, and the arithmetic of level-of-effort dilution.

Who this is for. Cost engineers and planners who collect progress; control account managers who sign it; construction and engineering managers who are asked for a percentage every month and would prefer the question to have a defined answer.

— 1. A percentage is a measurement, not an opinion

Ask five people what percentage complete a work package is and you will get five answers, all sincere, spanning thirty points. This is not a failure of honesty. It is what happens when a question has no defined method: each person answers a slightly different question — how much of the effort feels done, how much of the duration has elapsed, how much of the money has been spent, how much of the quantity is installed — and all four are legitimate answers to questions nobody asked.

Rules of credit remove the ambiguity by fixing, in advance and in writing, exactly what physical state earns exactly what credit. After that, progress is read off a rule rather than judged, the monthly conversation moves from what percentage are you? to has this step been completed and is the evidence there?, and the number becomes something a control account manager can sign.

The consequence runs further than tidiness. Earned value, the cost performance index, the forecast at completion and every recovery argument built on them all rest on this measurement. [BPG-08 – Earned value in practice](#) covers what to do with the number; this guide covers whether the number means anything before you do it.

— 2. The rules-of-credit register

The register is a controlled document listing, for every work package: the measurement technique, the steps or units that earn credit, the weight attached to each, the evidence that closes each step, and the person authorised to confirm it.

Three properties make it work.

It is fixed before measurement begins. A rule agreed after the first progress claim is not a rule; it is a negotiation. If a rule must change — because it was genuinely wrong — the change applies

to the whole package retrospectively and is disclosed, because it will move earned value without anything physical happening.

It names evidence, not confirmation. "Progress confirmed by site" is not evidence. "Weld map signed by the quality assurance inspector, referencing the joint numbers" is. The difference is whether a reviewer six months later can reconstruct why the credit was taken.

It covers every package, including the awkward ones. Packages with no discrete output are the ones most often left out and most often abused. Give them a technique explicitly — usually level of effort — and record their share of the baseline, so the dilution in §7 is visible rather than accidental.

[TPL-05 – Progress measurement and rules of credit sheet](#) provides the instrument.

— 3. The techniques

Six techniques cover almost all work. They differ along one axis that matters more than any other: how much room they leave for judgement, and therefore how easy the resulting number is to move.

Units completed. Credit is the ratio of quantity installed to total quantity. Objective, verifiable, and the natural choice for repetitive measurable work — metres of cable, cubic metres of concrete, welds, drawings issued. Its weakness is that it treats all units as equal, which is wrong wherever the units differ in difficulty; the fix is to weight by a proxy for effort such as diameter-inch or tonnage rather than by count.

Incremental milestone. The package is divided into sequential steps, each carrying a share of the value, and credit is taken as each step completes. Objective if the steps are physically observable and the weights honest. This is the workhorse technique for anything with a defined production sequence, and §9 works one through.

Weighted milestone. Similar, but the milestones need not be sequential and are typically fewer and larger — suitable for engineering and procurement packages where the interim states are document issues or contractual events rather than physical steps.

Fixed formula (0/100, 50/50, 25/75). A fixed split between start and completion, with no interim judgement at all. Maximum objectivity, minimum resolution. Appropriate for short packages — as a rule of thumb, those spanning no more than one or two reporting periods — where the cost of measuring interim progress exceeds the value of knowing it. The 0/100 variant never over-credits, which is why it suits packages where optimism is the main risk.

Percent complete estimate. A judgement of physical completion by a competent assessor. The most flexible and the easiest to move, and therefore the technique that requires the most support: it should be used only where the assessor's judgement is anchored to something — a quantity take-off, a document count, a defined set of completion criteria — and never as the default because the package was hard to decompose.

Level of effort and apportioned effort. Two rules for work with no output of its own. **Level of effort** earns by the calendar: earned value is set equal to planned value as time passes, so the package can never show a schedule variance. Site supervision, project management and the controls function itself are the usual examples. **Apportioned effort** earns in proportion to another package's progress — quality inspection credited pro rata with the fabrication it inspects — which is the better choice whenever the support work genuinely tracks a measurable base, because unlike level of effort it can show a variance.

— 4. Choosing the technique: the gaming test

The choice is usually presented as a trade between accuracy and effort. The more useful frame is: **how much would the reported number move if the person reporting it were under pressure to move it?**

Run the test explicitly when the register is written. For each package, ask what the maximum defensible overstatement is under the proposed rule, and what evidence would be needed to sustain it.

Under units completed, overstatement requires claiming quantities that are not installed, which a physical count disproves. Under incremental milestone, it requires claiming a step that has not been completed, which the named evidence disproves. Under fixed formula, there is almost nothing to overstate. Under percent complete estimate, overstatement requires only a different opinion — and an opinion is not disprovable, which is why the technique needs an anchor.

Two practical rules follow. **Prefer the most objective technique the work will support**, not the most precise. And **where percent complete estimate is unavoidable, cap its share**: record what proportion of the baseline is measured by judgement, and treat a high figure as a control weakness in its own right, because it is the proportion of your reported progress that cannot be independently checked.

— 5. Setting the weights

Incremental and weighted milestone techniques are only as good as their weights, and the most common error is to set weights by **duration** rather than by **budget**.

The reasoning behind the error is intuitive: welding takes the longest, so welding should be worth the most. But earned value is a budget measure — the credit represents the budget of the work performed — so a step's weight should be the share of the package's budget that step consumes. Where duration and cost profiles differ, weighting by duration systematically misstates progress, and §9.3 shows the size of the effect.

Three disciplines keep weights honest. Derive them from the estimate build-up rather than from opinion, so they can be traced. Have them set by someone other than the person who will report against them. And keep the front-loading modest: a first step worth 25 % for "materials received" credits a quarter of the value for work that is procurement, not production, and it is the single easiest way to make a package look healthy early.

— 6. Evidence and who signs

A credit that nobody signed is a credit nobody owns. The register should name, per step, both the evidence and the role authorised to confirm it — and the two should not be the same person as the one claiming the progress.

The evidence should already exist for another reason. Quality records, inspection releases, delivery notes, document transmittals, survey reports and test certificates are all produced anyway, and using them as progress evidence costs nothing extra while making the progress claim auditable. Evidence invented for progress measurement alone tends to become a form-filling exercise that certifies itself.

One rule is worth stating flatly: **credit is not acceptance**. A package credited 100 % under its rules of credit has completed the physical work its rule describes. It has not necessarily been accepted by the client, and the two states must be tracked separately — a spool credited on installation that later fails a pressure test has not become 90 % complete; the rework is either scope that was always required and is now late, or it is a change. [BPG-04 – Baseline and baseline change control](#) covers which.

— 7. Rolling up, and the level-of-effort problem

Progress rolls up as a **budget-weighted average**, never as a simple mean of percentages. Averaging percentages across packages of different sizes produces a number that belongs to no one and is wrong in a direction nobody can predict.

The level-of-effort problem is the roll-up's characteristic distortion. Because level-of-effort packages earn by the calendar, their earned value equals their planned value by construction and their schedule performance index (SPI) is 1.000 whatever is happening. Aggregate them with discrete work and they pull the combined index towards 1.000 in proportion to their share. §9.4 quantifies this.

Two responses. **Segregate** level-of-effort packages into their own control accounts so the discrete performance can be read on its own. And **report the share**: state, every period, what proportion of the measured baseline earns by calendar. A reader who does not know that figure cannot interpret the index they are being shown.

— 8. How this goes wrong

The rule is chosen after the first claim. The foreman says 80 %, the cost engineer needs a rule that produces something defensible, and the rule that gets written is the one closest to 80 %. Every subsequent period inherits the precedent.

Percent complete estimate becomes the default. It is used for packages that could have been measured by units or milestones, because decomposing them was work. The proportion of the baseline measured by judgement climbs quietly past half, and no index computed from it can be independently checked.

Weights are set by the person who reports against them. Not usually dishonestly — but a production manager who believes the hard part is welding will weight welding heavily, and the schedule will then show rapid progress through the phase they consider difficult.

Credit is taken on delivery to site. Materials arriving earns a large share of a package's value. Progress looks strong for two months and then flattens completely during installation, and the flat period is read as a productivity problem when it is a measurement artefact.

Level of effort is used because the work was hard to define. Packages with genuine discrete output are put on level of effort to avoid the measurement work. They then cannot show a variance, and the project's schedule performance index becomes progressively less informative as their share grows.

The 90 % plateau. Packages reach 90 % quickly and stay there for months. This is the signature of a percent-complete rule with no defined final step: the last 10 % has no evidence attached, so nobody can say when it is earned. Every rule needs a defined closing condition.

Progress and acceptance are conflated. A package credited complete is reported as handed over. When the client's punch list arrives, the project has no budget left for work it has already claimed as earned, and the argument is about the measurement rather than about the work.

— 9. Worked example

Illustrative figures. One work package at one data date. Quantities are in diameter-inch (a conventional measure of pipework quantity that weights each joint by pipe diameter, so that a large joint counts for more than a small one). Currency in generic units. Percentages are stated against the package total. No real project is implied.

9.1 The package and its rules of credit

Work package: process pipework installation, one area. Budget **1,350,000**. Total quantity **5,400** diameter-inch. Technique: incremental milestone, weighted by budget share.

Value per diameter-inch = $1,350,000 \div 5,400 = 250.00$

Step	Weight	Evidence that closes it
Material received and verified	10 %	Goods receipt plus material certificate check, signed by materials control
Spool set and aligned	30 %	Survey confirmation of position and alignment, signed by the site engineer
Welded out	35 %	Weld map signed by the welding supervisor, joints listed
Non-destructive testing accepted	15 %	Test report accepted by quality assurance
Supports, insulation and punch-list clear	10 %	Completion certificate signed by the area superintendent
Total	100 %	

Check: $10 + 30 + 35 + 15 + 10 = 100$.

9.2 The measurement at the data date

Step	Weight	Quantity reaching this step (dia-in)	Earned dia-in
Material received and verified	0.10	5,400	540.0
Spool set and aligned	0.30	4,100	1,230.0
Welded out	0.35	3,250	1,137.5
Non-destructive testing accepted	0.15	2,480	372.0
Supports, insulation and punch-list clear	0.10	1,900	190.0
Total earned			3,469.5

Checks: $5,400 \times 0.10 = 540.0$; $4,100 \times 0.30 = 1,230.0$; $3,250 \times 0.35 = 1,137.5$; $2,480 \times 0.15 = 372.0$; $1,900 \times 0.10 = 190.0$. Sum = $540.0 + 1,230.0 + 1,137.5 + 372.0 + 190.0 = 3,469.5$.

Physical percent complete = $3,469.5 \div 5,400 = 0.6425 = 64.25 \%$
 Earned value = $0.6425 \times 1,350,000 = 867,375$
 or = $3,469.5 \times 250.00 = 867,375$ (the same number by the other route)

9.3 What the same package looks like without the register

The claim. The area superintendent reports "about 80 % — the spools are all up." The statement is sincere and partly true: setting is the visible milestone, and $4,100 \div 5,400 = 75.9 \%$ of the quantity has been set. But under the register, material and setting together are worth at most $10 + 30 = 40 \%$ of the package, and only some of the welding, testing and completion has been done.

Credit under the claim = $0.80 \times 1,350,000 = 1,080,000$
 Credit under the register = 867,375
 Difference = $1,080,000 - 867,375 = 212,625$
 As a share of the package = $212,625 \div 1,350,000 = 15.75 \%$

Nobody has lied. The claim answers "how far through the visible sequence are we?"; the register answers "what share of this package's budget has been earned?". Only the second belongs in a cost report.

Weights set by duration instead of budget. Suppose the weights had been assigned by how long each step takes — welding is the longest activity, testing the shortest — giving material 5 %, setting 25 %, welding 50 %, testing 5 %, completion 15 % (sum = 100 %). The same physical state then earns:

$5,400 \times 0.05 = 270.0$
 $4,100 \times 0.25 = 1,025.0$
 $3,250 \times 0.50 = 1,625.0$
 $2,480 \times 0.05 = 124.0$
 $1,900 \times 0.15 = 285.0$
 Total = 3,329.0 dia-inch equivalent
 Percent = $3,329.0 \div 5,400 = 61.65 \%$
 Earned value = $3,329.0 \times 250.00 = 832,250$

Check: $270.0 + 1,025.0 + 1,625.0 + 124.0 + 285.0 = 3,329.0$.

The difference from the budget-weighted register is $867,375 - 832,250 = 35,125$, or 2.6 % of the package budget. The direction of the error depends on the actual profile and would reverse at a different point in the sequence — which is exactly why weights must be derived from the estimate build-up rather than argued each period.

9.4 Level-of-effort dilution

Assumptions. A control account with planned value at the data date of 2,350,000, of which 350,000 is a level-of-effort package (site supervision). The discrete work has earned 1,640,000 against a planned value of 2,000,000. The level-of-effort package earns its planned value by construction.

```

Discrete SPI = 1,640,000 ÷ 2,000,000 = 0.820
Blended SPI  = (1,640,000 + 350,000) ÷ (2,000,000 + 350,000)
              = 1,990,000 ÷ 2,350,000
              = 0.847
Level-of-effort share of planned value = 350,000 ÷ 2,350,000 = 14.9 %

```

The reported index is 0.847 while the work that can actually be late is running at 0.820 — 2.7 index points of pure arithmetic, produced by 14.9 % of the baseline that cannot be late by construction. The larger the level-of-effort share, the closer the reported index sits to 1.000 regardless of performance. This is the reason to segregate, and the reason to publish the share every period.

— 10. Checklist

Take this into the meeting where progress is agreed, or into the review of a rules-of-credit register.

The register

- [] Does every work package have a technique recorded, including the awkward ones?
- [] Was every rule fixed before the first claim against it?
- [] Has any rule changed mid-package, and if so was the change applied retrospectively and disclosed?
- [] Does every step name evidence a reviewer could find six months later — a document, not a confirmation?
- [] Does every step name the role that confirms it, and is that role independent of the claimant?

Technique choice

- [] What proportion of the baseline is measured by percent complete estimate?
- [] For each such package, what is it anchored to, and could a more objective technique have been used?
- [] Is anything on level of effort that has a genuine discrete output?
- [] Where support work tracks a measurable base, has apportioned effort been considered instead?
- [] Does every rule have a defined closing condition, or does the package stop at 90 %?

Weights

- [] Were milestone weights derived from the estimate build-up, and can that derivation be shown?
- [] Were they set by someone other than the person who reports against them?
- [] How much value is earned before production starts — receipt, mobilisation, procurement?

Roll-up and reporting

- [] Is progress rolled up budget-weighted, not as a mean of percentages?
- [] Is the level-of-effort share of the measured baseline stated in the report every period?
- [] Are level-of-effort packages segregated so discrete performance can be read alone?
- [] Are credit and client acceptance tracked as separate states?

Sanity

- [] If a claim exceeds what the register can support, has the register been checked before the claim is disputed?
- [] Which packages have not moved for two periods, and is that production or measurement?

— Related

- [BPG-02 – The work breakdown structure](#) — the work packages these rules are written against, and how deep to decompose.
- [BPG-04 – Baselining and baseline change control](#) — what to do when a rule was wrong, and why rework is a scope question.
- [BPG-05 – Schedule quality – a practical review](#) — the duration checks that determine which packages need interim milestones.
- [BPG-08 – Earned value in practice](#) — what happens to the number this guide produces.
- [TPL-05 – Progress measurement and rules of credit sheet](#) — the register, with every field defined.
- [TPL-07 – Earned value calculation sheet](#) — the roll-up, with formulas stated and verified.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 6 (Earned Value Management and Forecasting) for earning rules and the distinction between physical and cost percent complete, Domain 5 (Cost Management and Cost Control) for control accounts and work packages, and Domain 10 (Project Scheduling) for progress updating and duration.

The measurement techniques described here are long-established practice across the profession and are explained in the Institute's own words. The gaming test in §4, the budget-versus-duration weighting rule in §5 and the recommendation to publish the level-of-effort share every period are PCI recommended practice. No external standard, table or scoring scheme is reproduced.

— Status and version

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BPG-07 · BEST PRACTICE GUIDES

Accruals and cut-off discipline

Why a missed accrual flatters this month and invents an overrun next month

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

How to close a reporting period so the cost figure describes the work done rather than the invoices processed: the three cost states, what must be accrued and from which source, how to size an accrual when the subcontractor has not certified, reversal discipline, the cut-off calendar and what happens when it is compressed. Includes a worked month-end showing the cost performance index with and without the accrual, the phantom overrun it creates in the following period, and the effect on the forecast.

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Accruals and cut-off discipline

arrive.

In one paragraph. How to close a reporting period so the cost figure describes the work done rather than the invoices processed: the three cost states, what must be accrued and from which source, how to size an accrual when the subcontractor has not certified, reversal discipline, the cut-off calendar and what happens when it is compressed. Includes a worked month-end showing the cost performance index with and without the accrual, the phantom overrun it creates in the following period, and the effect on the forecast.

Who this is for. Cost engineers and cost managers who close a period; project accountants and finance business partners who own the ledger side of it; project managers who have been told the overrun is "a timing thing" and want to know whether that is true.

— 1. The month does not end when the calendar does

A project's cost ledger is a record of invoices processed. A project's cost report is meant to be a record of resources consumed. These are different things, and the gap between them is a matter of weeks: a subcontractor works through March, certifies in early April, invoices mid-April and is paid in May. Read the ledger at the end of March and none of that work exists.

The accrual is the entry that closes the gap. It says: this work has been done, we owe for it, and the cost belongs to the period in which the work happened rather than the period in which the paperwork caught up.

Everything else in this guide follows from one observation. Because earned value measures work done at budgeted cost, comparing it to a cost figure that excludes work done but not invoiced compares two different things — and produces an index that is not slightly wrong but systematically flattering, in every period, by an amount that varies with invoice timing. §9 shows a twenty-four-point swing on a single control account from one month's accruals.

— 2. Three states, and where the accrual sits

Cost passes through three states, and control depends on tracking all three rather than only the last.

Commitment. A purchase order or subcontract is raised. Nothing has been received and nothing paid, but the organisation is exposed. Commitments are the earliest signal of future spend and the only one that arrives before the money does.

Accrual. Goods or services have been received, so the cost has been incurred, but no invoice has been processed. The accrual recognises the incurred cost in the period the work happened.

Actual. The invoice has been received and posted. The cost is in the ledger.

The cost figure a controls report needs is **actuals plus accruals** — everything incurred, whether or not it has been billed. Add open commitments and an estimate for uncommitted remaining scope and you have the forecast. Reporting invoices alone understates cost by the accrual and ignores the commitment entirely, which is a blind spot in both directions at once.

The transaction-state segment of the code of accounts is what makes these three separable in the first place; [BPG-03 – Cost breakdown structure and the code of accounts](#) covers its design.

— 3. What has to be accrued

Five categories account for nearly all of it.

Goods received not invoiced. Materials and equipment delivered and receipted before the cut-off, with the invoice still to arrive. This is the most reliable accrual because it comes straight from the goods receipt records, and it is the one most systems can produce without judgement.

Work performed by subcontractors but not yet certified or billed. The largest and hardest category. Certification cycles rarely align with the reporting cut-off, so at any month end there is a slice of subcontract work that is unambiguously done and entirely absent from the ledger.

Labour worked after the payroll extract. The last few days of the month, where the timesheet system's cut-off does not match the reporting cut-off. Small per day, material per month, and completely predictable — which means it can be computed rather than estimated.

Services consumed and not billed. Plant hire, temporary utilities, professional services, freight in transit under terms that have already transferred the cost.

Adjustments already known. A credit note agreed but not issued, a rejected delivery still sitting in the receipt records, a rate correction agreed but not yet applied. These accrue in the direction that makes the period true, which is sometimes negative.

What must not be accrued is work that has not been performed. An accrual for a cost that is expected but not yet incurred is a provision or a forecast, not an accrual, and mixing the two destroys the meaning of both. The accounting frameworks in use internationally draw this line carefully — the distinction between a liability whose timing or amount is uncertain and an accrual for goods or services already received is fundamental to standards such as IAS 37 — and the exact recognition point, along with the treatment of the resulting balances for tax, varies by framework and by jurisdiction. The controls professional's job is not to determine the accounting treatment but to supply an accurate statement of what was received before the cut-off, and to agree the boundary with finance rather than assume it.

— 4. Sizing the accrual

Sources, in descending order of reliability. Use the highest one available for each item and record which you used, because the reliability of the accrual is part of the information.

1. **A certified quantity or milestone.** The work has been measured and signed, only the invoice is missing. Effectively exact.
2. **A goods receipt.** Quantity received against a priced purchase order. Exact for materials.
3. **A rate multiplied by a measured quantity.** Contract rate against a surveyed quantity — the standard method where certification is behind.

4. **A rate multiplied by a recorded input.** Hours from timesheets or plant records at agreed rates. Reliable for labour, less so where the input does not translate directly into value.
5. **Progress applied to a package value.** The package's earned percentage multiplied by its value, where nothing more direct exists. Weakest, because it inherits any error in the progress measurement — see [BPG-06 – Progress measurement and rules of credit](#).
6. **A judgement by the responsible manager.** Acceptable only with a written basis, and it should be the exception. An accrual nobody can reconstruct is a number, not a measurement.

Two rules. **Accrue to the receipt, not to the document.** The date that matters is when the goods or services were received, not the date on the invoice or the date it was entered. An automated accrual driven by document date reproduces a cut-off error at scale. And **do not net accruals against retentions or contra-charges** unless the contract makes them a single obligation — netting hides both numbers and makes the cash forecast in [BPG-13 – Cash flow forecasting](#) unreliable.

— 5. Reversal discipline

Every accrual is reversed in the following period, and the actual invoice is posted in its place. This sounds mechanical and is where most accrual processes fail.

The failure mode is the accrual that is not reversed, so that when the invoice arrives the cost is recorded twice. It usually happens where accruals are entered manually into a controls spreadsheet rather than as reversing journals in a system, and it is detected by a discipline rather than by a control: maintain an **accrual register** with one row per accrual carrying the period raised, the basis, the source used, the value, the period reversed, and the actual value when it landed.

That last column is the one that earns the register its place, because it gives an **accrual accuracy** measure: compare each accrual to the invoice that replaced it. Persistent under-accrual on one subcontract is a signal about that subcontractor's certification lag; persistent under-accrual across the project is a signal about the process. Neither is visible without the comparison, and both change what you do next month.

— 6. The cut-off calendar, and the pressure on it

A cut-off is a date, published in advance, to which every source reports: cost extraction, goods receipts, timesheets, subcontract certification, schedule progress and field quantities. It works only if it is the same date for all of them.

The working-day sequence from cut-off to issue is designed backwards from the meeting that receives the report, and [BPG-01 – Building a project controls function from zero](#) works that calendar through. The relevant point here is what happens under compression. When the reporting deadline moves earlier, the steps with external deadlines — data extraction, the report issue itself — cannot move. The step with no external deadline is accrual determination, so that is the one that gets shortened, and the shortening takes the form of using a weaker source from §4 or skipping the smaller categories entirely.

The result is not a slightly less precise report. It is a report whose cost figure has moved systematically in one direction, because every omitted accrual understates cost and none overstates it.

Two defences are worth building in. **Pre-cut-off preparation:** subcontract quantities surveyed in the last week of the period rather than the first week of the next, and goods-receipt exceptions cleared before the cut-off rather than after. And a **standing de minimis**, agreed in advance, below which items are not individually accrued — which converts an implicit omission into an explicit, bounded and disclosed one.

— 7. Two accruals, one reconciliation

The accrual the project needs and the accrual the statutory accounts need are not always the same number, and pretending otherwise causes more argument than accepting it.

The project's accrual serves management control. It wants completeness at a fine granularity, at control-account level, on a fixed monthly rhythm, and it will accept a defensible estimate where an exact figure is unavailable. The statutory accrual serves financial reporting. It applies the recognition criteria of the governing framework, is subject to audit, and applies materiality thresholds set for the entity as a whole rather than for one project.

The two should be **reconciled, not forced to converge**. Where the project accrues something finance does not, the difference should be explainable in one line — usually a materiality threshold or a recognition timing difference. A reconciliation that cannot be explained is the finding; an unreconciled difference written off as "timing" every month for a year is how a real problem hides.

— 8. How this goes wrong

The accrual is skipped because the period was tight. One month's omission produces a flattering index and a decision taken on it. The following month, when the invoices land, the same account shows an overrun that nothing in that month caused. §9.3 works this through, and the phantom is severe: a period index below 0.44 in a month where performance was actually near 0.93.

The accrual is driven by invoice date. An automated rule accrues everything with a document date in the period. Work received in the period but invoiced with a later document date is missed; work received later but invoiced early is included. The rule looks systematic and is systematically wrong.

Accruals are estimated by the person whose performance they affect. A control account manager who sizes their own accrual has an incentive to be conservative. Not fraudulently — but conservative in one direction, every month, and the cumulative effect is a project that always looks slightly better than it is until the final account.

Accruals are not reversed. The cost is counted twice when the invoice lands. Detected, if at all, by a reconciliation that nobody has time to do. Prevented by reversing journals and an accrual register.

Retention is treated as an accrual. Retention withheld from a certified valuation is a payment timing matter, not a cost timing matter — the cost was incurred and should be recognised in full. Treating retention as though it reduces cost understates the project and misstates the cash forecast.

Accrual accuracy is never measured. The register exists but the "actual when it landed" column is never filled in, so nobody knows whether the accruals are systematically low. This is the cheapest improvement available to most cost functions and the one most often left undone.

The cut-off is held open for a large invoice. Somebody notices that a significant invoice will arrive on the second and delays the close. The period is now longer than a month, the comparison to every other period is broken, and the precedent is established that cut-off is negotiable when the number is unhelpful.

— 9. Worked example

Illustrative figures. Generic currency units. One control account, monthly reporting cycle, values cumulative to the data date unless stated. Indices to three decimal places; currency to the nearest 1,000 where a rounded figure is presented. No real project, jurisdiction or accounting framework is implied — the treatment of any specific item follows the framework the entity reports under.

9.1 Month 9 at cut-off

Item	Value
Budget at completion (BAC)	5,600,000
Earned value at the data date (EV)	3,720,000
Invoices processed to date (ledger actuals)	3,140,000
Goods received not invoiced (from the goods-receipt report)	268,000
Subcontract work surveyed in the period, not yet certified or billed	412,000
Labour worked after the payroll extract: 1,850 hours at a blended 62 per hour	114,700

Timesheet accrual = $1,850 \times 62 = 114,700$
 Total accrual = $268,000 + 412,000 + 114,700 = 794,700$
 True actual cost = $3,140,000 + 794,700 = 3,934,700$

9.2 The same month, reported two ways

Writing actual cost as AC, the cost performance index as CPI and the cost variance as CV:

Without the accrual: $\text{CPI} = \text{EV} \div \text{AC} = 3,720,000 \div 3,140,000 = 1.185$
 $\text{CV} = \text{EV} - \text{AC} = 3,720,000 - 3,140,000 = +580,000$

 With the accrual: $\text{CPI} = \text{EV} \div \text{AC} = 3,720,000 \div 3,934,700 = 0.945$
 $\text{CV} = \text{EV} - \text{AC} = 3,720,000 - 3,934,700 = -214,700$

The index swings by $1.185 - 0.945 = \mathbf{0.240}$, twenty-four index points. The variance swings by $580,000 - (-214,700) = \mathbf{794,700}$, which is exactly the accrual — the arithmetic check that the two statements describe the same project.

What this does to the forecast. Forecasting on the assumption that performance to date persists — one method among several, and [BPG-09 – Estimate at completion](#) owns the choice — the estimate at completion (EAC) and variance at completion (VAC) diverge by more than the accrual itself:

Without the accrual: $EAC = BAC \div CPI = 5,600,000 \div 1.184713 = 4,727,000$
 $VAC = 5,600,000 - 4,727,000 = +873,000$ (an underrun)

With the accrual: $EAC = BAC \div CPI = 5,600,000 \div 0.945434 = 5,923,000$
 $VAC = 5,600,000 - 5,923,000 = -323,000$ (an overrun)

The forecast moves by $5,923,000 - 4,727,000 = \mathbf{1,196,000}$, which is $1,196,000 \div 5,600,000 = \mathbf{21.4\%}$ of budget at completion. One month's accruals, worth 794,700, change the project's forecast outcome from a 873,000 underrun to a 323,000 overrun. The unrounded indices are used in the divisions and the results rounded once, at the end.

9.3 The phantom overrun in month 10

In month 10 the 794,700 of month-9 accruals arrive as invoices, a further 690,000 of genuine month-10 work is incurred, and earned value rises by 640,000.

Cumulative actual cost = $3,140,000 + 794,700 + 690,000 = 4,624,700$
 Cumulative earned value = $3,720,000 + 640,000 = 4,360,000$

On the un-accrued path, month 9 reported cost of 3,140,000, so the period cost recorded in month 10 is everything since:

Period AC = $4,624,700 - 3,140,000 = 1,484,700$
 Period EV = 640,000
 Period CPI = $640,000 \div 1,484,700 = 0.431$

On the accrued path, the month-9 accrual reverses, the invoices replace it, and the month-10 accrual is booked. The period cost is the 690,000 genuinely incurred in month 10:

Period AC = 690,000
 Period EV = 640,000
 Period CPI = $640,000 \div 690,000 = 0.928$

Month 10 was a period of mildly disappointing performance — a period index of 0.928. The un-accrued report shows 0.431, which would in most organisations trigger an intervention, a recovery plan and a difficult conversation about a month in which nothing unusual happened.

And the cumulative index converges anyway. At month 10 the cumulative figure is the same under both treatments: $4,360,000 \div 4,624,700 = \mathbf{0.943}$. That is the trap. The cumulative index eventually tells the truth, which is why missing accruals so often go undetected — but every decision is taken on a period number, and the period number is destroyed. In month 9 the project was told it was 580,000 under budget. In month 10 it is told it has collapsed. Neither statement was true.

10. Checklist

Take this into the month-end close, or into the review of a period that produced a surprise.

Completeness

- [] Goods received not invoiced: pulled from the receipt records, or estimated? (It should never be estimated.)
- [] Subcontract work performed and not certified: has the quantity been surveyed, or assumed?
- [] Labour after the payroll extract: how many days, how many hours, at what rate?
- [] Plant hire, utilities, freight and professional services: is anything consumed and unbilled?
- [] Known credits, rejections and rate corrections: accrued in the correct direction?
- [] Is anything being accrued that has not actually been received? (That is a provision, and it belongs elsewhere.)

Basis and evidence

- [] For each accrual, which of the six sources in §4 was used, and is it recorded?
- [] Is any accrual sized by the person whose reported performance it affects?
- [] Is any accrual driven by document date rather than receipt date?
- [] Is retention being treated as a reduction in cost? (It is not one.)
- [] Is anything netted that should be reported gross?

Reversal

- [] Is there an accrual register with the period raised, the basis, the value and the reversal period?
- [] Have all prior-period accruals actually reversed?
- [] Is the "actual when it landed" column filled in, and what is the accrual accuracy trend?
- [] Which supplier or subcontract is consistently under-accrued, and why?

Cut-off

- [] Is the cut-off the same date for cost, goods receipts, timesheets, certification and schedule progress?
- [] Was the cut-off held open for anything this period? For what, and who authorised it?
- [] Was the accrual step compressed because the reporting deadline moved?
- [] Is there an agreed de minimis, written down, and is the total below it disclosed?

Reconciliation

- [] Does the project accrual reconcile to the finance accrual, with the difference explainable in one line?
- [] Has that difference been "timing" for more than two consecutive periods?
- [] Has the boundary between a controls accrual and a statutory provision been agreed with finance in writing?

— Related

- [BPG-01 – Building a project controls function from zero](#) — the cut-off calendar this depends on, and why it precedes reporting.

- [BPG-03 – Cost breakdown structure and the code of accounts](#) — the transaction-state coding that makes commitment, accrual and actual separable.
- [BPG-06 – Progress measurement and rules of credit](#) — the earned value figure that the cost figure here is compared against.
- [BPG-08 – Earned value in practice](#) — how the indices in §9 are read and reported.
- [BPG-13 – Cash flow forecasting](#) — why accrual and payment timing must be kept distinct.
- [TPL-07 – Earned value calculation sheet](#) — the calculation, with the accrual input made explicit.

— Sources and standards

Drawn from the Institute's Body of Knowledge: Domain 5 (Cost Management and Cost Control) for the commitment-accrual-actual cycle and cost-to-date, Domain 1 (Foundations of Accounting for Project Controls) for accrual-basis recognition, and Domain 2 (Financial Reporting and the Standards) for the relationship between project cost records and statutory reporting.

Accounting standards are named where relevant and their principles described in the Institute's own words; no standard text, table or clause numbering is reproduced. The distinction between an accrual for goods or services already received and a provision for a liability of uncertain timing or amount is treated by frameworks such as IAS 37, and the recognition point, the materiality thresholds and any tax consequences vary by reporting framework and by jurisdiction. Nothing in this guide should be read as determining the accounting treatment of any item; that is a matter for the entity's finance function under the framework it reports under.

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BPG-08 · BEST PRACTICE GUIDES

Earned value in practice

Making the three measures mean something on a real control account

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Earned value management fails at the measurement, not at the arithmetic. This guide sets out what planned value, earned value and actual cost each have to be measured against before any index is worth reading, how the control account holds them together, how to read cost variance, schedule variance and the two performance indices, why the schedule performance index becomes actively misleading late in a project, and the questions earned value cannot answer at all.

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Earned value in practice

In one paragraph. Earned value management fails at the measurement, not at the arithmetic. This guide sets out what planned value, earned value and actual cost each have to be measured against before any index is worth reading, how the control account holds them together, how to read cost variance, schedule variance and the two performance indices, why the schedule performance index becomes actively misleading late in a project, and the questions earned value cannot answer at all.

Who this is for. Cost engineers, control account managers, planners and project controls managers who already produce an earned value report every month and want it to survive being challenged.

— 1. The measurement problem comes before the formula

The arithmetic of earned value management (EVM) is three subtractions and two divisions. Nobody has ever failed at that. Projects fail at EVM because one of the three inputs was not measured against the thing it claims to measure, and the indices then describe a project that does not exist.

Of the three quantities, two are supposed to be found and one is created:

- **Planned value (PV)** is read from the time-phased baseline. If the baseline was never properly time-phased — if a lump of budget was dropped into the month it was approved rather than the months the work is planned — then PV is fiction and the schedule variance derived from it is fiction too. Baseline integrity is the subject of [BPG-04 – Baselining and baseline change control](#).
- **Actual cost (AC)** is read from the ledger. If accruals are not booked at cut-off, AC is not the cost of the work performed; it is the cost of the invoices that happened to arrive. [BPG-07 – Accruals and cut-off discipline](#) owns that problem.
- **Earned value (EV)** is created by a person, every month, by applying an earning rule to physical progress. It is the only one of the three that a project manufactures rather than retrieves, which is exactly why it is the one that gets bent. Which rule applies to which work package is the subject of [BPG-06 – Progress measurement and rules of credit](#).

The consequence is a hierarchy of trust that most reports never state. A cost performance index is only as honest as its earning rules, and only as complete as its accruals. Before you defend an index, you should be able to say which rule earned the value and whether the ledger was closed against the same date.

— 2. The control account is the unit of control

A **control account** is the point in the work breakdown structure where scope, schedule and budget intersect and one named person — the **control account manager (CAM)** — is accountable for the result. It is the level at which earned value is managed. It is not the level at which

earned value is measured: value is earned bottom-up in work packages, each under its own fixed earning rule, and rolled up.

Three disciplines make a control account usable:

1. **One owner.** If two people can explain a variance differently and both be right, the account is drawn wrong.
2. **One cause per variance, as far as possible.** An account that mixes civil works, a specialist subcontract and a currency-exposed equipment purchase will produce a cost variance that nobody can act on. The test is not size; it is whether the CAM can name the cause within a day.
3. **A fixed boundary shared by the schedule and the ledger.** The same scope must define the activities that generate PV, the packages that earn EV, and the cost codes that collect AC. Where those three boundaries differ even slightly, the variance is partly an artefact of coding.

The commonest structural failure in an otherwise competent EVM system is a control account that is a reporting convenience rather than a management unit — sized to fit a page rather than to isolate a cause.

— 3. The three measures, stated precisely

Measure	Definition	Read from
Planned value (PV)	The budget for the work scheduled to be complete by the data date	The time-phased cost baseline
Earned value (EV)	The budget for the work actually performed by the data date	Physical progress × earning rule
Actual cost (AC)	The cost actually incurred for that work, including accruals	The ledger, at cut-off

All three are expressed in the same currency, over the same period, for the same scope boundary, at the same data date. Every word in that sentence is load-bearing. A report that compares progress measured on the 28th with costs closed on the 25th is comparing two different projects, and the difference shows up as a variance that no action will fix.

Note what EV is not. It is not the value of the work to the client, not the amount billed, and not the cost incurred. It is budget, re-earned by performance — the only common currency in which planned work and performed work can be compared.

— 4. Variances and indices

Four derived numbers do almost all the work:

$CV = EV - AC$	Cost variance – currency ahead of (+) or behind (–) budget
$SV = EV - PV$	Schedule variance – currency of work ahead of (+) or behind (–) plan
$CPI = EV \div AC$	Cost performance index – value earned per unit of cost
$SPI = EV \div PV$	Schedule performance index – value earned per unit of value planned

Variances answer how much; indices answer how efficiently, and because they are dimensionless they can be compared across accounts of wildly different size. A single work package with EV of

630,000 and AC of 665,000 gives $CPI = 630,000 \div 665,000 = 0.947$ — a 5 % cost inefficiency that is directly comparable with an account ten times its size.

Two habits separate a useful index from a decorative one.

Read cumulative and period indices together. The cumulative CPI is stable and damps the signal; the current-period CPI is noisy but leads. On a project of any size a cumulative CPI that moves three points in a month is a large movement, because it means the period performance was far worse than everything that preceded it. Reporting only the cumulative figure is how a deteriorating trend stays invisible for two quarters.

Read the account beneath the rollup. A project-level CPI is an average weighted by spend. It will comfortably conceal one account in serious trouble behind three that are fine — which is precisely the condition under which management attention is worth most.

— 5. Why the schedule performance index is a poor schedule indicator

SPI is a cost-based measure wearing a schedule's name, and it fails as a schedule indicator in three distinct ways.

It is denominated in currency, not time. An SPI of 0.80 does not mean the project is 20 % late. It means 20 % of the value planned to have been earned by now has not been. Whether that represents two weeks or five months depends entirely on the shape of the baseline curve at that point.

It is blind to the critical path. Value earned is value earned, whether the work sat on the driving path or on an activity with four months of float. A project can hold an SPI near 1.00 while the one sequence that determines the completion date slips steadily, because the team is earning value on everything that is easy to reach. EVM must be read alongside a critical-path analysis, never instead of one — see [BPG-05 – Schedule quality: a practical review](#).

It converges to 1.00 at completion, by construction. When all the work is done, EV equals the budget at completion (BAC), and PV also equals BAC. The ratio is therefore exactly 1.00 no matter how late the project finished. Worse, it approaches 1.00 smoothly through the late stages, so it improves while the project is running out of time.

Illustrative figures. A control account with a BAC of USD 4,800,000, baselined to finish at Month 12, actually finishes at Month 15:

Point	EV	PV	$SPI = EV \div PV$
Month 12 (planned finish)	4,320,000	4,800,000	$4,320,000 \div 4,800,000 = 0.900$
Month 14	4,700,000	4,800,000	$4,700,000 \div 4,800,000 = 0.979$
Month 15 (actual finish)	4,800,000	4,800,000	$4,800,000 \div 4,800,000 = 1.000$

The index rises from 0.900 to 1.000 across the three months in which the project is delivering its lateness. A reader who watches SPI alone sees recovery.

The practical response is to stop asking SPI to be a schedule measure. Report schedule status in days against milestones and in critical-path terms, and if an index is wanted, use an **earned schedule** measure — the point in baseline time at which the current EV should have been earned,

compared with actual time — which does not collapse to 1.00 at completion. The method is developed in BoK Domain 6 (EVM/EAC).

— 6. What earned value cannot tell you

A professional states the limits of the instrument as clearly as the readings.

- **Whether the baseline deserved to be believed.** EVM measures conformance to a plan. A generous plan produces flattering indices; that is not performance, it is estimating.
- **Whether the remaining work resembles the work done.** Every index is a statement about the past. Turning it into a forecast requires an explicit assumption, which is the subject of [BPG-09 – Estimate at completion: choosing and defending a method](#).
- **Where on the network the trouble is.** See §5.
- **Anything about cash.** EV is budget re-earned; it is not billed, certified or collected. A control account with a CPI of 1.05 can sit inside a project that cannot pay its subcontractors — see [BPG-13 – Cash flow forecasting](#).
- **Anything about quality or rework not yet found.** Value earned under a units-completed rule counts metres installed, not metres that will survive inspection. Rework arrives later as AC with no corresponding EV, and reads as a cost variance months after the cause.
- **Scope that has changed but is not yet in the baseline.** Instructed variations not yet incorporated distort every measure at once: work is being performed and paid for that the baseline does not recognise. [BPG-11 – Change orders and variations](#) deals with the treatment.

There is one further distortion worth naming because it is silent. **Level of effort (LOE)** work packages — supervision, project controls, site management — earn value by the calendar, so EV is set equal to PV and their schedule variance is zero by construction. Mixed into a control account, LOE drags both indices towards 1.00 and makes the account look healthier than the discrete work inside it. The §9 example quantifies the effect.

— 7. Trend, tolerance and the conversation the report should start

Indices are inputs to a decision, not the decision. Two conventions make them operational.

Set **tolerances in advance** — the thresholds at which an account moves from normal reporting into exception reporting — and set them at the control account level rather than the project level, because that is where action is taken. Tolerances agreed after a bad month are not tolerances.

Then report **trend, cause and action** rather than status. "CPI 0.899" is a fact. "CPI 0.899, down from 0.94 two months ago, driven by cable-pulling productivity at 1.3 times budget hours per metre since access was released piecemeal; four supplementary crews mobilise next week" is a decision. [BPG-14 – Monthly reporting that gets read](#) owns the form of that narrative.

— 8. How this goes wrong

Earned value is derived from cost. The single most destructive error: setting EV as a percentage of budget equal to the percentage of budget spent. EV then tracks AC, CPI is 1.00 by construction, and the system reports perfect cost performance forever. It usually enters through a spreadsheet formula nobody has opened in two years.

The 90 % plateau. Percent-complete claims advance briskly to about 90 % and then stall for months while the last of the work is finished. The pattern is a symptom of subjective earning rules on long work packages; the fix is objective rules — units, weighted milestones or 0/100 on short packages — fixed before work starts.

Level of effort quietly dominating a control account. Because LOE cannot show a variance, an account that is one-third LOE by value can only ever report two-thirds of the variance actually occurring in it. Segregate LOE into its own packages and cap its share.

Cost booked where no value can be earned. Mobilisation, standing time or early procurement charged to a package that has not started produces AC with no EV, and the resulting variance is attributed to the wrong cause. It is not always miscoding — sometimes the cost is real and the baseline simply never planned for it, which is a different and more serious finding.

Mismatched data dates. Progress cut on one date and the ledger on another. The variance that results is a timing artefact, and chasing it wastes a control account manager's month.

Re-baselining to remove a variance. Adjusting the baseline so that PV agrees with EV eliminates the schedule variance and, with it, the only record that anything went wrong. Baseline changes are legitimate and necessary; a baseline change made because a variance is uncomfortable is not one. The change log, not the index, is the audit trail.

Reporting the rollup only. The project CPI is an average. If nobody reads the account detail beneath it, the system is producing a number rather than a control.

— 9. Worked example

Illustrative figures. Currency USD; all values at the Month 7 data date; earning rules fixed at baseline; the ledger is closed at the same date with accruals booked; indices to three decimal places.

Control account CA-4200, cabling and terminations. BAC = USD 4,800,000.

Earned value is measured package by package, each under its own rule:

Work package	Budget	Earning rule	Status at data date	EV
WP-1 Containment	900,000	Units completed	6,300 of 9,000 m	$(6,300 \div 9,000) \times 900,000 = 630,000$
WP-2 Cable pulling	1,600,000	Units completed	18,000 of 40,000 m	$(18,000 \div 40,000) \times 1,600,000 = 720,000$
WP-3 Terminations	1,200,000	Weighted milestones	Milestones worth 25 % achieved	$0.25 \times 1,200,000 = 300,000$
WP-4 Testing and commissioning	700,000	0/100	Not started	0
WP-5 Supervision	400,000	Level of effort	Earns with time	$= PV = 260,000$
Total	4,800,000			1,910,000

Planned value and actual cost against the same packages:

Work package	PV	AC (incl. accruals)
WP-1 Containment	810,000	665,000
WP-2 Cable pulling	960,000	845,000
WP-3 Terminations	360,000	330,000
WP-4 Testing and commissioning	0	20,000
WP-5 Supervision	260,000	265,000
Total	2,390,000	2,125,000

The four derived measures.

$$\begin{aligned}
 CV &= EV - AC = 1,910,000 - 2,125,000 = (215,000) \\
 SV &= EV - PV = 1,910,000 - 2,390,000 = (480,000) \\
 CPI &= EV \div AC = 1,910,000 \div 2,125,000 = 0.899 \\
 SPI &= EV \div PV = 1,910,000 \div 2,390,000 = 0.799
 \end{aligned}$$

Percent complete by value: $EV \div BAC = 1,910,000 \div 4,800,000 = 39.8 \%$.

What the numbers say, and what they hide.

WP-4 carries USD 20,000 of actual cost against zero earned value. It is not a rounding matter: either the cost is miscoded, or testing equipment was procured earlier than the baseline planned. Both are findings. Until it is resolved, USD 20,000 of the cost variance has nothing to do with cabling productivity, and the CAM should say so in the narrative rather than let it be absorbed into a single number.

Now remove the level-of-effort package and recompute the discrete work only:

$$\begin{aligned}
 EV \text{ (discrete)} &= 1,910,000 - 260,000 = 1,650,000 \\
 PV \text{ (discrete)} &= 2,390,000 - 260,000 = 2,130,000 \\
 AC \text{ (discrete)} &= 2,125,000 - 265,000 = 1,860,000 \\
 \\
 CPI \text{ (discrete)} &= 1,650,000 \div 1,860,000 = 0.887 \\
 SPI \text{ (discrete)} &= 1,650,000 \div 2,130,000 = 0.775
 \end{aligned}$$

Supervision is 8.3 % of the account by budget ($400,000 \div 4,800,000$), and it moves the reported CPI from 0.887 to 0.899 and the reported SPI from 0.775 to 0.799 — a 2.4-point improvement in the schedule index bought entirely by including work that is incapable of reporting a schedule variance. The account is performing worse than its headline says, and the difference is structural rather than behavioural.

The schedule reading. The SV of (480,000) is 480,000 of budget not yet earned. It is not a duration. Converting it into a completion date requires the schedule: which of WP-2's remaining 22,000 m sit on the driving path to the testing start, and what does the network say about the finish. The index frames the question; it does not answer it.

Assumptions this example depends on. Earning rules were fixed before work started and have not changed; the ledger and the progress measurement share the Month 7 data date; accruals for delivered but uninvoiced subcontract work are booked; no variation has been added to the baseline during the period. Change any one of these and every figure above moves.

— 10. Checklist

Before you publish an earned value report, confirm each of the following. Anything you cannot answer in one sentence is the thing to fix before the meeting.

Measurement

- [] Every work package has an earning rule recorded in the baseline, and the rule applied this month is the rule recorded.
- [] Percent-complete claims above 85 % are supported by an objective basis, not an assessment.
- [] Level of effort is segregated in its own packages, and its share of each control account is known.
- [] Progress and the cost ledger share one data date.
- [] Accruals are booked for work performed but not invoiced.

Structure

- [] Each control account has one named manager who can state the cause of its variance.
- [] The scope boundary is identical across the schedule activities, the work packages and the cost codes.
- [] No control account contains actual cost against a package that cannot yet earn value — or, where it does, the reason is stated.

Reading

- [] Cumulative and current-period CPI are both reported.
- [] The rollup is accompanied by the two or three accounts driving it.
- [] SPI is reported with a schedule status in days, never on its own.
- [] Any account past 85 % complete has its SPI read with the convergence effect stated.
- [] Tolerances were set before the period, not after.

Integrity

- [] Every baseline change in the period is in the change log with an approval reference.
- [] No variance was removed by a baseline change during the period.
- [] Instructed but unincorporated variations are disclosed separately, so the reader knows the baseline is incomplete.

— Related

- [BPG-06 – Progress measurement and rules of credit](#) — the earning rules that determine whether EV means anything; read it before this one if your rules are not fixed.
- [BPG-07 – Accruals and cut-off discipline](#) — the other half of an honest CPI, on the actual cost side.
- [BPG-04 – Baseline and baseline change control](#) — where planned value comes from, and what a legitimate baseline change looks like.
- [BPG-09 – Estimate at completion: choosing and defending a method](#) — turning these indices into a forecast you can defend.

- [TPL-07 – Earned value calculation sheet](#) — the working instrument for the calculations in §9.

— Sources and standards

Earned value practice is described in several published frameworks — among them the AACE International Total Cost Management framework and the PMBOK Guide, alongside national and sector earned value management system standards. Their principles are explained here in our own words; no text, table or figure from any of them is reproduced. The internal reference for this guide is BoK Domain 6 (EVM/EAC) and BoK Domain 4 (Performance Management, Variance Analysis & Management Reporting). All figures in §5 and §9 are illustrative and were computed for this document.

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BPG-09 · BEST PRACTICE GUIDES

Estimate at completion — choosing and defending a method

The formula was never the hard part

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Every estimate at completion is a claim about the work that has not been done yet, and the formula only encodes which claim you are making. This guide runs the four standard methods over one shared data set so the spread is visible, gives the criteria for selecting between them, shows how to test a forecast for credibility before a board does, and sets out the four sentences a defensible forecast has to survive.

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Estimate at completion — choosing and defending a method

In one paragraph. Every estimate at completion is a claim about the work that has not been done yet, and the formula only encodes which claim you are making. This guide runs the four standard methods over one shared data set so the spread is visible, gives the criteria for selecting between them, shows how to test a forecast for credibility before a board does, and sets out the four sentences a defensible forecast has to survive.

Who this is for. Cost engineers, project controls managers and project managers who own a monthly forecast, and the sponsors and portfolio managers who have to decide whether to believe one.

— 1. A forecast is a decision, not an output

Three competent people can take the same control account data, apply three standard methods, and produce three estimates at completion (EAC) that differ by more than ten per cent of the budget. None of them has made an arithmetic error. They have made different assumptions about the remaining work, and the formulae are simply the shorthand in which those assumptions are written down.

This is the point most forecasting practice misses. The question a forecast answers is not "what does the data say" — the data says several things at once — but "what do we believe about the work ahead, and on what grounds". A team that cannot state its assumption in a sentence has not made a forecast. It has run a spreadsheet.

The professional obligation follows directly: **choose the method for a stated reason, disclose the reason, and say what would change it.** A forecast you cannot defend is a guess with decimals.

— 2. The identity underneath every method

Every EAC obeys one identity:

$$EAC = AC + ETC$$

where **AC** is actual cost to date and **ETC** is the estimate to complete — the forecast cost of the work remaining. Nothing else in the family is new. The methods differ only in how ETC is derived, and each derivation is an assumption about how the remaining work will behave relative to the work already done.

Two supporting quantities:

$VAC = BAC - EAC$ Variance at completion; negative = forecast overrun
 $TCPI = (BAC - EV) \div (BAC - AC)$ The cost efficiency the remaining work must achieve to land on BAC

BAC is the budget at completion and EV the earned value; both are defined and measured in BPG-08 — Earned value in practice, which this guide assumes.

— 3. The four methods, and the claim each one makes

Method	ETC derivation	The claim being made
A — Remaining work at budget	$ETC = BAC - EV$	The variance to date came from a discrete, closed event. The rest of the job runs to budget.
B — Remaining work at current cost performance	$ETC = (BAC - EV) \div CPI$, so $EAC = BAC \div CPI$	The inefficiency is systemic and persists at its present rate for the remainder.
C — Remaining work dragged by cost and schedule	$ETC = (BAC - EV) \div (CPI \times SPI)$	Being late is itself inflating cost — extended preliminaries, disruption, acceleration — so the two compound.
D — Bottom-up re-estimate	ETC estimated directly from the remaining scope	Past performance is not representative of what is left, because the remaining work is different in kind.

CPI is the cost performance index and SPI the schedule performance index.

Two remarks that save arguments later.

Method B has two forms that are algebraically identical. $AC + (BAC - EV) \div CPI$ reduces to $BAC \div CPI$ whenever CPI is computed as $EV \div AC$ on cumulative figures, because $AC = EV \div CPI$. If your two forms disagree, you have mixed cumulative and period indices, or you are dividing by a CPI that was rounded before use.

Method C is not "the conservative one". It embeds a specific causal story — that schedule slippage is converting into cost — and it is wrong, not merely cautious, on a project that is behind schedule for reasons that cost nothing extra. Reaching for C because it gives the largest number is as unprofessional as reaching for A because it gives the smallest.

— 4. Criteria for choosing

The method should match the **cause** of the variance, and cause is established by investigation, not by inspecting the index. Four questions, in order:

1. What actually caused the variance to date, and is it closed? If the variance came from a discrete event that has finished — a rework episode now complete, a rate spike now locked by a signed order, a one-off remediation — then the remaining work has no reason to inherit it, and method A is defensible. The test is evidence of closure: an executed contract, a completed scope, a signed-off inspection. "We think it was a one-off" is not closure.

2. Is the inefficiency structural? Productivity shortfalls, rate pressure in a tight labour market, an under-resourced supervision model and poor buildability are all conditions rather than events. They persist until something changes them, and method B is the natural forecast. Ask what specific, funded intervention would break the pattern; if nobody can name one, B is the honest answer.

3. Is lateness costing money? Where the project is behind and the delay is drawing time-related cost — extended site establishment, standing plant, retained supervision, acceleration measures — cost and schedule performance genuinely compound and method C reflects the mechanism. Where the project is behind on float-rich work with no time-related consequence, C overstates.

4. Does the remaining work look like the work done? This is the question that overrides the other three. A project moving from bulk construction into commissioning, from civils into fit-out, or from a first-of-a-kind unit into repeat units has a remainder whose cost behaviour is not predicted by the past at all. Extrapolation methods are then structurally unsuited, whatever their arithmetic, and a bottom-up ETC (method D) is the only sound basis.

A fifth consideration cuts across all four: **the stability of the CPI trend**. A cumulative CPI that has been flat for three or four periods is a settled measurement, and a forecast that assumes it will improve is making a claim that needs evidence. Published research on large programmes has examined how early cumulative cost performance stabilises and how rarely it recovers materially thereafter [\[CONFIRM: citation and date for the cumulative-CPI stability research, before publication\]](#). Whatever the precise finding, the practical rule is defensible on its own: once a CPI has settled, "the variance was atypical" is the claim that carries the burden of proof, not the default.

— 5. Reporting a range, not a point

A single EAC implies a precision the data does not support. Better practice is to report:

- the **selected** EAC and its method;
- the **assumption** it rests on, in one sentence;
- a **range** bounded by the plausible methods, with the assumption at each end;
- what would move the forecast, and by roughly how much.

A range is not evasion. It is a statement about which uncertainty is real and which is resolvable. If the range is wide because nobody has re-estimated the remaining scope, say so — that is a resource decision the sponsor can take. If it is wide because a major commercial position is unresolved, that is a different decision and belongs to a different person.

The **to-complete performance index (TCPI)** is the credibility test that should accompany any forecast that is better than method B. It states the cost efficiency the remaining work must achieve. Comparing it with the efficiency actually achieved so far converts an optimistic forecast into a specific, answerable claim: this forecast requires the team to perform 20 % better for the rest of the project than it has to date — here is why that is possible. If no such answer exists, the forecast is not a forecast.

— 6. Defending it to a board

A board does not want the method; it wants to know whether to act. Four sentences carry a forecast:

1. **The number and its movement.** "The forecast at completion is USD 19.2 million, up 0.4 million on last month."
2. **The method and the assumption in the same breath.** "It is a bottom-up re-estimate of the remaining scope, because the commissioning phase ahead does not resemble the construction behind us."

3. **The credibility test.** "It assumes the remaining work runs about 7 % more efficiently than the work to date, which the recovery plan is built to deliver."
4. **The trigger.** "If the commissioning schedule slips beyond the end of next quarter, extended preliminaries add roughly 0.6 million and the forecast moves to the top of the range."

What must not happen is the reverse order — the method explained first, the number last, the assumption never. And a forecast should never be presented as having "improved" because the method changed. Changing method between periods is legitimate when the cause changed; it is a reporting failure when it is not disclosed. State the previous method, the new method, and the reason, every time.

Automation makes this discipline more important, not less. A tool can compute the whole family in milliseconds and rank the results; it cannot know whether the commissioning phase resembles the civils. AI proposes; the professional disposes — and signs.

— 7. How this goes wrong

One method, applied forever, because it is in the template. The most common failure. $BAC \div CPI$ becomes the house forecast, and the assumption embedded in it is never revisited even when the project changes character entirely.

Method chosen by the answer it gives. Selecting downwards to protect a position, or upwards to build private contingency, is the same defect in opposite directions. Both are visible over time as forecast bias — a project whose EAC only ever moves in one direction was not forecasting, it was conceding.

Rounding the index before dividing by it. On a budget of 18,000,000, a CPI of 0.8949 gives $18,000,000 \div 0.8949 = 20,113,979$. Round the index to 0.89 first and the same calculation gives $18,000,000 \div 0.89 = 20,224,719$ — about 111,000 of forecast created by rounding. Carry full precision through the calculation and round only the presented result.

Forecasting the whole project as one number. EAC computed on project totals hides offsetting movements: a control account 400,000 under and another 900,000 over produce a comfortable-looking aggregate. Forecast at control account level, then aggregate.

Mixing the reserves into the forecast. Contingency drawn down for a materialised risk belongs in the forecast; contingency still held against open risk does not belong in an EAC as though it were spent, and management reserve is outside the baseline altogether. [BPG-10 – Contingency and management reserve](#) sets out the treatment; confusing the two is how a project reports an overrun it does not have, or hides one it does.

A bottom-up ETC that was not actually built bottom-up. "We re-estimated" frequently means the previous number was adjusted by a judgement. A genuine bottom-up ETC has a scope list, quantities, rates and a basis-of-estimate note. Anything else is method A with a different label.

Silent method changes. The forecast improves month on month and the reason is that the method rotated. This destroys the credibility of the whole reporting line, because a reader who spots it once will discount every subsequent forecast.

Ignoring what the forecast implies about the schedule. An EAC that assumes recovery of cost efficiency while the programme assumes the current durations is two documents describing different projects.

— 8. Worked example

Illustrative figures. Currency USD; data date the end of Month 10 of an 18-month baseline; all figures cumulative; indices carried at full precision and presented to four decimal places; a single control account portfolio.

The shared data set.

Quantity	Value
Budget at completion (BAC)	18,000,000
Planned value (PV)	7,680,000
Earned value (EV)	7,200,000
Actual cost (AC), including accruals	8,000,000

Derived:

```

CPI      = EV ÷ AC = 7,200,000 ÷ 8,000,000 = 0.9000
SPI      = EV ÷ PV = 7,200,000 ÷ 7,680,000 = 0.9375
CPI × SPI = 0.9000 × 0.9375 = 0.84375
Percent complete = EV ÷ BAC = 7,200,000 ÷ 18,000,000 = 40.0 %
Work remaining at budget = BAC - EV = 18,000,000 - 7,200,000 = 10,800,000
Budget remaining in cash = BAC - AC = 18,000,000 - 8,000,000 = 10,000,000

```

Method A — remaining work at budget.

```

ETC = BAC - EV = 10,800,000
EAC = AC + ETC = 8,000,000 + 10,800,000 = 18,800,000
VAC = BAC - EAC = 18,000,000 - 18,800,000 = (800,000)

```

Method B — remaining work at current cost performance.

```

ETC = (BAC - EV) ÷ CPI = 10,800,000 ÷ 0.9000 = 12,000,000
EAC = AC + ETC = 8,000,000 + 12,000,000 = 20,000,000
Cross-check: EAC = BAC ÷ CPI = 18,000,000 ÷ 0.9000 = 20,000,000 ✓
VAC = 18,000,000 - 20,000,000 = (2,000,000)

```

Method C — cost and schedule compounding.

```

ETC = (BAC - EV) ÷ (CPI × SPI) = 10,800,000 ÷ 0.84375 = 12,800,000
EAC = 8,000,000 + 12,800,000 = 20,800,000
VAC = 18,000,000 - 20,800,000 = (2,800,000)

```

Method D — bottom-up re-estimate of the remainder. The remaining scope is re-estimated from quantities and rates rather than extrapolated:

Remaining scope element	ETC
Remaining installation work	6,900,000
Commissioning and handover	2,400,000
Extended site preliminaries for the forecast two-month slip	1,300,000
Escalation on subcontract packages not yet let	600,000
ETC (bottom-up)	11,200,000

$$\begin{aligned} \text{EAC} &= \text{AC} + \text{ETC} = 8,000,000 + 11,200,000 = 19,200,000 \\ \text{VAC} &= 18,000,000 - 19,200,000 = (1,200,000) \end{aligned}$$

The spread, side by side.

Method	Assumption	EAC	VAC
A — remaining work at budget	The variance was a closed, one-off event	18,800,000	(800,000)
D — bottom-up re-estimate	The remainder is different work, priced directly	19,200,000	(1,200,000)
B — current cost performance persists	The inefficiency is structural	20,000,000	(2,000,000)
C — cost and schedule compound	Lateness is converting into cost	20,800,000	(2,800,000)

The four forecasts span $20,800,000 - 18,800,000 = 2,000,000$, which is $2,000,000 \div 18,000,000 = 11.1\%$ of the budget. That spread is not imprecision. It is the range of futures consistent with the same past, and choosing within it is the professional act.

The credibility test.

$$\text{TCPI to BAC} = (\text{BAC} - \text{EV}) \div (\text{BAC} - \text{AC}) = 10,800,000 \div 10,000,000 = 1.0800$$

Landing on budget requires the remaining 60 % of the work to run at a cost efficiency of 1.08 when the first 40 % achieved 0.90 — an improvement of $1.0800 \div 0.9000 = 1.20$, or 20 %, sustained to completion. Absent a named and funded change, that is not a forecast; it is a wish.

Apply the same test to the selected forecast:

$$\begin{aligned} \text{TCPI to EAC (method D)} &= (\text{BAC} - \text{EV}) \div (\text{EAC} - \text{AC}) \\ &= 10,800,000 \div (19,200,000 - 8,000,000) \\ &= 10,800,000 \div 11,200,000 = 0.9643 \end{aligned}$$

Even the bottom-up forecast assumes the remaining work runs at 0.9643 against 0.9000 achieved — an improvement of $0.9643 \div 0.9000 = 1.0714$, about 7 %. That is a much smaller claim than 20 %, but it is still a claim, and it belongs in the narrative. A forecast whose embedded improvement is never stated is optimistic by default.

Selecting, and saying why. The remaining scope includes commissioning and handover — work of a different character from the installation performed to date — so extrapolation methods are structurally unsuited, and method D is chosen. The bottom-up ETC lands at 19,200,000, inside the

range set by the formula methods, which is a reassurance rather than a validation: it means the re-estimate has not drifted away from what performance to date implies.

The board paragraph. "Forecast at completion USD 19.2 million against a budget of 18.0 million, a projected overrun of 1.2 million. The basis is a bottom-up re-estimate of the remaining scope, chosen because the commissioning phase ahead does not resemble the installation work behind us, so performance to date does not predict it. The forecast assumes the remaining work runs about 7 % more cost-efficiently than the work to date, which the revised supervision model is intended to deliver. If that improvement does not materialise, the persisting-performance forecast is 20.0 million; if the two-month slip extends, extended preliminaries take it towards 20.8 million. Recovery to the 18.0 million budget would require a 20 % sustained improvement in cost efficiency and is not credible on current evidence."

Assumptions this example depends on. Earning rules are objective and unchanged; accruals are booked so AC is complete at the data date; no instructed variation is awaiting incorporation into the baseline; the contingency held against open risk is excluded from all four EAC figures and reported separately; escalation is included only in the bottom-up ETC, where it is explicit.

— 9. Checklist

Use this before the forecast leaves your desk.

Before calculating

- [] The cost ledger is closed at the data date and accruals are booked.
- [] Instructed variations awaiting incorporation are identified and quantified separately.
- [] Contingency drawn down is separated from contingency still held.
- [] The cause of the cumulative variance has been investigated, not inferred from the index.

Calculating

- [] The forecast is built at control account level and aggregated, not computed on project totals.
- [] Indices are carried at full precision and rounded only for presentation.
- [] Where method B is used, both forms agree.
- [] Where method D is used, there is a scope list, quantities, rates and a basis-of-estimate note.
- [] Escalation and currency exposure are either in the ETC explicitly or excluded explicitly.

Testing

- [] TCPI to BAC computed, and the required improvement expressed as a percentage of performance to date.
- [] TCPI to the selected EAC computed, so the improvement embedded in your own forecast is visible.
- [] The forecast is consistent with the current schedule — the same slip, the same durations.
- [] A range is stated with the assumption at each end.

Reporting

- [] The method is named, and the assumption stated in one sentence.
- [] Any change of method since last period is disclosed with its reason.
- [] The movement since last period is explained by cause, not by method.

- [] The trigger that would move the forecast is named, with its approximate value.

— Related

- [BPG-08 – Earned value in practice](#) — the measurement discipline every EAC method depends on; read it first if your CPI is not trusted.
- [BPG-10 – Contingency and management reserve](#) — what belongs in the forecast, what sits beside it, and what sits outside the baseline entirely.
- [BPG-16 – Risk registers that work](#) — where the quantified exposure behind a forecast range comes from.
- [BPG-14 – Monthly reporting that gets read](#) — how the forecast and its assumption reach a decision-maker intact.
- [TPL-08 – Estimate at completion scenario comparison](#) — the instrument for running §8 on your own data.

— Sources and standards

The EAC family and the to-complete performance index are described in published earned value frameworks, including the AACE International Total Cost Management framework and the PMBOK Guide; their principles are explained here in our own words and no text or table is reproduced. The internal reference is BoK Domain 6 (EVM/EAC) and BoK Domain 3 (Budgeting & Forecasting). All figures in §8 are illustrative and were computed for this document. One citation remains outstanding, marked in §4.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

BPG-10 · BEST PRACTICE GUIDES

Contingency and management reserve

Two reserves, two owners, two entirely different governance events

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Contingency and management reserve are not two sizes of the same pot. This guide sets out what each covers, who owns it, where it sits relative to the cost baseline, how contingency is derived from and kept traceable to the risk register, how a drawdown is approved and recorded without moving the budget at completion, and how to read the drawdown curve as an early-warning indicator months before the forecast catches up.

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Contingency and management reserve

In one paragraph. Contingency and management reserve are not two sizes of the same pot. This guide sets out what each covers, who owns it, where it sits relative to the cost baseline, how contingency is derived from and kept traceable to the risk register, how a drawdown is approved and recorded without moving the budget at completion, and how to read the drawdown curve as an early-warning indicator months before the forecast catches up.

Who this is for. Project controls managers, cost engineers, risk managers and project sponsors — and anyone who has been asked "how much contingency is left" and found that the honest answer took a week to assemble.

— 1. Two reserves answer two different questions

The distinction is not about size or seniority. It is about what kind of uncertainty each reserve is funding.

Contingency reserve funds **identified** risk — the entries in the risk register, plus quantified estimating uncertainty. Someone has written the risk down, assessed it and priced it. Contingency sits **inside** the cost baseline, is controlled by the project, and is drawn down as those identified risks materialise.

Management reserve funds **unidentified** risk — the events nobody wrote down, and scope not yet foreseen. It sits **outside** the cost baseline, is controlled by the sponsor or the governing body, and its release is a change to the baseline rather than a movement within it.

The budget structure follows directly:

Cost baseline (BAC) = Σ control account budgets + contingency reserve
Total authorised budget = BAC + management reserve

Everything that matters operationally falls out of that arithmetic. Earned value performance is measured against BAC, so contingency is inside the measurement and management reserve is not. Drawing contingency does not change BAC. Releasing management reserve does.

Note also what the two reserves are not. Neither is an accounting provision. How reserves, provisions and onerous-contract positions are recognised in financial statements is governed by the applicable reporting framework and varies; a project budget reserve is a control instrument, and the accounting treatment is a separate question for the finance function.

— 2. Where contingency comes from

Contingency has one legitimate parent: **analysed risk**. A percentage applied to the estimate is not contingency, it is a habit. The moment anyone asks "what is this covering", a percentage has no answer, and the reserve becomes indefensible in exactly the meeting where it needs defending.

Two derivations are standard, and they answer different questions.

Expected monetary value (EMV) sums *probability × impact* across the register. It gives the mean outcome — the amount you would need if this project were run many times. It is simple, transparent and traceable to individual risks, and it systematically underfunds any single project, because a single project does not experience the mean.

A probabilistic P-level, typically from a Monte Carlo simulation of the register, gives the amount at which the project has a stated confidence of not overrunning — commonly P80, meaning an 80 % chance the outturn falls at or below the funded level. It captures how risks combine, including the effect of shared drivers, which the EMV sum cannot.

The gap between the two is the price of confidence, and stating it is good practice: "the mean exposure is 1.74 million; funding to P80 costs 2.60 million; the 0.86 million difference buys the organisation's stated risk appetite." That sentence turns a reserve from a number into a decision the sponsor has actually taken.

Correlation matters here more than most registers admit. Risks driven by a single shared cause — one supplier, one permitting authority, one commodity — do not behave independently, and treating them as independent understates the tail without changing the mean. Quantification method belongs to *BPG-17 – Quantitative schedule risk analysis* and the register itself to *BPG-16 – Risk registers that work*; what this guide insists on is the link: **every unit of contingency should be traceable to a risk identifier**.

— 3. Traceability, and why it is the whole control

A contingency line that cannot be decomposed into risks is a slush fund, and it will be spent like one. Traceability delivers four things that no aggregate figure can:

1. **A drawdown test.** A request to draw contingency is answerable — does it correspond to a risk in the register, and did that risk occur? If not, the cost is an overrun, and calling it a contingency draw conceals a performance problem inside a risk provision.
2. **A release mechanism.** When a risk passes without occurring, its contingency can be released back to the pot or to the sponsor, deliberately and visibly.
3. **A re-assessment path.** When a risk's probability or impact changes, the required contingency changes with it, and the reserve is re-derived rather than argued about.
4. **A defence.** "This is what it covers, risk by risk" is the only answer that survives an assurance review.

The practical form is unglamorous: a contingency register keyed to risk identifiers, showing for each risk the assessed exposure, the amount allocated, the amount drawn, the residual, and the status. It is one extra column set on the risk register rather than a separate document, and it should reconcile to the total contingency figure in the budget every month.

— 4. Holding it, and drawing it down

Hold contingency centrally. Contingency distributed into control account budgets stops being contingency; it becomes budget, and budget is always eventually consumed. The control account manager who has 60,000 of "risk allowance" inside a 900,000 package has no incentive to hand it back and no mechanism to do so. Hold the reserve at project level and release it against evidence.

Define the drawdown mechanism before the first draw. It needs four elements:

- **A trigger** — the risk has occurred, or has become certain to occur.
- **Evidence** — what happened, which risk identifier it corresponds to, and the cost of the consequence.
- **An approval level** — thresholds by value, with the sponsor's involvement rising with the amount.
- **A record** — a numbered drawdown entry, posted through the change log so the movement is auditable.

Record the draw as a transfer, not a variance. This is the mechanical point most often muddled. When contingency is drawn, budget moves within the baseline: the contingency pot decreases and a control account budget increases by the same amount. BAC is unchanged. Planned value for the affected account increases from the point of transfer forward, so the account's future earned value measurement is against the revised budget. Nothing about the past is restated. If your process instead increases BAC when contingency is drawn, contingency was never inside the baseline, and every historical index is irreconcilable.

— 5. The drawdown curve as an early-warning indicator

Plot cumulative contingency drawn against time, with two comparators on the same axes: physical progress (percent complete by earned value) and, separately, the remaining assessed exposure. The shape tells you things the cost report will not say for another two quarters.

Healthy. Drawdown tracks the retirement of risk. Contingency falls as risks either occur (drawn) or pass (released), and the remaining reserve stays at or above the remaining exposure throughout.

Front-loaded. Contingency consumed rapidly in the first third. Either the early risks were the severe ones — legitimate, and the register should show it — or contingency is subsidising ordinary estimating error, which will continue. A front-loaded curve on a project whose register still carries its major risks is one of the strongest early-warning signals in project controls, because it means the reserve will be gone before the exposure is.

Flat then cliff. Nothing drawn for months, then a large single draw. Almost always a reporting failure rather than a risk profile: costs were absorbed into control accounts and only escalated when they became impossible to absorb. Look for the corresponding CPI deterioration in the preceding periods.

Never drawn. Either the risks are not materialising — say so, and consider releasing reserve — or draws are being avoided because the approval process is painful, in which case the same costs are hiding inside control accounts and the contingency figure is meaningless.

The test that matters is not the burn ratio. Comparing "percentage of contingency drawn" with "percentage complete" is the intuitive check and it is close to useless, because contingency is not consumed in proportion to progress; it is consumed in proportion to risk occurrence, which is lumpy. The defensible test is:

Remaining contingency $\geq$ Remaining exposure (at the same confidence level as it was set)

Run it every month. §8 shows a project that passes the burn-ratio test comfortably and fails this one.

— 6. Management reserve: release is a baseline change

Management reserve is used when something occurs that the register did not and reasonably could not contain — a new statutory requirement, a client-directed change of a kind nobody anticipated, a systemic event. Three consequences follow, and all three are frequently missed.

It is a re-baselining event. Releasing reserve into the baseline increases BAC. Every earned value measure computed against the old BAC is now on a different basis, and the change must be logged with its date, so a reader of the trend knows where the discontinuity is. [BPG-04 – Baselining and baseline change control](#) owns the procedure.

It is a governance decision, not a controls decision. The project may request; the sponsor or governing body releases. A project that can draw management reserve on its own authority does not have a management reserve.

It is not a solution to an exhausted contingency. When the register has outgrown the reserve, the correct response is to re-derive contingency from the current register, show the shortfall, and ask for a decision — funding it, de-scoping, or accepting the exposure. Silently drawing management reserve to cover identified risks converts a visible funding gap into an invisible one, and destroys the distinction that makes either reserve meaningful.

— 7. Schedule contingency, briefly

The same logic applies in time. Schedule contingency is the float deliberately held against identified schedule risk, owned by the project and drawn down as risks occur; it is not the incidental float that falls out of the network, and it should be visible as a named buffer rather than hidden in inflated durations. Padding individual activities is the schedule equivalent of distributing contingency into control accounts: the protection is consumed silently and cannot be managed. Quantification of schedule risk is the subject of [BPG-17 – Quantitative schedule risk analysis](#).

Where the two interact — a delay that draws both time and time-related cost — take care not to fund the same event twice, once as schedule contingency and again as cost contingency for prolongation. Price it once, in the register, against one risk identifier.

— 8. How this goes wrong

Contingency set as a percentage. Ten per cent, because it has always been ten per cent. It is undefendable, it bears no relation to this project's risk profile, and it is the first thing an assurance reviewer will ask about.

Contingency distributed into control accounts. It is spent. Always.

Drawdown without a risk identifier. Ordinary overruns re-labelled as risk events. The reserve depletes, the CPI looks better than it should, and the register never changes — a combination that should trigger immediate investigation.

Double-counting risk and escalation. A commodity price risk priced in the register and covered by an escalation allowance in the estimate. Both are legitimate instruments; funding the same exposure twice is not, and it inflates the reserve in a way that is easy to attack.

Treating unspent contingency as a saving. At completion, contingency not drawn should be explained, not celebrated. Either the register overstated the exposure, which is an estimating lesson worth capturing, or the risks genuinely did not materialise, which is luck worth distinguishing from performance.

Reporting the pot without the exposure. "Contingency remaining: 1.65 million" is not information. "1.65 million remaining against 1.90 million of assessed remaining exposure" is.

Using contingency to fund scope. Scope growth is a change, and it belongs in the change control process with a decision about who pays. Absorbing it in contingency hides the growth and defunds the risks the reserve exists for.

No release discipline. Risks pass, and their contingency stays in the pot without being re-assessed. Over time the reserve stops corresponding to the register in either direction, and nobody trusts the figure.

— 9. Worked example

Illustrative figures. Currency USD; a 20-month project; risk impacts are the assessed cost consequence should the risk occur; the confidence level for contingency is P80, taken from a Monte Carlo model of the register; all figures at the stated data dates.

At sanction — assembling the budget.

Element	Amount
Σ control account budgets	24,000,000
Contingency reserve (set at P80)	2,600,000
Budget at completion (BAC)	26,600,000
Management reserve (sponsor-held, outside baseline)	1,400,000
Total authorised budget	28,000,000

Deriving the contingency. The register carries five priced risks:

Risk	Description	Probability	Impact	EMV = P × I
R-01	Consent granted later than programmed	0.40	900,000	$0.40 \times 900,000 = 360,000$
R-02	Ground conditions worse than survey	0.30	1,500,000	$0.30 \times 1,500,000 = 450,000$
R-03	Steel price movement beyond the hedged volume	0.50	700,000	$0.50 \times 700,000 = 350,000$
R-04	Specialist subcontractor default	0.15	2,000,000	$0.15 \times 2,000,000 = 300,000$
R-05	Rework arising from late design change	0.35	800,000	$0.35 \times 800,000 = 280,000$
Total EMV (the mean)				1,740,000

The Monte Carlo model of the same register, capturing the shared exposure between R-02 and R-05 (both driven by the same late design package), returns a P80 of **2,600,000**. Contingency is set there.

$$\text{Cost of confidence} = \text{P80} - \text{EMV} = 2,600,000 - 1,740,000 = 860,000$$

That 860,000 is the amount the organisation is paying to move from funding the average outcome to funding an 80 % confidence outcome. It is a sponsor's decision, and it is recorded as one.

At Month 9 — the drawdown position.

Two risks have occurred. R-01 materialised and the consent delay cost 640,000 in standing time and resequencing — less than the 900,000 assessed impact. R-05 materialised in part, drawing 310,000.

$$\begin{aligned} \text{Contingency drawn} &= 640,000 + 310,000 = 950,000 \\ \text{Remaining contingency} &= 2,600,000 - 950,000 = 1,650,000 \end{aligned}$$

Both draws were posted as transfers into the affected control accounts. BAC remains 26,600,000; the contingency pot inside it fell by 950,000 and control account budgets rose by the same amount.

The comfortable test. Earned value shows the project 40.0 % complete. Contingency drawn is $950,000 \div 2,600,000 = 36.5 \%$. Drawing 36.5 % of the reserve at 40.0 % complete looks disciplined, and this is the figure that appears on most reserve reports.

The test that matters. The register is re-assessed at the same date:

Risk	Status at Month 9	Probability	Impact	EMV
R-01	Occurred and closed; drawn	—	—	—
R-02	Open; trial pits raised the probability	0.45	1,500,000	$0.45 \times 1,500,000 = 675,000$
R-03	Closed — remaining volume price-fixed; contingency released	—	—	0
R-04	Open, unchanged	0.15	2,000,000	$0.15 \times 2,000,000 = 300,000$
R-05	Occurred in part; residual closed	—	—	—
R-06	New: commissioning resource availability	0.25	1,200,000	$0.25 \times 1,200,000 = 300,000$
Remaining EMV				1,275,000

Re-running the model on the remaining register at the same P80 confidence gives a **remaining exposure of 1,900,000**.

$$\begin{aligned} \text{Remaining contingency} &= 1,650,000 \\ \text{Remaining exposure (P80)} &= 1,900,000 \\ \text{Shortfall} &= 1,900,000 - 1,650,000 = (250,000) \end{aligned}$$

The project is 250,000 short of the confidence level it was funded to, at Month 9 of 20 — while the burn ratio says it is fine. The gap opened because R-02's probability rose after the trial pits and a

new risk (R-06) entered the register; both are ordinary events, and neither is visible in the drawdown percentage.

What the report should say. "Contingency remaining 1.65 million against remaining assessed exposure of 1.90 million at P80 — a shortfall of 0.25 million. Driven by R-02, whose probability rose to 0.45 following the trial pit results, and by new risk R-06. Options: fund the gap from management reserve as a baseline change, reduce exposure by advancing the ground investigation to convert R-02 into a known quantity, or accept a reduced confidence level and record the decision. Recommendation: advance the ground investigation at an estimated 140,000, which retires the largest single exposure. Decision required this month; the option closes when the foundation sequence starts."

If management reserve were released. Suppose the sponsor releases 800,000 of management reserve to fund the gap and a small scope addition:

```
New BAC = 26,600,000 + 800,000 = 27,400,000
Management reserve remaining = 1,400,000 - 800,000 = 600,000
Total authorised budget = 27,400,000 + 600,000 = 28,000,000 (unchanged)
```

The total authorised budget does not move — the money was always authorised. What moves is the baseline, and therefore every earned value comparison from that date forward. The change log records the release, its date and its reason, and the performance trend is annotated so no reader mistakes the step change for a performance improvement.

Assumptions this example depends on. Probabilities and impacts are the assessed values recorded in the register at each date; the Monte Carlo outputs are model results at the stated confidence level; risk impacts are cost only, with any time-related consequence priced once and not duplicated in schedule contingency; no escalation allowance covers the same exposure as R-03.

10. Checklist

Setting it

- [] Contingency is derived from the risk register, not from a percentage.
- [] The derivation method and confidence level are stated (EMV, or a named P-level).
- [] The gap between the mean and the funded level is disclosed as the cost of confidence.
- [] Shared drivers between risks have been considered, not assumed away.
- [] Escalation, currency and risk allowances do not cover the same exposure twice.

Holding it

- [] Contingency is held at project level, not distributed into control account budgets.
- [] Management reserve is held outside the baseline, by the sponsor.
- [] The budget structure reconciles: control accounts + contingency = BAC; BAC + management reserve = total authorised budget.

Drawing it

- [] Every draw cites a risk identifier and the evidence that the risk occurred.
- [] Approval thresholds by value are defined and followed.
- [] Draws are posted as transfers within the baseline; BAC is unchanged.

- [] Every draw is in the change log with a number, date and approver.
- [] Contingency for risks that have passed is released deliberately, not left in the pot.

Reading it

- [] Remaining contingency is compared with remaining exposure at the same confidence level, monthly.
- [] The drawdown curve is plotted against risk retirement, not against percent complete.
- [] Any shortfall is reported with options and a recommendation, not as a status.
- [] Management reserve releases are annotated on every performance trend they affect.
- [] At completion, undrawn contingency is explained rather than presented as a saving.

— Related

- [BPG-16 – Risk registers that work](#) — the register that contingency must be traceable to; without it, nothing in this guide is possible.
- [BPG-17 – Quantitative schedule risk analysis](#) — where a defensible P-level comes from, and how correlation moves the tail.
- [BPG-04 – Baselining and baseline change control](#) — the procedure for a management reserve release and for logging drawdown transfers.
- [BPG-09 – Estimate at completion: choosing and defending a method](#) — what belongs in a forecast and what sits beside it as held reserve.
- [TPL-10 – Risk register](#) — the instrument, including the contingency allocation columns used in §9.

— Sources and standards

The two-reserve structure and the derivation of contingency from analysed risk are described in published cost and risk management frameworks, among them the AACE International Total Cost Management framework, the PMBOK Guide and the ISO 31000 risk management principles; they are explained here in our own words and no text or table is reproduced. Accounting treatment of provisions and reserves is governed by the applicable financial reporting framework and is outside the scope of this guide. The internal references are BoK Domain 3 (Budgeting & Forecasting) and BoK Domain 12 (Risk Management for Project Controls). All figures in §9 are illustrative and were computed for this document.

— Status and version

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BPG-11 · BEST PRACTICE GUIDES

Change orders and variations

A change to the work and a change to the baseline are two separate events

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

A change to the work and a change to the baseline are two separate events with different authorities and different dates, and the gap between them is where control is lost. This guide covers who may instruct a change, why notice provisions decide entitlement before anyone argues about value, the pricing bases and what each does to productivity risk, how instructed-but-unpriced and disputed variations should sit in a forecast, why a variation dropped into the baseline at the wrong date corrupts the performance indices, and the log discipline that holds it together.

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Change orders and variations

In one paragraph. A change to the work and a change to the baseline are two separate events with different authorities and different dates, and the gap between them is where control is lost. This guide covers who may instruct a change, why notice provisions decide entitlement before anyone argues about value, the pricing bases and what each does to productivity risk, how instructed-but-unpriced and disputed variations should sit in a forecast, why a variation dropped into the baseline at the wrong date corrupts the performance indices, and the log discipline that holds it together.

Who this is for. Quantity surveyors, commercial managers, cost engineers and project controls managers on either side of a contract, and the project managers who sign the instructions.

This guide is not legal advice. Entitlement to additional payment or time, the effect of failing to give notice, and the valuation rules that apply all depend on the specific contract and on the law governing it. Nothing here describes the position under any particular jurisdiction or standard form. Take advice on your own contract.

— 1. Two changes, two clocks

When a client instructs additional work, two distinct things need to happen and they almost never happen together:

1. **The scope changes.** From the moment of instruction, the work required is different from the work priced. Resources are committed and cost begins to accrue.
2. **The baseline changes.** At some later point — after the value is agreed and a change order is issued — the budget, the schedule and the measurement yardstick are updated to include the new work.

The interval between them is often months. Inside it, the project performs work its baseline does not recognise, incurs cost its budget does not contain, and reports indices that measure new work against an old plan. Nothing is wrong with the project; the reporting is describing a project that no longer exists.

Three cases are worth separating, because they are governed differently:

- **Scope change with a baseline change** — the ordinary, well-run case.
- **Scope change without (yet) a baseline change** — the instructed but unpriced variation. Legitimate, temporary, and dangerous if not disclosed.
- **Baseline change without a scope change** — re-phasing, transfers between control accounts, contingency drawdown. These are internal and must never be described as variations, because they carry no commercial entitlement.

The controls consequence: a change register that only records agreed change orders is recording the second event and missing the first. The register must start at the instruction, not at the agreement.

— 2. Instruction, authority and the notice trap

Authority. Contracts name who may instruct a change. Work performed on the say-so of someone without that authority is, commercially, work performed for free until proved otherwise. This is not a theoretical risk: it is the single most common source of unrecovered cost on a busy site, because the instruction usually arrives as a reasonable-sounding request from someone senior standing next to the work.

The practical control is unglamorous and effective: **no change is worked until it has an instruction reference in the log.** Where site conditions make that impossible, the person who accepted the direction records it the same day, in writing, to the named contract representative, describing what was directed and by whom.

Notice. Most contracts require the party seeking additional time or money to notify within a stated period of the event, or of becoming aware of it. What happens if that period is missed varies enormously: under some contracts and some governing laws the failure bars the entitlement outright; under others it affects only the evidence or the quantum; under others still it has little effect. Standard forms differ from each other, and the same form can operate differently under different governing law. **Never assume the position on your project — read the contract and take advice.**

What is universal is the control. Notice periods are short, they run from an event date rather than from the date somebody noticed, and they are missed for administrative reasons rather than commercial ones. A project therefore needs:

- **A notice diary** with one owner, listing every potential entitlement event, its event date, the applicable period and the due date.
- **A weekly review** of items approaching their due date, at a meeting where somebody can actually issue the notice.
- **A default to notify.** The cost of an unnecessary notice is a small amount of goodwill. The cost of a missed one can be the whole claim.

Change by conduct. Work can also be changed without an instruction — by a drawing revision, by a specification clarification that clarifies nothing, by acts that alter the basis on which the work was priced. Whether that carries entitlement depends entirely on the contract and the governing law; the controls discipline does not change. Record it as a potential change on the day it is noticed, price it, and give notice.

— 3. Pricing bases and who carries the productivity risk

Contracts set out how varied work is valued, and the hierarchy typically moves from "use the rates you already agreed" towards "price it afresh" as the work departs further from what was priced. Four bases cover most situations.

Contract or bill rates. Where the varied work is of similar character and executed under similar conditions to work already priced, the existing rates apply. Fast, uncontentious, and it leaves productivity risk where the original contract put it.

Pro-rata or "star" rates. Where the work is similar in kind but different in conditions — higher, deeper, smaller quantities, worse access — a rate is derived from the contract rates and adjusted. The adjustment is the argument; the derivation should be shown element by element.

A new rate built up from first principles. Where the work bears no useful relation to anything priced, the rate is constructed from materials, labour hours and rate, plant, on-costs and profit. Agreed before work starts, it becomes a lump sum and the contractor carries the productivity risk.

Daywork or cost reimbursement with defined additions. The work is recorded as it happens and reimbursed at daywork rates or at cost plus a stated percentage. This moves productivity risk to the paying party, which is why paying parties resist it and why daywork records must be signed every day by someone with authority. An unsigned sheet produced three months later is worth very little.

Two points that save money:

- **The basis is determined by the contract, not by preference.** Choosing daywork because it is easier to administer, or bill rates because they are favourable, is not valuation; it is negotiation dressed as valuation, and it collapses under scrutiny.
- **Time-related cost is a separate question from the direct works.** A variation that extends the duration of the works may attract prolongation cost, and that is a different assessment with different evidence. Pricing the direct works and assuming the time cost is "in the rates" is how contractors lose the second half of a variation. Where a delay is involved, see [BPG-12 – Claims and extension of time](#).

— 4. Where variations sit in a forecast

At any data date a project's change population falls into three buckets, and each is treated differently.

Bucket 1 — instructed and agreed. The value is fixed and a change order has been issued. The budget is added to the baseline, time-phased to when the work is planned (see §5), and the work then earns value normally. Both cost and recovery are in the numbers.

Bucket 2 — instructed but not yet priced or agreed. The work is proceeding; the value is not settled. The cost belongs in the forecast at the best current estimate, because it is going to be incurred. The budget does **not** belong in the baseline, because there is no agreed value to add. The result is an account whose estimate at completion legitimately exceeds its budget at completion, and the report must say why — otherwise a reader sees an overrun where there is extra scope.

Bucket 3 — notified, claimed or disputed. Entitlement is asserted but not accepted. Cost still belongs in the forecast where it will be incurred. Recovery is the difficult half, and the discipline is asymmetric: **recognise the cost when it is probable; recognise the recovery only when it meets the applicable recognition threshold.** Revenue standards work on this principle — IFRS 15, for example, constrains variable consideration such as a claimed variation — but the exact threshold, and the treatment in management accounts as opposed to statutory ones, depends on the reporting framework in use. Take the accounting position from the finance function; take the control position from this rule: **never forecast cost net of an assumed recovery.**

Report the estimate at completion in two lines: the forecast for the baselined scope, and the forecast effect of unincorporated change. A single blended number hides which of the two is moving, and they move for entirely different reasons.

— 5. Time-phasing a variation into the baseline

When an agreed variation enters the baseline, its budget must be spread across the periods in which the varied work is planned to be performed. This sounds obvious and is routinely got wrong, because the simplest thing a cost system can do is add the budget at the date of the change order.

The effect is immediate and entirely artificial. Adding planned value in the current period for work not yet started leaves earned value unchanged and the denominator of the schedule performance index larger, so the index falls. Nothing about performance has changed; the project has simply been credited with a plan it was never given a chance to meet. §9 quantifies the effect: an index drop of 0.030 created by a filing decision.

The same distortion runs the other way at the end. A variation phased into months that have already passed inflates planned value retrospectively and creates schedule variance that no action can recover.

The rule: **the variation's budget is phased to the varied work's planned dates, and the schedule is updated in the same cycle.** If the varied work has not been programmed, the variation is not ready to be incorporated — and that, not the value, is usually the thing holding it up.

— 6. The change order log

One log, one owner, reviewed weekly. It is the project's commercial memory, and it is also — although nobody calls it this at the time — the first draft of any claim file that may be needed later. The columns that earn their place:

Column	Why it is there
Change identifier	The permanent citation key; never reused
Event date	When the change arose — the date notice periods run from
Notice date and reference	Evidence that notice was given, and when
Instruction reference and issuer	Establishes authority (\$2)
Description and originating cause	Client change, design development, site condition, error
Bucket (agreed / instructed-unpriced / claimed)	Drives the forecast treatment (\$4)
Pricing basis	Bill rates, star rate, quotation, daywork (\$3)
Submitted value	What was asked for
Assessed or certified value	What was accepted
Time effect claimed and granted, in days	Kept separate from money, always
Affected control accounts and cost codes	So cost can be isolated as it is incurred
Status, next action, owner, next action date	Turns a register into a live control
Age in current status	The single most useful column in the log

Ageing is the control. Set a threshold — nothing sits in "submitted, awaiting assessment" for more than a stated number of weeks — and report the ageing profile monthly alongside the values. A log where the average age of unpriced variations is rising is a project whose commercial position is deteriorating whatever the cost report says, because the unpriced book is growing faster than it is being closed.

Code the cost as it is incurred. Each change needs cost codes that capture its cost separately from the day work starts. Costs never coded to the event cannot be isolated a year later, which is precisely when someone will need them.

— 7. Cash, and why an unpriced book hurts twice

Even when a variation goes well it is paid late: instructed, worked, priced, assessed, certified, then paid on ordinary terms. Until agreement, the work is being funded with nothing certified against it, so a growing unpriced book deepens the funding trough exactly when base scope is at peak spend. See [BPG-13 – Cash flow forecasting](#).

— 8. How this goes wrong

Working on an instruction that does not exist. Verbal direction, a marked-up drawing, an email that reads like a request. Six months later the question is not whether the work was done but whether it was instructed by someone entitled to instruct it.

Missing a notice period for administrative reasons. The entitlement may be real, the cost evidenced, and the outcome still nil. Notice discipline costs almost nothing and protects everything.

Pricing the direct works and forgetting the time. Prolongation, disruption and extended preliminaries are separate heads with separate evidence. A variation "priced in full" that ignored the two-week delay it caused is priced in half.

Netting recovery against cost in the forecast. A forecast that shows a disputed variation at zero net cost, on the assumption of full recovery, is a forecast of a negotiation. The cost is real now; the recovery is contingent.

Dropping the variation budget into the current period. See §5 and §9. It corrupts every schedule index in the account and hides the corruption inside a legitimate-looking change.

A change register that starts at the change order. Everything instructed-but-unpriced is invisible, and the project's exposure is understated by the entire size of the unpriced book. The same register, left without an ageing column or an owner, stops being a control and becomes a filing cabinet.

No cost coding for the change. The cost is incurred inside the ordinary control accounts, becomes indistinguishable from base scope, and the eventual submission has to be reconstructed by estimate. This is also how a well-founded claim becomes a global one — see [BPG-12 – Claims and extension of time](#).

Double-counting between a variation and a claim. The same disruption recovered once inside a star rate and again in a disruption claim. It is the most damaging quantum error there is, because finding one instance of it invites the assessor to distrust everything else in the submission.

— 9. Worked example

Illustrative figures. Currency USD; a single control account within a construction contract; rates and percentage additions are those stated in the contract for the purposes of this example; quantities remeasured on site; all values exclude any indirect tax, whose treatment varies by jurisdiction.

9.1 One variation, three pricing bases

The client instructs an additional **1,200 m²** of cladding at high level. The contract's valuation rules must decide which basis applies; the three candidate outcomes are:

(a) Contract bill rates. The bill carries a cladding item at **420.00 per m²**.

$$\text{Value} = 420.00 \times 1,200 = 504,000$$

(b) A star rate derived from the bill rate and adjusted for conditions. The additional area is at height, needing mast climbers and slower installation, so a rate is built up:

Element	Rate
Materials	210.00 / m ²
Labour: 1.6 h/m ² at 68.00/h	$1.6 \times 68.00 = 108.80$ / m ²
Access equipment	91.20 / m ²
Direct cost	410.00 / m²
Site on-costs and overhead at 10 %	$0.10 \times 410.00 = 41.00$ / m ²
Subtotal	451.00 / m ²
Profit at 5 %	$0.05 \times 451.00 = 22.55$ / m ²
Star rate	473.55 / m²

Value = $473.55 \times 1,200 = 568,260$

Difference from the bill rate route = $568,260 - 504,000 = 64,260$

(c) Daywork at contract daywork rates. Recorded as executed, with the contract's stated percentage addition of 15 % on materials and plant:

Labour:	$1,200 \times 1.6 = 1,920$ h at 82.00/h	= 157,440
Materials:	$210.00 \times 1,200 = 252,000$, + 15 %	= 289,800
Plant:	$91.20 \times 1,200 = 109,440$, + 15 %	= 125,856
Total		= 573,096

The spread. $573,096 - 504,000 = 69,096$, which is $69,096 \div 504,000 = 13.7$ % of the lowest figure. All three are arithmetically correct valuations of the same physical work. Which one applies is a question about the contract — whether the work is of similar character executed under similar conditions — and it is decided by reading the contract, not by preference.

Note where the productivity risk lands. Under (a) and (b) the contractor is paid a rate and absorbs the consequence if installation takes longer than assumed. Under (c) the paying party absorbs it, hour by hour, which is why daywork records must be signed the day the work is done.

9.2 Incorporating the agreed variation into the baseline

Assume basis (a) applies and **504,000** is agreed. The control account before incorporation, at the Month 6 data date:

Quantity	Value
Budget at completion (BAC)	6,000,000
Planned value (PV)	3,600,000
Earned value (EV)	3,300,000
Actual cost (AC)	3,500,000

$CPI = EV \div AC = 3,300,000 \div 3,500,000 = 0.943$

$SPI = EV \div PV = 3,300,000 \div 3,600,000 = 0.917$

The varied work is programmed to start in two months' time. The new BAC is $6,000,000 + 504,000 = 6,504,000$ either way; what differs is the phasing.

Done wrongly — 120,000 of the variation's budget is credited as planned value in the current period because that is when the change order was issued:

$$\begin{aligned} PV &= 3,600,000 + 120,000 = 3,720,000 \\ SPI &= 3,300,000 \div 3,720,000 = 0.887 \end{aligned}$$

The schedule performance index falls by $0.917 - 0.887 = 0.030$ because of a filing decision. No work changed, no productivity changed, nobody was late.

Done correctly — the 504,000 is phased into the two months in which the varied work is planned:

$$\begin{aligned} PV \text{ at the data date} &= 3,600,000 \text{ (unchanged)} \\ SPI &= 3,300,000 \div 3,600,000 = 0.917 \text{ (unchanged)} \end{aligned}$$

From the month the varied work starts, planned value and earned value move together and the index measures performance on the enlarged scope, which is what it is for.

9.3 Reporting the unpriced book

Suppose that at the same data date the account also carries **780,000** of instructed-but-unpriced variations (bucket 2). The forecast for the baselined scope, on a persisting cost performance basis:

$$\begin{aligned} EAC \text{ (baselined scope)} &= BAC \div CPI = 6,000,000 \div 0.942857 = 6,363,636 \\ EAC \text{ (including unincorporated change)} &= 6,363,636 + 780,000 = 7,143,636 \\ \text{Apparent variance against the original BAC} &= 7,143,636 - 6,000,000 = (1,143,636) \end{aligned}$$

Reported as one number, that is a 1.14 million overrun and the conversation is about performance. Reported as two lines — 363,636 of cost performance variance and 780,000 of instructed scope not yet in the baseline — it is two conversations with two different owners, and only one of them is a controls problem.

Assumptions this example depends on. Rates and percentage additions are as stated in the contract for this example; the 1,200 m² is a remeasured quantity, not an estimate; the cost performance index is computed on the baselined scope only, with the cost of unincorporated variations excluded from AC and EV alike; no prolongation or disruption effect is included in any of the three valuations, and any such effect would be a separate assessment.

— 10. Checklist

At the event

- [] The change is logged on the day it arises, with its event date — not the day it is priced.
- [] The instruction reference and the name of the person who issued it are recorded; where no instruction exists, the direction has been confirmed in writing to the contract representative the same day.
- [] The notice position has been checked against the contract, and notice given or diarised.
- [] Cost codes for the change are open before work starts.

Pricing

- [] The valuation basis is the one the contract requires, and the reasoning is recorded.
- [] Star rates show their build-up element by element.
- [] Daywork sheets are signed contemporaneously by an authorised person.
- [] Time-related cost is assessed separately from the direct works.
- [] The submission does not recover the same cost twice under two heads.

Forecast

- [] Every change is assigned to a bucket: agreed, instructed-unpriced, or claimed/disputed.
- [] Cost is forecast gross; recovery is never netted off.
- [] The estimate at completion separates baselined scope from unincorporated change.

Baseline

- [] Agreed variations are time-phased to the varied work's planned dates, and the schedule is updated in the same cycle as the budget.
- [] The baseline change is in the change log with an approval reference.
- [] Performance trends are annotated where the baseline changed, so a step is not read as performance.

Log health

- [] One log, one owner, reviewed weekly.
- [] Ageing by status is reported monthly, with a threshold that triggers escalation.
- [] Submitted values and assessed values are both held, so the assessment gap is visible.
- [] Time effects are recorded in days, separately from money.

— Related

- [BPG-04 – Baselining and baseline change control](#) — the procedure a change order runs through, and what a legitimate baseline change looks like.
- [BPG-12 – Claims and extension of time](#) — where a variation stops being a valuation and becomes an entitlement argument, and how not to double-count between the two.
- [BPG-08 – Earned value in practice](#) — why an unincorporated variation distorts every index in the account.
- [BPG-09 – Estimate at completion: choosing and defending a method](#) — reporting a forecast that separates performance from scope growth.
- [TPL-12 – Change order log](#) — the instrument, with the columns set out in §6.

— Sources and standards

Valuation hierarchies, notice provisions and change procedures are set out differently in each contract and in the standard forms published by bodies such as FIDIC and the NEC family; this guide describes the general principles in our own words and reproduces no clause, wording or numbering from any of them, and asserts no position under any jurisdiction's law. The IFRS 15 constraint

on variable consideration is named in §4 and described in principle only; the applicable treatment depends on the reporting framework in use. Internal references are BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 5 (Cost Management & Cost Control). All figures in §9 are illustrative and were computed for this document.

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BPG-12 · BEST PRACTICE GUIDES

Claims and extension of time

Cause, effect and quantum — and the records that have to exist a year before anyone needs them

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Delay and disruption are different problems needing different proof, and most submissions fail on causation rather than on quantum. This guide separates the two, sets out the causal chain a claim has to establish, explains what an extension of time does and does not buy, describes the competing approaches to concurrency and to delay analysis without asserting any jurisdiction's position, works a measured-mile disruption calculation, and lists the records that have to exist long before anyone knows a claim is coming. It is not legal advice.

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Claims and extension of time

In one paragraph. Delay and disruption are different problems needing different proof, and most submissions fail on causation rather than on quantum. This guide separates the two, sets out the causal chain a claim has to establish, explains what an extension of time does and does not buy, describes the competing approaches to concurrency and to delay analysis without asserting any jurisdiction's position, works a measured-mile disruption calculation, and lists the records that have to exist long before anyone knows a claim is coming. It is not legal advice.

Who this is for. Project controls managers, planners, quantity surveyors and commercial managers on either side of a contract, and project managers who will be asked, eventually, what the records show.

This guide is not legal advice, and it does not state the law of any jurisdiction. Whether a party is entitled to additional time or money, what notice is required, how concurrent delay is treated, and which delay analysis is acceptable are all determined by the specific contract and by the law governing it — and these differ materially between contracts, between countries and between legal traditions. Published industry guidance exists on delay and disruption, of which the Society of Construction Law's Delay and Disruption Protocol is the best known; such guidance is not law, is not binding unless the parties have adopted it, and does not override the contract. Take advice on your own contract from someone qualified to give it.

— 1. What a claim has to establish

A claim is an assertion of entitlement arising from an event: additional time, additional money, or both. Whatever the contract, the assertion has three parts, and a submission that is strong on one and weak on another fails as a whole.

Cause. An event occurred, and the contract allocates its risk to the other party. This is a contractual question before it is a factual one: an unforeseen ground condition is a cause under some contracts and a contractor risk under others, and the same rainfall is a compensable event, a neutral event or nobody's problem depending on what was agreed.

Effect. The event produced a consequence — days of critical delay, degraded productivity, or both. Most submissions treat this as self-evident, and it is where they are attacked. The effect must be demonstrated by analysis of contemporaneous records, not asserted by proximity in time.

Quantum. The consequence had a cost, evidenced from records, calculated on a stated basis, and not recovered anywhere else.

The order matters. Quantum built before causation is established is a number looking for a reason, and assessors recognise the shape immediately.

— 2. Delay and disruption are different animals

They are routinely conflated, and they need different proof.

Delay is lateness. The completion date, or a contractual sectional date, moves. The proof is a schedule argument: the event affected the sequence of work that determined the completion date. The remedy is time — an extension — and, separately and conditionally, the time-related cost of the extended period.

Disruption is loss of productivity. The work may finish exactly on time and still cost far more than it should have, because it was performed out of sequence, in congested conditions, with repeated remobilisation, or in short bursts as access was released piecemeal. The proof is a productivity argument, not a schedule one. The remedy is money.

A single event frequently causes both, and the two must then be assessed separately and added carefully — a disrupted, delayed activity can generate a prolongation head and a disruption head, and the risk of counting the same hours twice is severe.

The commonest failure in disruption claims is trying to prove them the way delay is proved. A critical-path analysis says nothing about whether crews were half as productive as they should have been; it only says when things happened. Different question, different evidence.

— 3. Extension of time: what it buys, and what it does not

An extension of time (EOT) adjusts the contractual completion date. On most contracts it relieves the contractor of liability for liquidated damages (LDs) for the extended period, and it preserves the contractual mechanism, since a completion date that has become impossible is not a date anyone can manage to.

What an EOT does **not** automatically do is pay for the extended period. Time and money are separate questions with separate tests on most contracts, and the outcome depends on the event, the contract and the governing law. It is entirely possible — and on some contracts and some facts, entirely correct — to be granted time and recover nothing.

This asymmetry has a commercial consequence that is often missed by non-commercial readers of a report: an EOT with no money is still valuable, sometimes very valuable, because it removes an exposure that accrues at a fixed rate per day. §9 quantifies both sides of that on the same facts.

Two further points:

Time-related cost is not the same as the LD rate. The LD rate is what the contractor pays for lateness; the prolongation cost is what the contractor actually incurs while remaining on site. They are different numbers with different bases and they should never be substituted for one another.

Acceleration is a third thing. Where an extension is refused or delayed and the contractor accelerates to protect the date, whether the acceleration cost is recoverable — and under what description — varies markedly by contract and jurisdiction, and is contentious in most of them. It should never be assumed, and the decision to accelerate should be documented at the time with its reasons and its instruction status.

— 4. Concurrency, described without taking sides

Concurrent delay arises when delays for which different parties bear the risk affect the same period. The client's late instruction and the contractor's own resource shortfall each, independently, drive the completion date across the same weeks. The question is what the contractor is entitled to for that period.

Practice recognises several approaches, and different contracts, jurisdictions and forums resolve it differently. Described neutrally:

- **Dominant cause.** The analysis identifies the effective or dominant cause of the delay in the period, which then carries the entitlement. Attractive in principle; frequently impossible in fact, because two genuinely concurrent causes may have no dominant one.
- **Apportionment.** Time, money, or both are divided between the parties in proportion to their causative contribution. Requires a basis for the division that survives challenge.
- **Time but no money.** The contractor receives the extension — relieving the LD exposure — but not the prolongation cost, on the reasoning that the time-related cost would have been incurred anyway because of its own concurrent delay.

There is a fourth question that precedes all of these: **what counts as concurrent at all.** Whether two delays must be truly simultaneous, or merely overlapping in effect, and whether the test is applied to the causes or to their effects, are themselves contested, and different contracts define the term differently or not at all.

Do not assume any of these applies to your project. Some contracts address concurrency expressly; many do not, and then the governing law supplies the answer — and the answer is not the same everywhere. The practical instruction for a controls professional is narrow and useful: **establish the facts, week by week, so that whichever approach applies can be applied to real evidence.** Whichever test a tribunal uses, it will be applied to the records. The party with the better records shapes the analysis, and that advantage is built during the works, not afterwards.

— 5. Delay analysis at concept level

Several families of method exist for demonstrating the effect of an event on the completion date. Described in principle, and without endorsing any:

- **As-planned versus as-built.** Compare the baseline with what actually happened and identify where the critical sequence diverged. Intuitive and evidence-rich; weak at attributing cause where several things changed at once.
- **Impacted as-planned.** Insert the event into the baseline programme and observe the effect on the computed completion date. Simple; criticised because it tests a plan rather than the project as it was.
- **Time impact analysis.** Insert the event into the programme as it stood immediately before it, updated with actual progress. More faithful to reality; depends entirely on the quality of programme updates.
- **Collapsed as-built.** Remove the delay events from the as-built programme and observe when the works would have completed without them. Powerful where the as-built record is complete; contentious where it is not.

- **Windows or time-slice analysis.** Divide the project into periods and analyse what drove the critical path in each. Usually the most informative where good periodic updates exist, and the most laborious.

Three observations that matter more than the taxonomy:

1. **Different methods can produce different answers on the same facts.** That is a property of the methods, not evidence of manipulation — but it means the choice of method must be justified, on the record, by reference to the contract and the data available.
2. **Every method consumes programme data.** A project with no baseline, or with updates that were never properly statused, cannot run any of them credibly. Schedule quality is therefore a claims asset years before it is a claims issue — see [BPG-05 – Schedule quality: a practical review](#).
3. **Some contracts specify the method.** Where they do, that is the end of the discussion.

— 6. Disruption and the measured mile

Disruption is proved by comparing productivity, and the strongest comparison is against the claimant's own performance on the same work in an un-impacted period — the **measured mile**. The logic is simple and it is why the method persuades: the benchmark is not a published norm or a tender allowance that the other party can attack as optimistic, but what these crews actually achieved on this project.

Mile productivity	= hours ÷ quantity, in the un-impacted period
Impacted productivity	= hours ÷ quantity, in the impacted period
Excess hours	= (impacted productivity – mile productivity) × impacted quantity
Quantum	= excess hours × demonstrated cost rate

The method stands or falls on the comparability of the mile. Four tests before using a period as the mile:

- **Same work** — same activity, specification and materials.
- **Same resources** — same or equivalent crews, same supervision, same shift pattern.
- **Same conditions, apart from the impact** — not the winter period against the summer one.
- **Not the learning curve.** A mile taken from the first work performed is penalised by inexperience, so it understates productivity and overstates the claim — precisely the weakness an assessor looks for. Choose a mature period.

Where no clean mile exists — and often none does — the fallback is a bottom-up build of the lost hours from contemporaneous records, event by event. What must not happen is a leap to a global figure (see §8).

— 7. Records that survive scrutiny

By the time anyone needs them, the records either exist or they do not. There is no retrospective fix, and the difference between a project with records and one without is usually the difference between a settlement and a write-off.

Programme records. The accepted baseline, every periodic update with its data date, and the statusing that produced it — actual dates, remaining durations, logic changes with reasons. An update re-logged without a note is worth much less than one that explains itself.

Labour and productivity records. Timesheets or allocation sheets coded to area, activity and — the crucial one — to a change or event reference where applicable, with quantities installed per period in the same coding. Coding at source is what makes a measured mile or a bottom-up build possible later.

Daily records. Site diaries recording resources, work fronts available, weather, access restrictions, instructions received and interruptions. Contemporaneous, dated, and preferably signed.

Correspondence and notices. A notices register showing what was notified, when, under what provision and with what response; instructions with their references and issuers.

Photographs, survey records and meeting minutes — dated and located, showing conditions rather than achievements, and recording access dates and undertakings given.

Two rules govern all of it. **The record is made for the project, not for the claim** — records manufactured once a dispute is in sight are visibly different and are treated as such. And **contemporaneous means contemporaneous**: a diary written up weekly from memory is a reconstruction.

— 8. How this goes wrong

The global claim. The total cost overrun is presented as the composite consequence of many events, without linking each cause to its own effect and its own quantum. It is tempting where events genuinely interact, and it is fragile: because the quantum is one undifferentiated sum, demonstrating that any part of it was caused by the claimant, or that the tender was underpriced, undermines the whole. There is no mechanism inside a global claim for removing the bad element. The antidote is structural and early — cost coding fine enough to isolate each event's cost as it is incurred, which is a controls decision made a year before anyone contemplates a claim.

Notice missed. Covered in [BPG-11 – Change orders and variations](#) §2. The consequence of late notice varies by contract and by governing law, and on some contracts it is fatal. It is nearly always an administrative failure rather than a commercial one.

Causation assumed from proximity. "The instruction was issued in Week 14 and we finished eight weeks late" is not an argument. The analysis must show that the event affected the sequence that determined the completion date, in the period it occurred.

Prolongation priced at the LD rate. Convenient, and wrong. Prolongation is what was actually incurred, evidenced; the LD rate is a different number agreed for a different purpose.

Double recovery between variations and claims. The most damaging quantum defect. Hours recovered inside an instructed variation and again in a disruption claim; preliminaries recovered in a star rate and again as prolongation. One instance found invites the assessor to distrust the entire submission, including the parts that were correct.

A measured mile chosen because it produces the best number. An early, short or unrepresentative mile will not survive comparison with the rest of the record. Choose the mile on comparability grounds and record why.

Head office overhead asserted by formula. Formula-based approaches to unabsorbed head-office overhead exist and are used, and their acceptability varies substantially between jurisdictions and forums. Presenting one as though it were settled entitlement invites a challenge that can discredit better-founded heads alongside it.

Records assembled retrospectively. The claim is written first and the evidence is searched for afterwards. It shows: gaps in the coding, diaries with uniform handwriting, and quantities that do not reconcile to the valuations submitted at the time.

Nobody owns the file. Claims are lost by projects where three people each held part of the record and none of them held the chronology.

— 9. Worked example

Illustrative figures. Currency USD; a single work front on a construction contract; the example assumes that entitlement under the contract has been established for the events described and quantifies the consequence only. **The arithmetic below does not create entitlement; whether these events give any entitlement at all is a question about the contract and the governing law.**

9.1 Prolongation for a 24-day critical delay

A window analysis of the period shows that late release of a work front delayed the sequence determining the completion date by **24 calendar days**. Time-related costs actually incurred and evidenced during the extended period:

Time-related cost head	Rate per calendar day
Site establishment and supervision	9,400
Tower crane hire and operation	2,150
Site accommodation and services	780
Total	12,330

$$\text{Prolongation quantum} = 12,330 \times 24 = 295,920$$

Every element is a cost the project actually incurred because it remained on site, evidenced from invoices, hire records and payroll — not an allowance from the tender, and not the liquidated damages rate.

9.2 The value of the extension itself

Liquidated damages under this illustrative contract run at **18,000 per calendar day**.

$$\text{LD exposure removed by a 24-day extension} = 18,000 \times 24 = 432,000$$

Set the two side by side, because they answer different questions:

Outcome	Prolongation cost recovered	LD exposure removed	Net position
Extension granted, with cost	295,920	432,000	+727,920
Extension granted, no cost	0	432,000	+432,000
Extension refused	0	0	(432,000) exposure retained

The middle row is the one that surprises people. On facts where the contract gives time but not money — a neutral event, or a concurrency outcome of that kind — the extension is still worth 432,000, because it switches off an exposure priced per day. A report that presents "no money recovered" as a failure has misread the commercial position.

9.3 Disruption: a measured mile

Separately, pipe support installation on the same contract was disrupted by piecemeal release of work fronts. Two periods, same crews, same support type, same specification:

Period	Supports installed	Labour hours	Productivity
Un-impacted (the mile)	2,100	6,720	$6,720 \div 2,100 = 3.20 \text{ h/support}$
Impacted	1,450	6,090	$6,090 \div 1,450 = 4.20 \text{ h/support}$

Excess productivity = $4.20 - 3.20 = 1.00 \text{ h/support}$
 Excess hours = $1.00 \times 1,450 = 1,450 \text{ hours}$
 Demonstrated cost rate (from payroll, all-in) = 62.00 / hour
 Gross quantum = $1,450 \times 62.00 = 89,900$

Now remove what has already been recovered. Of those 1,450 excess hours, **180** relate to work that was also the subject of an instructed variation, priced on a built-up rate that included the additional labour. Recovering them again here would be double recovery:

Net excess hours = $1,450 - 180 = 1,270$
 Net quantum = $1,270 \times 62.00 = 78,740$

This deduction is the single most valuable paragraph in a disruption submission. Making it yourself, and showing the workings, tells an assessor that the rest of the numbers have been through the same test.

9.4 The total, and what it rests on

Prolongation (\$9.1) = 295,920
 Disruption, net (\$9.3) = 78,740
 Total quantified = 374,660

Assumptions this example depends on. Entitlement under the contract is assumed, not demonstrated; the 24-day critical delay is the output of a window analysis of contemporaneous programme updates, not an assertion; the mile period is comparable on work type, crews, shift pattern and conditions, and is not the first work performed; the cost rate is demonstrated from payroll

rather than tendered; no head-office overhead is claimed; and no element of the disruption quantum overlaps with the prolongation head, which covers time-related site costs only.

— 10. Checklist

Before anything happens — the standing position

- [] The baseline programme is accepted and issued, and updates are produced on a fixed cycle with data dates and statusing recorded.
- [] Timesheets are coded to area and activity, and can be coded to an event reference on the day one arises.
- [] Site diaries are written daily, recording resources, work fronts, access and interruptions.
- [] A notices register exists with one owner.
- [] Someone is named as owner of the records chronology.

When an event occurs

- [] The event is logged with its date on the day it is identified.
- [] The contract has been read for the applicable notice requirement, and notice given or diarised.
- [] Separate cost codes are opened before the affected work proceeds.
- [] Photographs, minutes and correspondence relating to the event are collected into one file immediately.
- [] The potential effect — delay, disruption, or both — is stated at the outset, because they need different evidence.

Building the case

- [] Cause is established against the contract, not merely as a matter of fact.
- [] Effect is demonstrated by an analysis whose method is stated and justified.
- [] Where concurrency may arise, the week-by-week factual position is documented before any legal characterisation is attempted.
- [] The delay analysis and the disruption analysis are kept separate.
- [] Any measured mile is justified against the four comparability tests.

Quantum

- [] Prolongation is priced from costs actually incurred, not from the LD rate or tender allowances.
- [] Every head is checked against variations already valued, and overlaps deducted with the workings shown.
- [] Acceleration, if claimed, has a contemporaneous record of the decision and its reasons.
- [] Nothing is claimed that the records cannot support, at any value.

Before submission

- [] Each event has its own cause, effect and quantum — nothing is aggregated for convenience.
- [] The submission states what it assumes and what it does not.
- [] Legal advice has been taken on entitlement and on the governing law.

— Related

- [BPG-11 – Change orders and variations](#) — the valuation route that resolves most changes before they become claims, and the double-counting risk between the two.
- [BPG-05 – Schedule quality: a practical review](#) — the programme discipline every delay analysis consumes.
- [BPG-04 – Baselining and baseline change control](#) — why an accepted, controlled baseline is the starting point of any delay argument.
- [TPL-13 – Claim and extension-of-time narrative structure](#) — the structure for setting out cause, effect and quantum.
- [TPL-12 – Change order log](#) — the register that becomes the first draft of a claim file.

— Sources and standards

Contracts and standard forms — including those published by FIDIC and the NEC family — allocate risk, require notice and define valuation differently from one another, and their operation also depends on the governing law; no clause, wording or numbering from any of them is reproduced here and no jurisdiction's position is asserted. Published industry guidance on delay and disruption exists, of which the Society of Construction Law's Delay and Disruption Protocol is the best known; it is guidance rather than law, is not binding unless adopted by the parties, and its substantive positions are not stated here. Internal references are BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 10 (Project Scheduling). All figures in §9 are illustrative and were computed for this document.

— Status and version

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BPG-13 · BEST PRACTICE GUIDES

Cash flow forecasting

The gap between doing the work and having the money



VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

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— Summary

A cost S-curve says nothing about whether a project can pay its people next month. This guide sets out the four mechanisms that separate progress from cash — measurement and certification, payment terms, retention and advance recovery — builds a monthly forecast through the certificate cascade, identifies the peak funding requirement, shows why a profitable project can still fail on cash, and runs the payment-timing sensitivity that turns a forecast into a decision about facilities.

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Cash flow forecasting

In one paragraph. A cost S-curve says nothing about whether a project can pay its people next month. This guide sets out the four mechanisms that separate progress from cash — measurement and certification, payment terms, retention and advance recovery — builds a monthly forecast through the certificate cascade, identifies the peak funding requirement, shows why a profitable project can still fail on cash, and runs the payment-timing sensitivity that turns a forecast into a decision about facilities.

Who this is for. Cost engineers, commercial managers and project controls managers who produce or rely on a project cash forecast, and the finance and treasury colleagues who have to fund the answer.

— 1. The S-curve is a cost curve

The cumulative cost curve every project draws describes when value is earned and cost is incurred. Cash is a different curve with a different shape, and on most contracts it sits well below and well behind the cost curve for most of the project's life.

The distinction is not academic. A project can be exactly on budget, exactly on programme, and unable to pay its subcontractors, because being owed money and having money are different states. Every mechanism that separates them is contractual, predictable and modellable — which means a funding shortfall is almost always a forecasting failure rather than a surprise.

Two curves are therefore needed, and they answer different questions:

Curve	Question it answers	Driven by
Cumulative cost / earned value	Are we going to be within budget?	Progress and productivity
Cumulative net cash	Can we pay for the work while we do it?	Certification, terms, retention, advances

— 2. The four gaps

Gap 1 — measurement and certification. Work performed is not work valued. The contractor applies for what it believes the work is worth; the certifier assesses; the certified figure, not the applied figure, is what becomes payable. Persistent under-certification is a commercial issue, but for the forecast it is a simple discipline: **forecast on expected certified value, and hold the application-to-certification gap as a stated assumption** rather than assuming full certification and treating shortfalls as bad luck.

Gap 2 — payment terms. Certified sums are paid after a contractual period running from the certificate or the application. On a monthly cycle the practical lag between performing work and re-

ceiving cash for it is usually two months, sometimes three, and the difference between those two is enormous — \$9 quantifies it.

Gap 3 — retention. A percentage of each certified amount is withheld as security, released in stages, often long after the work is done and partly after a defects period. Retention converts margin into a deferred receivable. A project can finish, hand over, be paid everything payable, and still be cash-negative until release.

Gap 4 — advance recovery. Where a mobilisation advance is paid, it is a loan repaid through the measure — recovered as a deduction on each certificate, usually at a stated percentage of the gross value. The forecast must carry both the early inflow and the reduced net certificates that repay it, or the advance looks like free money in the first month and an unexplained shortfall thereafter.

Together these four produce the shape every project cash curve has: a trough, whose depth and timing are the things that actually need managing.

— 3. Building the forecast: the certificate cascade

The inflow side of the forecast is one repeated calculation, per period:

Gross value certified	(from the valuation of work done)
– retention	(stated % of gross)
– advance recovery	(stated % of gross, until the advance is repaid)
– previous payments	(where the valuation is cumulative)
= net amount due	
→ received on the payment date given by the contract terms	

Three modelling rules make it reliable:

Model the value, then the deductions, then the date — separately. Collapsing them into one net figure per month makes the forecast impossible to interrogate when it goes wrong, and it will go wrong.

Phase from the schedule, not from the calendar. Gross value per period comes from the time-phased baseline and the current forecast of progress. A cash forecast built by spreading the contract value evenly across the programme is not a forecast.

Carry retention forward explicitly, with its release dates. Retention is not a cost and it is not lost; it is cash held. It belongs on the forecast as a receivable with dates, and somebody must be accountable for recovering it. Retention that nobody chases is a real and recurring loss.

— 4. The outflow side

Outflows are governed by different terms from inflows, and the mismatch is the whole problem.

- **Labour** is paid weekly or monthly, in arrears of days rather than months. It is close to uncompressible.
- **Subcontractors** are paid on their own terms. Where those terms are shorter than the terms on which the main contract pays, every period of work is being funded by the contractor.

- **Materials and equipment** are paid on supplier terms, sometimes with deposits or stage payments before delivery — an outflow that precedes any possibility of certifying the corresponding value.
- **Plant and preliminaries** run at a rate per period regardless of progress, which is why they dominate the cash consequence of any delay.

The two levers on this side are supplier payment terms and the timing of procurement commitments. Both have limits: stretching supplier terms transfers a funding problem down the supply chain, has commercial consequences, and in some jurisdictions is regulated. Treat it as a lever with a cost, not as free financing.

— 5. The funding trough

The deepest point of the cumulative net cash curve is the **peak funding requirement** — the amount of money the project must have available, from a facility, from group treasury, or from partners, in order to keep operating. It is a single number, it has a date, and it is the most useful output of the whole exercise.

Its drivers are precisely the levers a commercial and controls team manages: certification performance, payment terms both ways, retention percentage and release profile, billing cadence, advance payments and their recovery rate, and the margin itself.

Two consequences follow that reports rarely state:

The trough is a decision, not a fact. Every driver in that list is negotiable at some point — at tender, at award, or in the running of the job. A forecast that presents the trough as a given has skipped the conversation that matters.

The trough moves for reasons that have nothing to do with performance. A perfectly performing project whose client moves from 30-day to 60-day payment has a materially worse funding position and an unchanged cost report. That divergence is exactly why cash is reported separately.

— 6. Why a profitable project fails on cash

Profit is measured on value earned against cost incurred. Cash is measured on money received against money paid. On a contract of any length the second is far more volatile than the first, for a structural reason: cost is incurred continuously and revenue is collected in lumps, late, net of deductions.

Three patterns account for most failures.

Growth. A contractor winning more work funds each new project's trough before the previous one has recovered. The business is profitable and getting steadily closer to insolvency. This is why the aggregate funding profile across a portfolio matters as much as any single project's.

Front-loaded cost, back-loaded value. Design, mobilisation, temporary works, procurement deposits and long-lead equipment are paid early and certified late — or, in some structures, not certified as value at all until incorporated into the works.

Retention plus a long defects period. The entire margin can be smaller than the retention held. Where that is true, the project's profitability is contingent on a release that happens a year after everyone has moved on.

— 7. Sensitivity: change one thing at a time

A cash forecast that is presented as a single line is not decision-ready. Run a small, disciplined set of sensitivities, each moving exactly one variable, and report the effect on two numbers only: the depth of the trough and its date.

The four worth running on almost every project:

1. **Payment terms slip by one period.** The single most common and most damaging change.
2. **Certification at less than application.** A stated percentage, with the balance certified later.
3. **No advance, or a slower advance.** Tests how much of the funding position depends on it.
4. **Retention released late.** Moves the end of the curve rather than the trough, but decides when the margin becomes real.

Moving one variable at a time is not a stylistic preference. A forecast that changes three assumptions simultaneously produces a number nobody can attribute, and attribution is the entire purpose — the point is to know which lever to pull.

— 8. Timing effects that are not costs

Several flows move cash without changing project cost, and they belong in the forecast for their timing alone.

Where an indirect tax such as a value-added or goods-and-services tax applies, invoices are issued and paid gross of it, and the tax then leaves again on its own remittance calendar. Whether such a tax applies, at what rate, on what basis and with what remittance timing depends entirely on the jurisdiction and is a question for the finance function — but where it applies, the gross flows can be material at the trough. Similarly, where withholding applies to cross-border payments, the total cash is unchanged but its counterparties and dates are not. Neither affects project cost; both affect the funding requirement.

The discipline is to model gross flows with their remittance dates, and never to let a tax-inclusive invoice value reach the cost ledger.

— 9. How this goes wrong

The cash forecast is the cost curve with a lag applied. A single blanket delay applied to the S-curve, with no retention, no advance recovery and no certification assumption. It will be wrong by the size of the deductions, which is the size of the problem.

Forecasting on applications rather than certificates. Optimistic by exactly the assessment gap, every month, cumulatively.

Retention modelled as lost or forgotten entirely. Either the forecast understates the eventual position or it overstates the current one. Model it as held cash with release dates and an owner.

The advance treated as income. It arrives, the curve lifts, and nobody models the recovery deductions that follow. The second and third months then look inexplicably poor.

Ignoring the outflow terms. Modelling receipts carefully and assuming all costs are paid in the month incurred understates the position on projects with long supplier terms and overstates it where subcontractors are paid quickly.

No sensitivity on payment timing. The one assumption most likely to change, and the one most often presented as fixed.

Unpriced variations invisible in the cash line. Work instructed but not agreed is being funded with nothing certified against it. A growing unpriced book is a cash problem before it is a commercial one — see [BPG-11 – Change orders and variations](#).

One project at a time. The funding requirement that matters to a business is the aggregate of its projects' troughs at each point in time, which is not the sum of their peak requirements and is never visible from any single project's report.

— 10. Worked example

Illustrative figures. Currency USD; monthly periods; a six-month package of contract value 6,000,000; retention 5 % of gross certified value; mobilisation advance 8 % of contract value, paid at the start of Month 0 and recovered at 8 % of each gross certificate; the certifier is assumed to certify the full applied value; payment is received two months after the month in which the work is performed; all outflows are assumed paid in the month the cost is incurred; total cost 5,400,000, giving a margin of 600,000, which is 10 % of contract value. Figures are rounded to the nearest whole unit and are exact as stated.

10.1 The certificate cascade

Month	Gross value certified	Retention 5 %	Advance recovery 8 %	Net amount due
1	500,000	25,000	40,000	435,000
2	900,000	45,000	72,000	783,000
3	1,300,000	65,000	104,000	1,131,000
4	1,500,000	75,000	120,000	1,305,000
5	1,100,000	55,000	88,000	957,000
6	700,000	35,000	56,000	609,000
Total	6,000,000	300,000	480,000	5,220,000

Two reconciliations to run every time, because they catch most modelling errors:

Retention total	= 5 % × 6,000,000 = 300,000 ✓
Advance recovered	= 8 % × 6,000,000 = 480,000 = the advance paid ✓
Net certified total	= 6,000,000 – 300,000 – 480,000 = 5,220,000 ✓

The second one matters: the recovery percentage must be set so that the advance is fully repaid by the end of the works. Set it too low and the final certificates carry a balance nobody has planned for.

10.2 The cash position, month by month

Costs are $0.9 \times \text{gross value}$ in each month, paid in the month incurred. Receipts arrive two months after the work month.

Month	Cash in	Cash out	Net in month	Cumulative cash
0	480,000 (advance)	—	480,000	480,000
1	—	450,000	(450,000)	30,000
2	—	810,000	(810,000)	(780,000)
3	435,000	1,170,000	(735,000)	(1,515,000)
4	783,000	1,350,000	(567,000)	(2,082,000)
5	1,131,000	990,000	141,000	(1,941,000)
6	1,305,000	630,000	675,000	(1,266,000)
7	957,000	—	957,000	(309,000)
8	609,000	—	609,000	300,000

Peak funding requirement = (2,082,000), at the end of Month 4
 Total cash in = 480,000 + 5,220,000 = 5,700,000
 Total cash out = 5,400,000
 Closing position at Month 8 = 300,000

The closing 300,000 is not the margin. The other 300,000 is sitting in retention, released in two tranches — half at completion and half after the defects period — at which point the cumulative position reaches $300,000 + 150,000 + 150,000 = 600,000$, the margin, in cash, roughly a year after the work finished.

10.3 Profitable, and needing 2.08 million

The package makes 600,000 on 6,000,000 of value. To make it, the business must fund 2,082,000 at Month 4:

Peak funding ÷ margin = $2,082,000 \div 600,000 = 3.47$

The funding requirement is about three and a half times the entire profit on the job. Nothing has gone wrong; this is what a normally performing contract looks like.

10.4 Sensitivity 1 — payment one month later

Everything unchanged except that receipts arrive three months after the work month rather than two:

Month	Cash in	Cash out	Cumulative cash
0	480,000	—	480,000
1	—	450,000	30,000
2	—	810,000	(780,000)
3	—	1,170,000	(1,950,000)
4	435,000	1,350,000	(2,865,000)
5	783,000	990,000	(3,072,000)
6	1,131,000	630,000	(2,571,000)
7	1,305,000	—	(1,266,000)
8	957,000	—	(309,000)
9	609,000	—	300,000

New peak funding requirement = (3,072,000), at the end of Month 5
 Increase = 3,072,000 – 2,082,000 = 990,000
 As a proportion = 990,000 ÷ 2,082,000 = 47.6 %

One month of payment timing raises the funding requirement by nearly half, and moves the trough a month later. Scope, programme, cost and margin are all identical. This is the number that should be in front of whoever negotiates payment terms.

10.5 Sensitivity 2 — no mobilisation advance

Remove the advance and its recovery. Net certificates rise to gross less retention only — 475,000, 855,000, 1,235,000, 1,425,000, 1,045,000 and 665,000 — still received two months in arrears:

Month	Cash in	Cash out	Cumulative cash
0	—	—	0
1	—	450,000	(450,000)
2	—	810,000	(1,260,000)
3	475,000	1,170,000	(1,955,000)
4	855,000	1,350,000	(2,450,000)
5	1,235,000	990,000	(2,205,000)
6	1,425,000	630,000	(1,410,000)
7	1,045,000	—	(365,000)
8	665,000	—	300,000

Peak without the advance = (2,450,000)
 Benefit of the advance = 2,450,000 – 2,082,000 = 368,000

The advance was 480,000, but it improves the funding position by only 368,000, because by the trough at Month 4 two certificates have already repaid part of it:

Recovery deducted from the Month 1 and Month 2 certificates = $40,000 + 72,000 = 112,000$
 $480,000 - 112,000 = 368,000$ ✓

An advance is worth less than its face value to the funding position, and how much less depends on the recovery rate and on where the trough falls. A forecast that credits the full advance overstates the funding benefit by 112,000 — which is $112,000 \div 480,000 = 23.3$ % of the advance.

Assumptions this example depends on. Full certification of applied value; a fixed two-month (or, in §10.4, three-month) payment lag; all outflows paid in the month incurred; no variations, escalation or indirect tax; retention released half at completion and half after the defects period, both outside the eight-month window shown; cost at a constant 90 % of value in every month. Relaxing any one of these moves the trough, which is exactly why sensitivities are run one variable at a time.

— 11. Checklist

Structure

- [] The cash forecast is a separate model from the cost forecast, driven by the same schedule.
- [] Gross value is phased from the time-phased baseline and the current progress forecast, not spread evenly.
- [] Retention, advance recovery and previous payments are modelled as separate lines, not netted.
- [] The payment date rule is stated explicitly, in periods, and matches the contract.
- [] Outflow terms are modelled by category — labour, subcontract, materials, plant — not as a single lag.

Reconciliations, every month

- [] Retention deducted to date equals the retention percentage times gross certified to date.
- [] Advance recovered to date is on track to repay the advance by completion.
- [] Total cash in over the life equals contract value, once retention is released.
- [] The closing cash position equals the margin, once all deductions have unwound.

Reporting

- [] The peak funding requirement is stated with its date.
- [] Sensitivities are run one variable at a time, and reported as depth and date of the trough.
- [] The payment-terms sensitivity is on the page every month, not on request.
- [] Retention outstanding is reported as a receivable with release dates and a named owner.
- [] Instructed-but-unpriced work is shown as cash exposure, not omitted because it has no certificate.

Judgement

- [] The certification assumption is stated, and compared against actual certification history.

- [] The forecast has been tested against the aggregate position of the portfolio, not just this project.
- [] Where the trough exceeds the available facility, the date it does so is escalated before it arrives.

— Related

- [BPG-07 – Accruals and cut-off discipline](#) — the difference between cost incurred, cost invoiced and cost paid, which this guide depends on.
- [BPG-11 – Change orders and variations](#) — why an unpriced variation book is a funding problem as well as a commercial one.
- [BPG-09 – Estimate at completion: choosing and defending a method](#) — the cost forecast this model takes as its input.
- [BPG-14 – Monthly reporting that gets read](#) — how the peak funding requirement reaches a decision-maker in time to matter.
- [TPL-09 – Cash flow forecast](#) — the instrument implementing §10's cascade.

— Sources and standards

The valuation, certification, retention and advance-recovery mechanisms described here follow the general structure used across construction and engineering contracts, including the standard forms published by bodies such as FIDIC and the NEC family; no clause, wording or numbering is reproduced, and the specific terms of any contract prevail over this description. Indirect tax and withholding treatment is jurisdiction specific and is a matter for the finance function. Internal references are BoK Domain 3 (Budgeting & Forecasting), BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 11 (Business Process Cycles). All figures in §10 are illustrative and were computed for this document.

— Status and version

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BPG-14 · BEST PRACTICE GUIDES

Monthly reporting that gets read

Lead with the decision, not the data

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

A monthly report exists to cause a decision, and most of them do not. This guide sets out the one-page structure that leads with the decisions required rather than the data, the four-part variance narrative that names cause, effect, action and owner, how to report a control account whose indices are undefined, the arithmetic that makes a recommendation defensible, why reporting bad news early is a control discipline rather than a courtesy, and why a report nobody reads is a control failure rather than a communications one.

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Monthly reporting that gets read

In one paragraph. A monthly report exists to cause a decision, and most of them do not. This guide sets out the one-page structure that leads with the decisions required rather than the data, the four-part variance narrative that names cause, effect, action and owner, how to report a control account whose indices are undefined, the arithmetic that makes a recommendation defensible, why reporting bad news early is a control discipline rather than a courtesy, and why a report nobody reads is a control failure rather than a communications one.

Who this is for. Project controls managers and cost engineers who write the monthly report, and the project managers and sponsors who have to act on one.

— 1. A report is an instrument, not a record

The test of a monthly report is not whether it is accurate, complete or on time. Those are entry requirements. The test is whether it caused the right decision to be taken at the right moment.

That reframing changes what belongs on the page. A report designed as a record answers "what happened". A report designed as an instrument answers four questions in order:

1. **What decisions do we need from you, and by when?**
2. **What is off track, why, and what is being done?**
3. **Where will this project finish, on what assumption?**
4. **What has changed since last time?**

Everything that does not serve one of those is supporting material. It is not deleted — it goes in the pack behind the page, where the reader who wants it can find it and the reader who does not is not slowed down by it.

— 2. The one page

The one-page report is not a summary of the pack. It is the report; the pack is the evidence. In order:

Decisions required. Two or three at most, each with the decision, the recommendation, the value, the consequence of not deciding, and the date the option expires. This block goes first, before any data, because it is what the meeting is for. A month with no decisions required says so in one line — that is information too.

Position. The current cost and schedule status in the smallest number of figures that support the rest of the page. Typically budget at completion, actual cost, earned value, the two indices, milestone status in days against plan, and the funding position. Trend arrows against last period, not just values.

Forecast. The estimate at completion, its method, its assumption, its movement since last period and why it moved. [BPG-09 – Estimate at completion: choosing and defending a method](#) sets out what makes that defensible.

Exceptions. Only the items outside tolerance, each with the four-part narrative of §3. Accounts inside tolerance are listed as such and not elaborated.

Actions. Open actions with owners and dates, and the ones that closed this period. An action list where the dates never move is not being managed; an action list where they always move is not being enforced.

Risk and change. Remaining contingency against remaining exposure (see [BPG-10 – Contingency and management reserve](#)), and the change position including instructed-but-unpriced work.

Tolerances decide what appears as an exception, and they are set in advance at control account level. Set after the fact, they are not tolerances; they are explanations.

— 3. The variance narrative

The narrative is where most reports fail, and the failure is consistent: they describe the number again, in words. "CA-30 M&E is showing an adverse cost variance driven by ongoing productivity challenges. The team is working to recover the position and will update next month." Nothing there is false and nothing there is usable.

A usable narrative has four parts and roughly four sentences.

Cause — named, specific, evidenced. Not "productivity challenges" but what happened, to which resource, since when, and how it is known. The evidence source belongs in the sentence: site returns, timesheets, delivery records, the certifier's assessment.

Effect — quantified, in the units the decision needs. Money for a cost variance, days for a schedule variance, and the rate at which exposure accrues if the situation continues. A variance in currency is not an effect; it is a measurement. The effect is what it does to the finish, the forecast or the funding.

Action — specific, resourced, with a date. What will be done, what it costs, when it starts, and what it is expected to achieve. "Monitoring" is not an action. "Recovery plan being developed" is an action only if it has a date and an author.

Owner — a person. Not a company, not a team, not "the project". One name.

Two supporting habits. Report **trend before status**, because direction is where the warning lives — an index of 0.94 that was 0.98 two months ago is a different report from an index of 0.94 that was 0.90. And **say what you do not know**: an honest "the cause of the M&E variance is not yet established; investigation reports on the 12th" is far stronger than a confident attribution that turns out to be wrong, because the second one costs you every subsequent narrative.

— 4. Reporting what the arithmetic cannot say

Two situations recur and both are handled badly by default.

Undefined indices. A control account with no planned value and no earned value cannot produce a performance index — the division is undefined. Reporting a blank, a zero or a spurious 1.00 are

all worse than reporting the fact. Where an account has cost booked against it but no value earned, that is itself the finding, and it belongs in the narrative rather than in the index column.

Rollups that conceal. A project-level index is a spend-weighted average and will comfortably hide one account in trouble behind three that are not. The convention that fixes it costs one line: report the project index and the two or three accounts driving it, every month, whether or not they breach tolerance. A reader who only ever sees the aggregate is not being informed; they are being averaged at.

— 5. Bad news early is a control, not a courtesy

The argument for early disclosure is usually made in ethical terms. It is stronger in arithmetic terms: **the value of a decision decays, and for most project problems it decays at a knowable rate.**

Once a slip is forecast, its cost accrues at the sum of the daily rates it triggers — liquidated damages, extended preliminaries, standing plant, retained supervision. That is a number, and it can be put on the page. Against it sits the cost of the intervention that would prevent it, which is generally a step cost — mobilising crews, changing shift patterns, adding supervision — and which usually only works if taken while enough work remains for it to bite.

Two consequences follow.

Deferring the decision does not reduce the exposure; it removes the options. The intervention that works with three months of remaining work does not work with three weeks, and the exposure is unchanged.

A report that reveals a problem the month after the option expired has failed, however accurate it is. This is why "no surprises" is a control requirement rather than a cultural aspiration, and why a controls function that softens a message to survive a meeting has stopped doing its job. The Institute's position on this is deliberate: the professional's obligation is to the accuracy of the number and the timeliness of the warning, and neither is negotiable against the comfort of the room.

The corollary is that a reporting culture which punishes early bad news guarantees late bad news. If red accounts produce interrogation rather than decisions, the next red account will be amber for two months first. Sponsors set this, not report writers — but report writers should say so when it happens.

— 6. Cadence, audience and the pack

The same underlying data serves different rhythms, and the aggregation should be automatic rather than re-keyed:

Audience	Cadence	What they need
Delivery team	Weekly	Granular, action-focused, forward two to four weeks
Project board	Monthly	Exceptions, forecast, decisions required
Portfolio or executive	Monthly or quarterly	Cross-project comparison, aggregate funding, escalations only

Automation deserves a clear boundary here. A tool can assemble the pack, detect the accounts breaching tolerance, compute the indices and draft the exception narratives; it cannot decide whether a cause has been correctly attributed, nor whether a recommendation is the right one, nor what to say when the data is ambiguous. AI proposes; the professional disposes — and signs, because a report drives decisions and sometimes external disclosure.

— 7. A report nobody reads is a control failure

If the report is not read, the control it was supposed to exert does not exist. That is not a communications problem to be solved with better formatting; it is the same class of failure as a missed accrual or an unmeasured variance, and it should be treated with the same seriousness.

The diagnostic questions are uncomfortable and worth asking annually:

- When did a decision last change because of something in the monthly report?
- Which sections has nobody asked a question about in six months?
- How long is the pack, and how long is the meeting?
- Does the report arrive with enough time before the meeting to be read, or is it tabled?
- Are the same three exceptions carried forward every month with the same action and a new date?

The last one is the most telling. A rolling exception is either not being acted on or not actually an exception, and both answers require someone to do something other than re-issue the report.

— 8. How this goes wrong

Leading with the data. Page one is a table of indices and the reader has to reverse-engineer what is wanted from them. Decisions go first.

Narrative that restates the number. "CPI is 0.86, which is below target." The reader can see that. Why, what it does, what is being done, by whom.

Cause without evidence. Attribution by intuition, presented with the same confidence as measurement. If the cause is not yet established, say so and give the date it will be.

Actions without owners or dates. An action list that is a list of intentions.

Green because red is unwelcome. The status that is set by the temperature of the last meeting rather than by the tolerance. It works exactly once.

The rolling amber. An account that has been amber for five months with the same narrative. Amber is a state of being about to become something; a permanent amber is a decision that has not been taken.

Everything in the pack, nothing on the page. Forty pages, no summary judgement. Volume is not rigour, and a reader who cannot find the decision will make one without the report.

Bad news timed to arrive with its solution. Waiting a month to report a problem so that a recovery plan can be presented alongside it is a defensible instinct and a serious error. The month is exactly what was needed to act.

Inconsistent basis between months. Method changed, tolerance changed, scope of the account changed, and none of it annotated. The trend line becomes meaningless and nobody says so.

No one owns the report. It is assembled from contributions and signed by nobody. The moment a figure is challenged, it turns out that no single person could defend the page.

9. Worked example

Illustrative figures. Currency USD; a project of five control accounts at the Month 8 data date; earned value measured under fixed earning rules; the ledger closed at the same date with accruals booked; indices to three decimal places.

9.1 The position block

Control account	BAC	PV	EV	AC	CPI	SPI
CA-10 Civils	5,200,000	4,100,000	4,050,000	3,980,000	1.018	0.988
CA-20 Structure	7,400,000	4,600,000	4,300,000	4,720,000	0.911	0.935
CA-30 M&E	6,800,000	2,900,000	2,520,000	2,940,000	0.857	0.869
CA-40 Fit-out	3,600,000	900,000	880,000	860,000	1.023	0.978
CA-50 Commissioning	2,000,000	0	0	40,000	n/a	n/a
Project	25,000,000	12,500,000	11,750,000	12,540,000	0.937	0.940

Project CPI = $EV \div AC = 11,750,000 \div 12,540,000 = 0.937$

Project SPI = $EV \div PV = 11,750,000 \div 12,500,000 = 0.940$

EAC (persisting cost performance) = $BAC \div CPI = BAC \times AC \div EV$
 $= 25,000,000 \times 12,540,000 \div 11,750,000 = 26,680,851$

VAC = $BAC - EAC = 25,000,000 - 26,680,851 = (1,680,851)$

Two things the aggregate is hiding, and both belong on the page:

- CA-30 M&E is at 0.857, well below the project figure of 0.937, and it is the account that determines the commissioning start.
- CA-50 has 40,000 of actual cost against zero earned value, so no index can be computed. The report says exactly that, and treats the 40,000 as an open item rather than absorbing it into a rollup.

9.2 The narrative, before and after

As usually written. "M&E is behind due to ongoing challenges with subcontractor resourcing. The team is working hard to recover and will provide an update next month."

As it should be written.

CA-30 M&E — red. CPI 0.857, SPI 0.869; both down from 0.91 and 0.92 at Month 6.

Cause. The containment subcontractor has averaged 11 of the 18 operatives committed since Week 22, verified from site access returns. Containment productivity is running at 0.81 h/m against a budget of 0.62 h/m — $0.81 \div 0.62 = 1.31$, or 31 % above budget hours per metre.

Effect. Commissioning handover is forecast to slip six weeks (42 calendar days). Exposure accrues at 35,500 per calendar day: liquidated damages 24,000 and extended site preliminaries 11,500. Total exposure if the slip is accepted: $35,500 \times 42 = 1,491,000$.

Action. Mobilise four supplementary crews and one additional supervisor; cost 620,000, starting within ten working days; recovery of the full six weeks assumes the added crews achieve current measured productivity.

Owner. M&E package manager. Decision required at this meeting.

9.3 The arithmetic that makes it a decision

Option	Cost now	Exposure avoided	Net
A — accelerate	620,000	1,491,000	$1,491,000 - 620,000 = 871,000$
B — accept the slip	0	0	(1,491,000)

Option A is worth 871,000 **if the acceleration delivers the full six weeks**. State the sensitivity rather than letting someone else find it:

If the supplementary crews achieve 80 % of current measured productivity:

Recovery achieved = 6 weeks $\times$ 0.80 = 4.8 weeks
 Residual slip = 6 – 4.8 = 1.2 weeks = 8.4 calendar days
 Residual exposure = 8.4 $\times$ 35,500 = 298,200
 Net benefit = 1,491,000 – 620,000 – 298,200 = 572,800

The recommendation survives its own sensitivity: even at 80 % effectiveness, accelerating is worth 572,800 more than accepting the slip. That sentence — not the index table — is the report.

9.4 What the decisions block looks like

Decisions required this month

1. **Approve 620,000 for M&E acceleration (CA-30).** Avoids exposure of 1,491,000 from a forecast six-week slip to commissioning handover; net benefit 871,000, or 572,800 if the added crews achieve only 80 % of current productivity. **The option expires when the commissioning front is handed over;** after that, crews cannot be added to the critical work. Recommendation: approve.
2. **Note the 40,000 of cost on CA-50** with no earned value. Investigation reports next period; treat as an open item, not as a cost variance, until the cause is established.

Assumptions this example depends on. Earning rules unchanged since baseline; ledger and progress share the Month 8 data date; the six-week slip is the output of a schedule analysis, not an extrapolation of the schedule performance index; the liquidated damages rate and preliminaries rate are as stated in the contract for this example; the acceleration cost is a quotation, not an estimate; and no part of the exposure is recoverable under an extension of time, which is a separate assessment — see [BPG-12 – Claims and extension of time](#).

— 10. Checklist

Before you write

- [] You can name the decisions this report needs to cause.
- [] Tolerances were set before the period.
- [] The data date is the same for progress, cost and schedule.
- [] Last month's actions have been checked for closure, not copied forward.

The page

- [] Decisions required appear first, with value, consequence and expiry date.
- [] The forecast states its method and its assumption, and explains its movement.
- [] Every exception has cause, effect, action and a named owner.
- [] Causes cite their evidence source.
- [] Effects are quantified in the units the decision needs — days, money, or a rate per day.
- [] Trend is shown alongside status.
- [] The project rollup is accompanied by the accounts driving it.
- [] Accounts with undefined indices are reported as such, never as zero or 1.00.
- [] Remaining contingency is shown against remaining exposure.
- [] Instructed-but-unpriced change is disclosed.

Integrity

- [] Any change of basis, tolerance or account scope is annotated on the trend.
- [] Nothing has been softened to suit the expected reception.
- [] Anything not yet known is stated as not yet known, with a date.
- [] One named person can defend every figure on the page.

Afterwards

- [] The report reached the reader with time to be read, not tabled at the meeting.
- [] Decisions taken are recorded against the items that requested them.
- [] Exceptions carried for more than two periods are escalated as a structural issue, not re-issued.

— Related

- [BPG-08 – Earned value in practice](#) — where the position block's numbers come from, and what they cannot say.
 - [BPG-09 – Estimate at completion: choosing and defending a method](#) — the forecast block, and how to explain a movement.
 - [BPG-15 – Dashboards and data visualisation for controls](#) — the visual layer over the same data, and the distortions to avoid in it.
 - [BPG-19 – Project controls assurance and health checks](#) — how to tell whether the reporting system is actually controlling anything.
 - [TPL-06 – Monthly project controls report](#) — the instrument implementing the structure in §2.
-

— Sources and standards

Management reporting principles — designing the report around the decision, management by exception, tolerance-based escalation — are common to published project management and cost engineering frameworks, including the PMBOK Guide and the AACE International Total Cost Management framework; they are described here in our own words and no text or table is reproduced. Internal references are BoK Domain 4 (Performance Management, Variance Analysis & Management Reporting) and BoK Domain 6 (EVM/EAC). All figures in §9 are illustrative and were computed for this document.

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BPG-15 · BEST PRACTICE GUIDES

Dashboards and data visualisation for controls

Designing a control instrument, not a wall decoration



VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

A controls dashboard is an instrument that changes decisions, and it should be engineered like one. This guide sets out the question a chart must answer before it is drawn, how to choose a chart form for the four questions controls actually asks, why an unreliable cut-off silently corrupts every tile above it, how to make a dashboard readable without relying on colour, and the discipline of one number per decision. It includes a fully worked axis-and-restatement example in which a dashboard shows an uptick while the arithmetic shows a fall into red.

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Dashboards and data visualisation for controls

In one paragraph. A controls dashboard is an instrument that changes decisions, and it should be engineered like one. This guide sets out the question a chart must answer before it is drawn, how to choose a chart form for the four questions controls actually asks, why an unreliable cut-off silently corrupts every tile above it, how to make a dashboard readable without relying on colour, and the discipline of one number per decision. It includes a fully worked axis-and-restatement example in which a dashboard shows an uptick while the arithmetic shows a fall into red.

Who this is for. Project controls managers and reporting leads who own a dashboard; cost engineers and planners whose numbers feed one; and PMO leads deciding what a portfolio view should contain.

— 1. The question comes before the chart

The commonest way to build a bad dashboard is to start from the data warehouse and ask what can be displayed. The result is technically accurate, visually busy and decisionally inert: nobody can say what any tile is for, so nobody acts on any of them, and the dashboard becomes something the controls team maintains rather than something the project uses.

Invert the order. Before a chart is drawn, write down three things.

The decision. Not a topic — a decision. "Whether to release the second tranche of contingency" is a decision. "Cost performance" is a topic. If no decision can be named, the chart is being drawn for completeness, and completeness is not a reason.

The threshold. The value at which the decision changes. If a reader can look at the chart and cannot tell whether they are on the acting side of a line, the chart has not been designed; it has been rendered. A threshold may be a tolerance band, a funded confidence level or a contractual date, but it exists, and it belongs on the chart.

The owner. The person who makes that decision, and the cadence at which they make it. A chart designed for a weekly site meeting and a chart designed for a quarterly investment board have different aggregation, different periods and often different measures, even when they describe the same work.

Write those three lines and most candidate charts delete themselves. That is the point: a dashboard's quality is set far more by what it excludes than by what it contains, because attention is the scarce resource in every meeting it was built to serve.

— 2. A dashboard is a control instrument

The word carries a bad analogy. A car's dashboard is passive: it reports what the engine is doing. A controls dashboard should be closer to an instrument on a process plant — a reading against a setpoint, with an alarm band, feeding a control action. Three properties follow.

It shows position against a threshold, not position alone. A cost performance index (CPI, earned value divided by actual cost) of 0.96 means nothing on its own. Against a tolerance of 0.98 it is an exception; against a tolerance of 0.90 it is noise. The threshold is part of the reading.

It shows direction, not only level. A measure sitting comfortably inside tolerance but moving one hundredth a month in the wrong direction is a different management situation from the same value moving nowhere, and a snapshot tile cannot tell them apart. Every measure carrying a threshold carries a trend.

It has a defined response. For each tile, someone should be able to answer: what happens when this goes amber? "We discuss it" makes the tile decoration. "The control account manager produces a recovery plan within five working days and the change is logged" makes it an instrument.

The corollary is worth stating plainly: a tile with no defined response should be removed, however interesting it is. Interesting is a property of the analyst, not of the project.

— 3. Choosing the form for the question

Controls asks a small number of questions repeatedly. Matching form to question is most of the craft.

Question the reader has	Form that answers it	Common wrong choice
How are we tracking against plan over time?	Line or cumulative S-curve, with the plan, the actual and the forecast as three series	A bar chart per period, which hides cumulative position
What moved the number since last period?	Waterfall (variance bridge), ordered by contribution	A table of variances, which makes the reader do the addition
How do items compare, and which are worst?	Horizontal bar, sorted by value, labelled with the number	An unsorted bar chart, which turns comparison into a search
What is the spread of possible outcomes?	Histogram or cumulative distribution curve with the funded confidence level marked	A single point estimate presented as fact
What is this total made of?	Stacked bar, few categories, consistent order across periods	A pie chart with more than about five slices
Are two measures related?	Scatter, with the relationship stated in words	A dual-axis line chart, which manufactures relationships

Two rows mislead most often. **Dual axes** let the designer choose two scales, and that choice — not the data — determines whether the lines appear to move together. If the point is that two measures are related, say so in words and show a scatter or a ratio; if the point is that both matter, stack two charts on a shared time axis. **Distributions** are generated constantly by controls — schedule risk output, cost risk output, estimate ranges — and then reported as one number. A cumulative curve with the funded confidence level marked lets a reader see how much the answer moves for a small change in confidence, and that slope is often the most decision-relevant thing on the dashboard. [BPG-17 – Quantitative schedule risk analysis](#) covers how those distributions are produced and what they may honestly be said to mean.

— 4. The cut-off problem: a dashboard is only as reliable as its data date

Every tile inherits the reliability of the data date beneath it. This is not a data-quality footnote; it is the single largest source of dashboard error, and it is systematic rather than random.

Cost arrives late. Subcontractor applications, goods-received records, remote timesheets and internal recharges all land after the period they belong to, while earned value is often computed on time because your own team generates it. At first publication, therefore, the numerator of every efficiency measure is more complete than the denominator, so the measure flatters — every month, in the same direction. That is a bias, not noise, and a bias can be measured. [BPG-07 – Accruals and cut-off discipline](#) covers the accrual practice; what matters here is what it does to the picture.

Three design consequences follow.

Mark provisional points. The most recent point on any trend is the least reliable point, and it is the one the meeting acts on. Distinguish it — a hollow marker, a dashed final segment, an explicit "provisional" label — so nobody reads it as equivalent to the settled history behind it.

Publish the restatement history. Keep, for each measure, the value as first published and the value as restated one period later. The average movement between them is the measure's completeness bias — the most useful diagnostic number a reporting function can hold. Section 8 works it through.

Snapshot, do not re-render. A dashboard that recomputes history from live data on every open changes its own past silently: two people looking at "the March chart" in April and in June see different charts, and neither knows it. Store each period's figures as issued, and make restatement a visible act.

— 5. Accessibility is a correctness requirement

Meaning encoded in colour alone is meaning some readers do not receive. Assume that part of your audience cannot reliably separate red from green, that someone will print the report in grey-scale before the meeting, and that a projector will flatten whatever palette you chose. Under those assumptions a red-amber-green status that is only a colour communicates nothing to a meaningful part of the room.

The fix is redundant encoding — every status carries at least two signals:

- a **shape or symbol** as well as a fill, so the category survives greyscale;
- a **label** with the actual value and the threshold it breached, so the reader is not decoding at all;
- **position**, which is the strongest visual channel available — a value plotted against a marked threshold line communicates status before any colour is processed.

Two further rules. Keep every category in the same order and the same encoding across every tile, because a reader who must relearn the key on each one will stop reading tiles. And give every chart a caption that doubles as its alternative text: what is shown, over what period, and what the reading is. "CPI trend" is a title. "Cost performance index by month, January to June, falling from 1.01 to 0.95 against a tolerance of 0.98" is an accessible description, and it is also the sentence the reader repeats in the meeting.

— 6. One number per decision

The strongest discipline in dashboard design, and the hardest to hold, is that each decision gets one headline number. Not a panel of five related indices from which the reader must synthesise a position — one number, with its threshold, its direction and its provisional status.

This is not simplification for its own sake. When several measures that can disagree support one decision, the meeting reconciles measures instead of deciding, and the person with the strongest view picks whichever measure supports it. Deciding in advance which number governs removes that choice, which is why it is resisted.

The supporting detail does not disappear; it moves. The structure that works is overview first, detail on demand: one screen answering where we are, where we are heading, what is outside tolerance and what is being done — with every red and amber traceable in one step to the control account and its narrative. Every promotion of detail onto the front page costs attention the headline numbers were competing for.

Where several measures genuinely bear on one decision, combine them explicitly and publish the combination rule. An explicit rule can be argued with and improved; an implicit one is re-invented at every meeting.

— 7. How this goes wrong

The dashboard that reports everything available. Forty tiles, none with a defined response. It survives because removing a tile requires someone to say it is not needed. The test that breaks the deadlock: ask each tile's owner what decision changed because of it in the last six months. Tiles with no answer come off.

The axis chosen after the data is seen. A truncated axis turns a movement of a few hundredths into a cliff; a zero-based axis on a ratio that lives between 0.9 and 1.1 turns a serious decline into a flat line. Both are defensible in isolation and both are choices about the message. Fix the axis to the decision thresholds before the period's data arrives, and keep it fixed across periods.

Comparing the provisional to the settled. The current month is compared with a fully restated prior month and the difference reported as movement. The worked example below quantifies this; it is the failure that most often causes a project to relax in the month it should have escalated.

The refreshing dashboard with no memory. Nobody can reconstruct what was reported when a decision was taken, which makes the dashboard useless as evidence and, on a project heading for a dispute, worse.

Colour doing all the work. Coloured circles with no labels, shapes or values: fast to build, unreadable in print, ambiguous to some readers, impossible to quote in minutes.

Two dashboards that disagree. A project view and a portfolio view built on separately maintained extracts. The meeting becomes a reconciliation. One source, aggregated through the coding structure, is the only cure; where reconciliation is unavoidable, publish the difference and its cause on both views rather than letting it be discovered.

The chart redrawn until it is acceptable. Rarely conscious — a series of individually reasonable choices about axis, window and grouping, each nudging the picture. The countermeasure is procedural: fix the chart specification with the tolerance, and record any mid-period change to it alongside the change to the data.

— 8. Worked example

Illustrative figures. Units are dimensionless index values unless stated. Period is monthly, months 1 to 6 of a delivery phase. Tolerance for this control account: amber below 0.98, red below 0.95. All arithmetic is shown; percentages of plot height are exact.

8.1 The setup

A control account's cost performance index (CPI) is reported monthly on a dashboard tile. The settled series — each month's value after one period of restatement — and the current month's provisional value are:

Month	Value on the tile	Status of the value
1	1.01	settled
2	0.99	settled
3	0.98	settled
4	0.96	settled
5	0.95	settled
6	0.96	provisional — first publication

Read naively, month 6 is an improvement: 0.96 against 0.95, up 0.01, and back above the red line.

8.2 Part A — what the axis does to the same six numbers

The full movement across the series is $1.01 - 0.95 = 0.06$.

On a zero-based axis of 0.00 to 1.20, the plot spans 1.20 index points, so the movement occupies $0.06 \div 1.20 = 0.05$, or **5 % of the plot height**. Six months of steady decline through two tolerance thresholds renders as a flat line.

On an axis truncated to 0.92-1.02, the plot spans 0.10, so the movement occupies $0.06 \div 0.10 = 0.60$, or **60 % of the plot height**, and month 5 sits at $(0.95 - 0.92) \div 0.10 = 0.30$, near the floor. The same decline renders as a collapse.

On an axis of 0.85-1.05, chosen because it contains both tolerance thresholds with margin, the plot spans 0.20 and the points sit at these heights:

Month	Value	Height on plot = (value – 0.85) ÷ 0.20
1	1.01	$0.16 \div 0.20 = 80\%$
2	0.99	$0.14 \div 0.20 = 70\%$
3	0.98	$0.13 \div 0.20 = 65\%$
4	0.96	$0.11 \div 0.20 = 55\%$
5	0.95	$0.10 \div 0.20 = 50\%$
6 (provisional)	0.96	$0.11 \div 0.20 = 55\%$

The amber threshold of 0.98 sits at $(0.98 - 0.85) \div 0.20 = 65\%$ and the red threshold of 0.95 at $(0.95 - 0.85) \div 0.20 = 50\%$ — both on the chart, both crossed by the series, and the movement occupies 30 percentage points of plot height between month 1 and month 5. **The axis is set by the thresholds, not by the data**, which is why it can be fixed before the period opens and does not need a decision after the numbers are known.

8.3 Part B — what the provisional point is actually worth

The reporting function has kept, for each month, the value as first published and the value after one period of restatement:

Month	First published	Restated one month later	Movement
1	1.05	1.01	-0.04
2	1.02	0.99	-0.03
3	1.00	0.98	-0.02
4	0.99	0.96	-0.03
5	0.98	0.95	-0.03

Mean restatement:

```
sum of movements = (-0.04) + (-0.03) + (-0.02) + (-0.03) + (-0.03) = -0.15
mean              = -0.15 ÷ 5 = -0.03
```

Every movement is negative. That is the signature of a completeness bias rather than random error: late cost lands in the denominator after publication, so first publication is systematically optimistic by roughly three hundredths.

Applying the measured bias to the provisional month-6 value:

```
expected settled month 6 = 0.96 + (-0.03) = 0.93
```

The reading changes. The tile shows 0.96, an apparent rise of 0.01 on the settled month-5 value of 0.95, sitting in amber. The bias-adjusted expectation is 0.93 — a **fall of 0.02** on month 5 ($0.93 - 0.95 = -0.02$) and **below the red threshold of 0.95**. The month the dashboard says has stabilised is, on the project's own restatement history, the month it went red.

8.4 What the numbers do and do not support

The adjusted figure is an expectation, not a measurement, and it does not license reporting 0.93 as the CPI. The correct treatment is to report 0.96 as provisional, state the measured bias of -0.03 on the face of the chart, and take the escalation the red threshold requires — recording that it was triggered by the adjusted expectation, so that if month 6 settles at 0.96 after all, the decision is reviewable rather than embarrassing.

Three assumptions carry the result: the history is five points, which is short; the bias is assumed stable, which fails if its cause is fixed or worsens; and it is specific to this control account's cost-capture pattern, so it may not transfer to another account, let alone another project.

— 9. Checklist

Take this into the design review, not into the reading of the report.

Per tile

- [] The decision this tile supports is written down, in decision form, not topic form.
- [] The threshold at which that decision changes is written down and drawn on the chart.
- [] The response when the threshold is breached is written down, with an owner and a timescale.
- [] The chart form matches the question — trend, variance, comparison, distribution or composition.
- [] The vertical axis is fixed to the thresholds and does not change when the data does.
- [] The most recent point is visually marked as provisional.
- [] Status is carried by at least two channels: never colour alone.
- [] The caption states what is shown, over what period, and what the reading is.

Per dashboard

- [] The top level fits one screen and answers where we are, where we are heading, what is outside tolerance, and what is being done.
- [] Every red and amber reaches its control account and its narrative in one step.
- [] Each decision has exactly one headline number; supporting measures sit at the level below.
- [] Categories, ordering and encoding are identical across every tile.
- [] Every tile that failed the "what decision changed" test has been removed.

Per period

- [] Published figures are snapshotted; history does not silently re-render.
- [] First-published and restated values are recorded for every measure carrying a threshold.
- [] The restatement bias is recomputed and shown on the face of the affected charts.
- [] Project and portfolio views reconcile, or the difference is published on both.
- [] Any change to a chart specification is recorded alongside the data change.

A dashboard built this way is slower to produce and considerably shorter. It also becomes quotable: when a number on it moves, the meeting knows what changes, who acts and by when — which is the only property that distinguishes an instrument from a picture of one.

— Related

- [BPG-14 – Monthly reporting that gets read](#) — the report the dashboard sits inside, and the narrative that must accompany it
 - [BPG-08 – Earned value in practice](#) — where the indices on the tiles come from and what they legitimately mean
 - [BPG-07 – Accruals and cut-off discipline](#) — the cost-capture practice that determines whether any tile can be trusted
-

- [BPG-16 – Risk registers that work](#) — the risk exposure that belongs on the dashboard, and the form it should take
- [TPL-06 – Monthly project controls report](#) — the reporting structure these tiles feed
- [AIG-03 – Data readiness: what AI needs before it is any use](#) — the upstream data conditions that determine what can be automated

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 4 — Performance Management, Variance Analysis and Management Reporting, first authored draft, August 2026: chart-to-question fit, common distortions, dashboard structure, exception reporting and cadence.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 6 — Earned Value Management and Forecasting, first authored draft, August 2026: the definition and behaviour of the performance indices used in the worked example.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

No external statistic, benchmark or vendor capability is cited, because none was verified for this guide. All figures in §8 are illustrative and were constructed for teaching.

— Status and version

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BPG-16 · BEST PRACTICE GUIDES

Risk registers that work

Turning a compliance artefact back into a management instrument

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Most risk registers are maintained and not used. This guide sets out what separates the two: risk statements built as cause, event and effect; why an effect such as "delay to project" cannot be managed as a risk; what qualitative probability-impact scoring can and cannot order; ownership and action dates as the register's live parts; the link between the register and contingency; and the health metrics that tell you, in a few minutes, whether your register is a management tool or wallpaper. The worked example shows two risks with identical matrix scores whose expected values differ by a factor of more than three.

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Risk registers that work

In one paragraph. Most risk registers are maintained and not used. This guide sets out what separates the two: risk statements built as cause, event and effect; why an effect such as "delay to project" cannot be managed as a risk; what qualitative probability-impact scoring can and cannot order; ownership and action dates as the register's live parts; the link between the register and contingency; and the health metrics that tell you, in a few minutes, whether your register is a management tool or wallpaper. The worked example shows two risks with identical matrix scores whose expected values differ by a factor of more than three.

Who this is for. Risk managers and project controls managers who own a register; cost engineers and planners who have to turn one into contingency or a schedule model; and project managers who chair the review that the register is supposed to serve.

— 1. The register that nobody uses

Almost every project has a risk register. A small minority have one that changes a decision in any given month. The gap is not effort — unused registers are often meticulously maintained — and it is not tooling. It is that the register was built to demonstrate that risk management is happening rather than to make risk decisions, and an artefact built for demonstration acquires a predictable shape: many entries, vague statements, team-level ownership, scores that never move, actions without dates, and a review that reads the top ten aloud and closes.

The diagnosis matters because the remedies differ. A register failing from neglect needs attention. A register failing from purpose needs rebuilding around the decisions it is meant to inform: what we fund, what we mitigate, what we escalate, what we tell the client.

— 2. Cause, event, effect — and why the structure is not pedantry

A risk entry that can be managed has three separable parts.

The cause is a condition or dependency that exists now. It is not uncertain; it is a fact about the project. "The permanent power connection is delivered by a utility outside the contract."

The event is the uncertain thing. It either happens or it does not, and a probability attaches to it. "Energisation is granted later than the commissioning window requires."

The effect is what the event does to an objective, measurable in time, money, scope or quality. "Systems commissioning is delayed and time-related site costs extend."

Written as a single entry: because the permanent power connection depends on a utility outside the contract, there is a risk that energisation is granted later than the commissioning window requires, which would delay systems commissioning and extend time-related site costs.

The structure earns its place because each part drives a different management act. The **cause** is what a mitigation attacks — you cannot reduce an event's probability except by acting on its cause,

which is why registers full of causeless risks generate mitigations that are really just monitoring. The **event** carries the probability and can be declared closed when the uncertainty resolves. The **effect** carries the impact, feeds quantification, and determines who should own the entry, because a risk belongs to whoever owns the objective it threatens, not to whoever first mentioned it.

Two tests. If the cause is not a present fact it is a second risk, and the two should be separated. If the event cannot be declared to have happened or not happened on a given date it is a condition, and it belongs in the assumptions or the issue log.

— 3. Why "delay to project" is not a risk

"Delay to project" is an effect. So is "cost overrun", "reputational damage" and "loss of client confidence". A register of effects cannot be managed, for three reasons that compound.

No action addresses it. Delay arises from dozens of unrelated causes with different owners. An entry aggregating them can only be assigned to the project manager, and the mitigation can only be "manage the project". That is a job description, not a control.

It can never be closed. The project is at risk of delay from first day to last, so the entry sits permanently with its score unchanged — teaching every reader that entries on this register do not move. One immortal entry is enough to establish the culture.

It double-counts. If the register also carries the specific events that would cause delay, the aggregate entry counts them again. Any quantification built on it is then wrong by an unknown amount, in the direction of overstating exposure — which sounds prudent until a contingency request must be defended line by line.

The same logic disposes of two other common entries. "Poor contractor performance" is an assessment, not an event; the events are the specific things a poorly performing contractor does — misses a delivery date, fails an inspection, under-resources a work front — each with its own cause and response. And something that has already happened is not a risk: it is an issue, and it belongs in the issue log with an action and a date. A register carrying issues has stopped distinguishing what might happen from what has, which is the one distinction it exists to make.

— 4. Qualitative scoring: what it orders, and what it cannot

Almost every register scores probability and impact on ordinal bands — typically one to five — and multiplies them into a matrix score. Used for what it is good at this is sound and fast: it triages a large register quickly, needs no cost model, works in a workshop with people who will not tolerate a quantitative exercise, and identifies which entries deserve quantification.

Its limitation is structural, not a matter of calibration. **The bands are ordinal and the impact scale is usually geometric, while the score is arithmetic.** Band 4 is not twice band 2 in money; it is often ten or twenty times. Multiplying two rank positions therefore produces a number with no monetary meaning, and two entries in the same cell can differ in expected value by an order of magnitude. Section 8 works this through.

Three consequences for practice.

Use the matrix to triage, not to rank for funding. It answers "which entries deserve more analysis", not "which entries drive the contingency". Any register whose top ten is selected by matrix score has the wrong ten in it.

Define the bands in the units of the project, and write them on the register. "Low, medium, high" is re-interpreted by everyone who uses it; a stated money range and a stated probability range are at least argued about consistently. Calibrate so that band 5 is genuinely material — a band-5 threshold set at a figure the project spends monthly makes the whole top row meaningless.

Score impact against a stated objective. A risk can be band 2 for cost and band 5 for schedule. One combined score hides which objective is threatened, and therefore who should own it.

Where a project has an appetite statement, the matrix is also where it is expressed — by drawing the escalation line on the grid rather than leaving each chair to apply their own tolerance. That is professional judgement made explicit, which is the only form in which it can be reviewed.

— 5. Ownership, actions and dates — the live parts

The columns that determine whether a register is alive are not the scoring columns.

Owner must be an individual. "Engineering" cannot receive a phone call. Where a risk genuinely spans functions, name one owner and list the others as contributors; a shared owner is an unowned risk.

Every response has an action, an owner and a date in the future. An action without a date is an intention. A register where a third of the action dates have passed has told you how much attention it receives, more reliably than any maturity assessment.

Responses are typed, and the type is honest. The threat strategies are avoid, transfer, reduce and accept; the opportunity strategies are exploit, share, enhance and accept. Most entries claiming to be mitigated are in fact accepted with monitoring, and calling that "mitigate" inflates the project's apparent control. Conscious acceptance with contingency held against it is legitimate and often correct; disguising acceptance as mitigation is not.

A response is justified when it reduces probability times impact by more than it costs. That test is arithmetic and should be shown: the register carries the pre-response and post-response assessment and the response's cost, so the decision is visible. Responses whose cost exceeds the exposure they remove are recorded as rejected with a reason, which is what stops the same proposal returning every quarter.

Every entry carries a review date. A risk whose score has not moved in three cycles is either genuinely stable — in which case say so and review it less often — or unexamined. Recording which takes one column.

— 6. From register to contingency, and to the schedule model

Summing the **expected monetary value (EMV)** of every entry — probability multiplied by impact, added across the register — gives a first-pass figure for the contingency needed against identified risks. That sum is valid whatever the correlations between entries, because the expected value of a sum is the sum of the expected values.

It is also not the answer, for two reasons that pull in the same direction.

The expected value is not a fundable number. No individual risk costs its expected value; each costs either its impact or nothing. What a project funds is a confidence level on the distribution of total outcomes, and that level sits above the expected-value sum whenever risks can coincide.

Correlation fattens the tail without moving the mean. Real registers are full of shared drivers: one weather season behind several delays, one overheated supply market behind several vendor slips. They make coincidence far more likely than independent arithmetic suggests, raising the funded confidence level while leaving the expected-value sum exactly where it was.

The practical discipline needs no mathematics. For each pair of significant entries, ask what would make both happen. A nameable shared cause is a correlation to declare — to the modeller if there is a model, and to the contingency basis if there is not. [BPG-10 – Contingency and management reserve](#) owns how the funded level is chosen and governed; [BPG-17 – Quantitative schedule risk analysis](#) owns what happens when the register is mapped onto the schedule and simulated.

One connecting discipline belongs to the register rather than the model: **risk loading**. Each entry that will be quantified names the specific activities and cost lines it would strike. Without that mapping the model runs on a parallel set of assumptions and the register merely resembles it; with it, the register is the model's input and a change to one is visible in the other.

— 7. How this goes wrong

The register is a deliverable, not a tool. Produced because the framework requires one, reviewed because the review is in the schedule, consulted by nobody in between. The tell is that nothing in the project plan changes as a result. A register that has never caused a schedule change, a contingency draw or a contract negotiation is not managing risk.

Entries are effects, so nothing can be closed. Covered in §3. The symptom is a register whose population only ever grows.

The top ten is chosen by matrix score. The heavy-tail entries — low probability, very high impact — sit below the line because the matrix rewards likelihood. Those are usually the entries that determine the contingency and, occasionally, the ones that end the project.

Mitigations are monitoring in disguise. "Monitor closely", "maintain dialogue with the supplier", "escalate if required". None changes probability or impact, so the residual assessment should be unchanged — yet the register almost always shows a reduced residual score once they are added.

Everything is owned by the project manager. A register with one owner is a to-do list belonging to someone who already has one.

Impact is scored on the project, not on the objective. "High" means high for schedule in one entry and high for reputation in another, and the two are then compared as though commensurable.

Opportunities are absent. A register with no upside entries is evidence about the workshop, not the project, and it biases the contingency conversation, because what would net against threats is not on the page.

The register outlives its assumptions. Scores set at sanction are still there at handover. The risk profile a project starts with is not the one it carries, so re-identification at each gate is not optional.

Quantification double-counts responses. The entry is quantified at its pre-response impact and the response's cost is also in the budget. The exposure is funded twice and the request cannot survive scrutiny.

8. Worked example

Illustrative figures. Currency is USD, whole dollars, no escalation and no discounting applied. Impacts are cost impacts on a single objective. Probabilities are point estimates from a workshop.

8.1 The scales in use

Band	Probability	Cost impact
1	below 10 %	below 25,000
2	10 % to 25 %	25,000 to 250,000
3	25 % to 50 %	250,000 to 750,000
4	50 % to 75 %	750,000 to 2,500,000
5	above 75 %	above 2,500,000

Matrix score is the product of the two band numbers, on a scale of 1 to 25. The project's escalation line is drawn at a score of 8.

8.2 Three entries

Ref	Event (abbreviated)	Probability	P band	Impact	I band	Matrix score
R-11	Late utility energisation	60 %	4	100,000	2	8
R-24	Single-source vendor fails to deliver the main package	20 %	2	1,000,000	4	8
R-31	Contaminated ground discovered on the east plot	5 %	1	1,200,000	4	4

R-11 and R-24 share a matrix score of 8 and sit on the escalation line. R-31 scores 4 and sits below it.

8.3 Expected value

EMV = probability × impact

R-11: $0.60 \times 100,000 = 60,000$

R-24: $0.20 \times 1,000,000 = 200,000$

R-31: $0.05 \times 1,200,000 = 60,000$

Total EMV = $60,000 + 200,000 + 60,000 = 320,000$

8.4 What the two orderings disagree about

Same score, different money. R-11 and R-24 both score 8. Their expected values are 60,000 and 200,000.

$$\text{ratio} = 200,000 \div 60,000 = 3.33 \text{ (to two decimal places)}$$

R-24 carries more than three times the expected exposure of R-11, and the matrix places them in the same cell. A register that funds by matrix position funds these two identically.

Different score, same money. R-11 scores 8 and R-31 scores 4 — R-11 is ranked at twice R-31's priority. Their expected values are identical at 60,000 each. The matrix has ordered by likelihood, and likelihood is not what the project pays.

Where the tail lives. R-31 can cost 1,200,000 in a single event — 3.75 times the register's entire expected-value sum ($1,200,000 \div 320,000 = 3.75$) — and the matrix places it below the escalation line. If the top ten were selected by matrix score, the project's largest single loss would not be in it.

8.5 Why this happens, in one line

The impact bands step by roughly an order of magnitude while the band numbers step by one. Multiplying band numbers multiplies rank positions; multiplying probability by impact multiplies a proportion by an amount of money. Only the second produces a quantity you can add up, compare or fund.

8.6 What follows for this register

Use the matrix to decide which entries go forward for quantification, with the forward criterion set on either dimension so a band-4 or band-5 impact goes forward regardless of probability. Order the funding conversation by expected value, and hold the heavy-tail entries separately because their expected value understates what one occurrence does. Then ask what R-24 and R-31 have in common: a shared driver — one contaminated-land condition that also delays the vendor's foundation works, say — is the pairing that moves the funded level.

The assumptions travel with the numbers: probabilities are workshop point estimates with no range around them; impacts are single figures, not distributions; nothing here nets opportunities against threats; and the expected-value sum is a starting point for contingency, not a contingency figure. [BPG-10](#) takes it on.

— 9. Register health in five minutes

Illustrative figures from a single register at one review date.

A register of **84 entries**, of which:

- **61** have a named individual as owner: $61 \div 84 = 72.6 \%$. Twenty-three entries are owned by a team, a function or nobody.
- **31** have a next action date that has already passed: $31 \div 84 = 36.9 \%$.
- **44** have not changed score in three or more review cycles: $44 \div 84 = 52.4 \%$.
- **2** were added and **0** closed in the last cycle.

Each figure is a question rather than a verdict, but together they are decisive. A register where more than half the entries have not moved in three cycles, more than a third of actions are overdue and nothing closed all cycle is not being used; it is being maintained. And 84 is itself a finding — a project board cannot manage 84 of anything, so either the register needs a working subset with a stated selection rule, or most entries should be closed, merged or moved to a watch list on a longer cycle.

Run these counts before the review meeting rather than after it. They take minutes, they are not disputable, and they change what the meeting is about.

— 10. Checklist

Every entry

- [] Cause is a present fact, event is a discrete uncertainty, effect is measurable in time or money.
- [] The event can be declared to have happened or not happened on a stated date.
- [] The entry is not an effect, an assessment or an issue.
- [] Owner is a named individual who owns the threatened objective.
- [] Probability and impact are assessed against published bands, in project units.
- [] Cost impact and schedule impact are scored separately.
- [] Response is typed honestly — avoid, transfer, reduce, accept, or the opportunity equivalents.
- [] Response has an action, an owner and a date in the future.
- [] Pre-response and post-response assessments are both recorded, with the response's cost.
- [] Where the entry will be quantified, the activities and cost lines it would strike are named.

Every review

- [] Overdue action count, unowned count and unchanged-score count computed before the meeting.
- [] Entries closed as well as entries added.
- [] Heavy-tail entries reviewed regardless of matrix position.
- [] Shared drivers between significant entries named and recorded.
- [] Opportunities reviewed on the same footing as threats.

Every gate

- [] Re-identification run, not just re-scoring, with the people doing the next phase's work.
- [] Assumptions behind the original scores tested against what is now known.
- [] The expected-value sum reconciled to the contingency held, with the difference explained.

A register that passes this list is smaller than the one it replaced and is read by people who are not paid to read it. That is the only durable evidence that risk management is happening: the register changed what the project did, and someone can name the month it happened.

— Related

- [BPG-10 – Contingency and management reserve](#) — how the quantified register becomes a funded number, and who may draw it
- [BPG-17 – Quantitative schedule risk analysis](#) — what happens when the register is mapped onto the schedule and simulated
- [BPG-19 – Project controls assurance and health checks](#) — how a reviewer tests whether the register is genuinely in use
- [TPL-10 – Risk register](#) — the column standard and definitions this guide assumes
- [AIG-06 – AI for risk identification and quantification](#) — where machine assistance helps, and what must stay with the professional

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 12 — Risk Management for Project Controls, first authored draft, August 2026: the cause-event-effect standard, qualitative and quantitative analysis, response strategies, correlation and shared drivers.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 3 — Budgeting and Forecasting, first authored draft, August 2026: estimating bias and the treatment of uncertainty in a budget.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

ISO 31000 is the international risk-management standard most often referenced in this area. This guide reproduces none of its text, structure or clause numbering; where its principles are relevant — that risk management is integrated into decisions rather than run alongside them, and that it addresses uncertainty in both directions — they are explained in the Institute's own words. No survey data, benchmark or industry average is cited, because none was verified for this guide.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

BPG-17 · BEST PRACTICE GUIDES

Quantitative schedule risk analysis

What a P80 date is confidence in, and what it is not

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Quantitative schedule risk analysis converts an uncertain network into a distribution of completion dates. This guide sets out what that distribution can honestly be said to mean, how three-point estimates are elicited without importing the estimator's anchor, why correlation is the assumption that most often determines the answer, the difference between duration uncertainty and register-driven risk events, how to read a P50 and a P80 as confidence in a model rather than in the world, and why running the analysis on a schedule that fails a quality review produces a distribution of the constraint set. Every figure in the worked example is hand-computed and checkable.

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Quantitative schedule risk analysis

In one paragraph. Quantitative schedule risk analysis converts an uncertain network into a distribution of completion dates. This guide sets out what that distribution can honestly be said to mean, how three-point estimates are elicited without importing the estimator's anchor, why correlation is the assumption that most often determines the answer, the difference between duration uncertainty and register-driven risk events, how to read a P50 and a P80 as confidence in a model rather than in the world, and why running the analysis on a schedule that fails a quality review produces a distribution of the constraint set. Every figure in the worked example is hand-computed and checkable.

Who this is for. Planners and schedulers who run or commission the analysis; project controls managers who must defend its output; and project managers asked to commit to a date that came out of it.

— 1. What the analysis actually does

Quantitative schedule risk analysis — **QSRA** — takes a logic-driven schedule, replaces fixed durations with ranges, adds discrete risk events from the risk register, samples the network many thousands of times and reports the distribution. Its output is completion dates with probabilities attached: the **P50**, reached on or before in half the runs; the **P80**, in eighty per cent of them; and the curve joining them.

Three things it does that a deterministic schedule cannot. **It exposes the optimism in the merge:** a forward pass takes the latest arrival at each convergence point, which is correct arithmetically and optimistic probabilistically, because the milestone is met only if every incoming path is met — and simulation samples all paths together. **It ranks the drivers:** what changes behaviour is rarely the date but the sensitivity ranking of which activities and register entries do the work in the tail, which is a priced mitigation shopping list. **It separates commitment from target,** letting a project hold an internal target and an external commitment at different confidence levels with the gap named.

And three it does not. **It does not tell you what the project will do** — only what the model does (§6). **It does not find risks:** whatever the workshop missed is absent from the distribution, invisibly. **It does not repair a schedule:** a network that cannot recalculate honestly cannot be simulated honestly (§5).

— 2. Three-point estimates and the elicitation problem

The input for duration uncertainty is a range per activity: optimistic, most likely and pessimistic values, written a, m and b. Getting them is an interviewing problem before it is a modelling problem, and the interviewing is where most QSRA quality is won or lost.

The dominant failure is **anchoring**. The estimator is shown the scheduled duration and asked for a range around it. That number becomes the anchor; the range comes back as roughly plus or minus ten or fifteen per cent, and the model inherits the schedule's optimism with a decorative band around it. Where the durations were themselves set by working backwards from a required completion date — common, and rarely admitted — the anchoring imports that too.

Countermeasures, in the order they matter. **Ask for the pessimistic value first, and as a story:** "what would have to happen for this to take twelve weeks?" requires a mechanism, and if none can be described the value is too low. **Elicit from the person who does the work, not the person accountable for the date** — accountability narrows ranges, not dishonestly, because someone who has committed to a date genuinely finds the pessimistic scenario less plausible than the crew supervisor does. **Do not show the scheduled duration until the range is given**, and where that is impractical, record whether it was shown. **Hold everyone to one definition of the pessimistic value. Calibrate against outturn** — showing estimators how their previous ranges compared with what happened is the most effective correction available, and [BPG-20 – Closeout, lessons learned and benchmarking](#) covers capturing that data.

Two conventions from the programme evaluation and review technique tradition summarise a three-point range: the weighted mean $(a + 4m + b) \div 6$ and the spread approximation $(b - a) \div 6$. Both are conventions rather than requirements of any standard, and both approximate one family of distributions. They are used in §8 because they can be checked by hand.

— 3. Duration uncertainty and risk events are different inputs

Two distinct sources of schedule uncertainty are frequently conflated, and conflating them either double-counts or omits.

Duration uncertainty is the variability in how long work takes when it goes ahead as planned: productivity varies, weather varies, the crew is better or worse than assumed. It applies to every activity, is modelled by the three-point range, and carries no probability of occurrence.

Risk events are discrete occurrences from the register that may or may not happen: a permit refused, a vendor failure, contaminated ground. They carry a probability of occurrence and an impact range, strike specific activities, and are modelled by mapping register entries onto the activities they would hit — the **risk-driver** approach, whose virtue is that the model inherits the register instead of running on a parallel set of assumptions.

The double-count happens when an estimator, asked for a pessimistic duration, includes a register entry — "twelve weeks if the permit is late" — and the modeller then loads the permit risk onto the same activity. The omission runs the other way: ranges elicited as "how long if nothing goes wrong", nothing loaded, and an analysis reported as showing a robust schedule. One instruction prevents both, given at elicitation and recorded on the input sheet: give the range assuming the identified risks do not occur; the register is loaded separately. [TPL-11 – Quantitative schedule risk analysis input sheet](#) carries the field structure, and [BPG-16 – Risk registers that work](#) the register discipline it depends on.

— 4. Correlation is the assumption that decides the answer

Simulation software will happily sample every activity independently. Real projects do not: activities share crews, a design authority, a weather season, a fabrication market, an approval process. When a shared driver moves, it moves everything it touches in the same direction at once.

Correlation does not move the mean. It fattens the tail. The expected value of a sum equals the sum of the expected values, correlated or not. What changes is the spread: independent components partly cancel — some run long while others run short — and that cancellation is what correlation removes. The P50 barely moves; the P80 and P90 move a lot.

Two structural cases matter, and they run in opposite directions. **Along a chain**, correlation increases the total's variability: four independent activities of equal spread give a chain whose spread is twice one activity's, because standard deviations add in quadrature, while perfectly correlated they give four times. **At a merge**, correlation reduces the penalty: paths driven by one shared cause tend to be ready or late together, so the milestone's probability approaches that of a single path, whereas independent paths multiply their probabilities. Section 8 computes both. Because they pull opposite ways, "assume some correlation to be safe" is not a conservative default.

The practical method needs no mathematics. List the shared drivers explicitly before the model is built — one weather season, one commissioning contractor, one approval authority, one specialist crew — and record which activities each touches. That list, not a coefficient chosen by feel, belongs in the basis document. Where the tool demands a coefficient, record its provenance and treat it as an assumption to test by sensitivity.

— 5. What the analysis needs before it is worth running

QSRA amplifies whatever is underneath it. Five preconditions, each checkable in an afternoon:

- **The schedule recalculates** — every activity except the first and last has a predecessor and a successor; date constraints are rare and justified; negative float is absent or explained; long lags are defended; progress is not out of sequence. [BPG-05 – Schedule quality – a practical review](#) and [TPL-14](#) own these.
- **The critical path is the work, not the constraints.** A network held together by imposed dates does not respond to sampled durations, so sampling it yields a distribution of the constraint set.
- **The level of detail suits simulation** — thousands of short activities give spurious precision and an elicitation effort nobody completes honestly; thirty summary bars hide the merges that drive the answer.
- **The register is current and loadable** — cause-event-effect entries, scored, mappable onto activities.
- **Someone owns the basis document** — distribution shapes, correlation drivers, what the ranges include and exclude, what was not modelled, written before the run.

The second is the most consequential and the most often skipped, because a heavily constrained schedule still produces a convincing curve. Without the fifth, a QSRA cannot be reviewed, only believed.

— 6. Reading a P-value honestly

A P80 completion date is a statement about a model: in eighty per cent of the simulated runs of this network, with these ranges, these loaded risks and these correlation assumptions, completion occurred on or before this date. Every clause is load-bearing.

A P80 is not an eighty per cent chance of finishing by that date. It is an eighty per cent chance within the model's world — a world excluding every risk not identified, every range not honestly given, every correlation not declared and every structural failure the schedule contains. Each omission moves the real probability down; none moves it up.

The gap between the deterministic date and the P50 is diagnostic. A deterministic date far below the P50 means optimistic durations, heavy merging, or both — a finding about the schedule, not about risk, and one to report before any conversation about contingency.

The shape matters more than the point. A curve nearly vertical between P50 and P80 says the date is fairly well determined; a long shallow tail says a small increase in required confidence costs a great deal of time. Report the curve, not two numbers off it. And a P80 reported to the day, on ranges elicited to the nearest fortnight, is false precision: round to a unit the inputs support.

The honest defence of a P-value is its basis document, not its confidence level. When a result is challenged — usually by someone who dislikes the date — what works is the list of assumptions and what happens when each is varied. What fails is "the model says so". The model proposes; the professional disposes.

— 7. How this goes wrong

Run on a schedule that fails its quality review. Constraints hold the dates, dangling activities never move, out-of-sequence progress has corrupted the critical path — and the model produces a smooth, plausible curve regardless. Nothing in the result announces that the network could not respond, which is exactly why the quality checks are run first and published alongside the distribution.

Ranges anchored on the deterministic durations. Plus or minus ten per cent on every activity tells you only that someone had a spreadsheet. Uniform ranges signal that elicitation did not happen.

No correlation, because nobody asked. Independence is most tools' default, is almost never true, and has a large effect, as §8 shows. A basis document silent on correlation made the assumption without recording it.

Modelling only the critical path. Near-critical paths become critical under sampling; that is much of the point, and a model built on the deterministic path cannot show it.

Presenting the P80 as the answer. The confidence level is a choice belonging to whoever carries the consequence. Present the curve, state what each level costs in time, record the decision. Holding a target and a commitment at different levels is often right; doing so by accident never is.

Attributing a threshold to a standard. The choice of level is organisational policy and appetite. Where a convention exists — P50 internally with a higher level for external commitment is widespread — describe it as a convention of the organisation that set it.

Running it once, at sanction. The distribution is a snapshot of an assumption set that ages immediately. Re-run at gates, at re-baseline, and when a major register entry resolves either way.

— 8. Worked example

Illustrative figures. Durations in working days. Two approximations are used deliberately so the arithmetic is checkable by hand, and both are named where they occur.

8.1 Part A — a merge, and what the correlation assumption is worth

Four work fronts converge on a commissioning readiness milestone, each assessed as **70 % likely** to be ready on or before the target date.

If the four are independent, the milestone is met only if all four are met:

$$\begin{aligned} P(\text{all four ready}) &= 0.70 \times 0.70 \times 0.70 \times 0.70 \\ &= 0.49 \times 0.70 \times 0.70 \\ &= 0.343 \times 0.70 \\ &= 0.2401 \rightarrow 24.0 \% \end{aligned}$$

If the four are perfectly correlated — one shared driver, say a single commissioning contractor whose resourcing determines all four fronts — they are ready together or late together:

$$P(\text{all four ready}) = 0.70 \rightarrow 70.0 \%$$

The honest answer to "what is the chance we make the milestone?" lies between **24 %** and **70 %**, and which end it sits nearer is decided entirely by an assumption that appears nowhere on the schedule. No amount of extra simulation resolves this; only naming the drivers does. Note the direction: at a merge, independence is the pessimistic assumption — the opposite of the chain case in Part B.

8.2 Part B — a chain, with and without correlation

A four-activity chain on the critical path, each activity with the same three-point estimate in working days:

$$a \text{ (optimistic)} = 20 \qquad m \text{ (most likely)} = 30 \qquad b \text{ (pessimistic)} = 70$$

The range is right-skewed, as construction and commissioning ranges typically are.

Step 1 — the per-activity mean and spread, using the conventional weighting described in §2:

$$\begin{aligned}
 \text{mean} &= (a + 4m + b) \div 6 \\
 &= (20 + 4 \times 30 + 70) \div 6 \\
 &= (20 + 120 + 70) \div 6 \\
 &= 210 \div 6 \\
 &= 35.0 \text{ days} \\
 \\
 \text{standard deviation} &\approx (b - a) \div 6 \\
 &= (70 - 20) \div 6 \\
 &= 50 \div 6 \\
 &= 8.333 \text{ days}
 \end{aligned}$$

Note that the **most likely value of 30 days is not the mean of 35 days**: a schedule built on most-likely durations is already five days optimistic on this activity before any risk event is considered.

Step 2 — the deterministic chain:

$$\text{deterministic duration} = 4 \times 30 = 120 \text{ days}$$

Step 3 — the chain mean:

$$\text{chain mean} = 4 \times 35.0 = 140.0 \text{ days}$$

Step 4 — the chain spread if the four activities are independent. Variances add:

$$\begin{aligned}
 \text{variance per activity} &= 8.333^2 = 69.44 \\
 \text{chain variance} &= 4 \times 69.44 = 277.78 \\
 \text{chain std deviation} &= \sqrt{277.78} = 16.667 \text{ days}
 \end{aligned}$$

Step 5 — the chain spread if the four are perfectly correlated. Standard deviations add directly:

$$\text{chain std deviation} = 4 \times 8.333 = 33.333 \text{ days}$$

Step 6 — read percentiles. Approximation: the sum of the four durations is treated as normally distributed, deliberately, so the arithmetic can be checked by hand. The underlying activity distributions are right-skewed and a real simulation samples them directly. The standard normal value for the 80th percentile is 0.8416.

Independent case:

$$\begin{aligned}
 P50 &= 140.0 \text{ days} \\
 P80 &= 140.0 + (0.8416 \times 16.667) \\
 &= 140.0 + 14.03 \\
 &= 154.0 \text{ days}
 \end{aligned}$$

Perfectly correlated case:

$$\begin{aligned}
 P50 &= 140.0 \text{ days} \\
 P80 &= 140.0 + (0.8416 \times 33.333) \\
 &= 140.0 + 28.05 \\
 &= 168.1 \text{ days}
 \end{aligned}$$

Step 7 — how likely is the deterministic date? Independent case:

$$\begin{aligned}
 z &= (120 - 140.0) \div 16.667 = -20 \div 16.667 = -1.20 \\
 P(\text{chain} \leq 120 \text{ days}) &= \Phi(-1.20) \approx 0.115 \rightarrow \text{about } 11.5 \%
 \end{aligned}$$

8.3 Reading the results

Measure	Independent	Perfectly correlated
Deterministic duration	120 days	120 days
Mean / P50	140.0 days	140.0 days
P80	154.0 days	168.1 days
Chance of achieving the deterministic 120 days	$\approx 11.5 \%$	$\approx 27.4 \%$

The correlated case's 27.4 % is $\Phi((120 - 140) \div 33.333) = \Phi(-0.60) \approx 0.274$. Higher spread makes the optimistic outcome likelier too — a point lost when correlation is described only as "making things worse".

The mean is identical in both cases at 140 days. Correlation moved the P80 by $168.1 - 154.0 = 14.1$ days and the mean by nothing. Correlation is a statement about the tail.

The deterministic date is not a forecast. At 120 days it has roughly an 11.5 per cent chance in the independent model — not a criticism of the planner, but the consequence of building a chain from most-likely values when the ranges are skewed. The 20 days from 120 to the P50 is pure estimating asymmetry.

The gap to a funded P80 is large. In the independent case $154.0 - 120 = 34.0$ days, or $34 \div 120 = 28.3 \%$ of the deterministic duration — a decision to be taken with that number in front of it.

8.4 What this example excludes, and why that matters

The exclusions are as instructive as the arithmetic:

- **One chain only** — merge bias is excluded entirely, and Part A shows how large that effect can be.
- **No loaded risk events** — register entries would add discrete jumps and typically lengthen the upper tail more than the body.
- **Identical activities**, so the independent-case spread simplifies to a clean multiple. Real chains do not.
- **Normality assumed at the sum** (step 6). A right-skewed input generally puts the true P80 slightly later than the approximation suggests, so 154.0 is a floor rather than a precise value.
- **No calendars, resource constraints or weather windows**, all of which a production model carries and all of which typically extend the result.

Reporting exclusions is not a disclaimer. It is the difference between a number a board can act on and one a board will regret.

— 9. Checklist

Before the model is built

- [] Schedule quality checks run and recorded — logic, constraints, negative float, lags, out-of-sequence progress; constraint and dangle counts published with the basis.
- [] Model level of detail agreed and reconciled to the working schedule's critical path if different.
- [] Register current, cause-event-effect compliant, mapped to activities.
- [] Shared drivers listed by name, with the activities each one touches.
- [] Elicitation instruction written: what the ranges include and exclude.
- [] Basis document opened, not left to be written afterwards.

During elicitation

- [] Pessimistic value asked first, with a mechanism, from the people doing the work.
- [] Whether the estimator saw the scheduled duration recorded per activity.
- [] One definition of optimistic and pessimistic applied by everyone.
- [] Ranges sense-checked against outturn data where it exists.

Before the output is issued

- [] Deterministic date, P50 and P80 all reported, with the gaps explained.
- [] The full curve issued, not two points off it.
- [] Correlation assumptions stated, and the answer re-run with them varied.
- [] Sensitivity ranking published — which activities and register entries drive the tail.
- [] No entry both loaded and embedded in a duration range.
- [] Output rounded to a unit the inputs support; exclusions listed explicitly.
- [] The confidence level chosen for commitment recorded as a decision, with its owner.

A QSRA that passes this list will sometimes produce a date the project cannot accept. That is the analysis working. The alternative is a date the project accepts and cannot achieve, discovered when nothing can be done about it.

— Related

- [BPG-05 – Schedule quality – a practical review](#) — whether the network can be simulated at all
- [BPG-10 – Contingency and management reserve](#) — how a confidence level becomes funded time and money
- [BPG-16 – Risk registers that work](#) — the register discipline the loaded risk events depend on
- [TPL-11 – Quantitative schedule risk analysis input sheet](#) — field structure for elicitation and loading
- [TPL-14 – Schedule quality review checklist](#) — the precondition checks in usable form

- [AIG-05 – AI in scheduling – and what must not be automated](#) — where machine assistance helps here

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 10 — Project Scheduling, first authored draft, August 2026: schedule risk analysis, merge bias, health checks and the target-versus-commitment posture.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 12 — Risk Management for Project Controls, first authored draft, August 2026: risk loading, correlation and shared drivers, register-to-model traceability.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

The weighting $(a + 4m + b) \div 6$ and the spread approximation $(b - a) \div 6$ used in §8 are conventions from the programme evaluation and review technique tradition, used here because they are hand-checkable; no published standard is cited as their source, and no confidence level named in this guide is attributed to one. No industry statistic, benchmark range or software capability is cited, because none was verified.

— Status and version

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BPG-18 · BEST PRACTICE GUIDES

Interface and subcontractor controls

Controlling the work you do not do yourself

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Most of the scope on a large project is delivered by organisations that do not report to you and whose data arrives on their calendar, not yours. This guide covers the interface register and why an interface is not a milestone, the cadence mismatch between a subcontractor's period end and your cut-off, keeping subcontract obligations back-to-back with the head contract, progress claimed versus progress verified, and what to do when a subcontractor's own controls cannot produce the data your reporting depends on. The worked example separates one claim into a payment decision and a cost decision, and shows a swing of 0.065 on the cost performance index caused by a cadence mismatch alone.

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Interface and subcontractor controls

In one paragraph. Most of the scope on a large project is delivered by organisations that do not report to you and whose data arrives on their calendar, not yours. This guide covers the interface register and why an interface is not a milestone, the cadence mismatch between a subcontractor's period end and your cut-off, keeping subcontract obligations back-to-back with the head contract, progress claimed versus progress verified, and what to do when a subcontractor's own controls cannot produce the data your reporting depends on. The worked example separates one claim into a payment decision and a cost decision, and shows a swing of 0.065 on the cost performance index caused by a cadence mismatch alone.

Who this is for. Project controls managers and cost engineers on projects with substantial sub-contracted scope; commercial managers and quantity surveyors who certify applications; and planners managing interfaces between contractors, clients and third parties.

— 1. The control problem when the work is somebody else's

Direct work gives a controls function primary data: your timesheets, your goods-received notes, your progress measurement, on your cut-off. Subcontracted work gives you reported data — a claim prepared by an organisation with a commercial interest in the number, covering a period that ends when their accounting month ends, measured against rules they applied. Three properties follow, and every failure in this area is a version of one of them.

The data is an assertion until you verify it. An application is a starting position in a commercial conversation. Treating it as a measurement transfers the subcontractor's optimism straight into your forecast.

The data arrives on their calendar — a systematic mismatch between numerator and denominator that recurs every month in the same direction.

Their control weakness becomes your reporting weakness. No clause converts data you never received into a number you can report. Decide in advance whether you will measure their work yourself or report an estimate.

— 2. The interface register, and why an interface is not a milestone

An **interface** is a point where a deliverable, information, an access, a service or a physical connection passes between two organisations. Interfaces are where integrated projects fail, because both sides believe they are managing them and neither owns the gap. The register that manages them is not the milestone list: a milestone has one owner and one date, while **an interface has two owners, two schedules and a tolerance.** That determines the columns:

Field	Why it exists
Interface reference	A permanent identifier both parties quote; never renumbered
Parties and direction	The two organisations named, and who provides and who receives
What crosses	The deliverable, information set, access or connection, defined tightly enough to check
Acceptance criterion	What the receiver must be able to do with it before it counts as delivered
Required-by date	The receiver's need date, from their schedule
Committed date	The provider's date, from theirs
Float between them	Required-by minus committed; the interface's health, in days
Owner, each side	A named individual on the provider side and on the receiver side
Consequence of late	What it costs, in days or money, against the receiver's critical path
Status and last confirmed	When both parties last agreed the dates, not when the register was edited

Two columns do the real work. **Float** turns an interface from a status into a number, and a register sorted by it is a management agenda. **Consequence of late** makes the provider's organisation care.

Bilateral confirmation keeps the register alive. A date taken from one party's schedule and never confirmed by the other is a hope with a date field. Flag any interface not jointly confirmed within an agreed period regardless of how comfortable its float looks — comfortable float is the condition under which nobody checks. Interfaces with third parties under no contract — utilities, authorities, neighbouring projects, the client's own operations teams — are the ones most often missing from the register entirely, and with no commercial lever behind them the only control is early identification, long lead times and explicit contingency.

— 3. Cadence mismatch: their period end and your cut-off

Subcontractors report on their own commercial cycle: typically a claim to a fixed day of the month — the twentieth, say — while your cost cut-off is the last day of the month and your progress is measured then.

If the claim is posted unadjusted, the month's cost covers work to the twentieth and the month's earned value covers work to the thirtieth. The cost performance index — earned value divided by actual cost — then compares thirty days of output with twenty days of input, and it flatters. Next month the missing ten days arrive on top of a full month and the index drops for a reason unrelated to performance. The resulting sawtooth costs the controls function its credibility, because everyone can see it and the explanation sounds like an excuse.

Three responses, of which only one is generally right.

Move their cut-off. Contractually cleanest: require the subcontract's valuation date to coincide with the project's cut-off. Negotiate it at award, when it costs nothing; it is usually impossible later. Where their accounting calendar makes this impractical, require a progress return on your cut-off, separate from the commercial application.

Accrue the gap. Estimate the value of work performed between their period end and your cut-off and post it as an accrual, so cost and earned value describe the same period. [BPG-07 – Accruals and](#)

[cut-off discipline](#) owns the accounting treatment; the refinement that belongs here is that **the gap is estimated on the verified run rate, not the claimed one**, which otherwise imports the over-claim and compounds it.

Do nothing and explain it every month. Common, and not a control. If it really is the only option, publish the mismatch on the face of the report — "cost to 20th, progress to 30th". The same mismatch runs the other way on schedule: their progress update is stale on arrival, and their float is calculated against their own network, not yours.

— 4. Back-to-back obligations, and the gaps that are not

"Back-to-back" describes a subcontract whose obligations mirror the head contract's, so the main contractor does not carry an obligation upstream that it cannot pass down. The failures are all in the details, and they are all arithmetic on dates and percentages.

Notice regimes. If the head contract requires notice of a delaying event within a stated period and the subcontract allows a longer one, there is a window in which you can be time-barred upstream before the subcontractor is obliged to tell you anything. The exposure is the difference between the two periods, worth computing for each subcontract rather than assuming the drafting handled it. Notice periods, time bars and their consequences vary substantially between contract forms and jurisdictions; the arithmetic of comparing two periods does not.

Retention. Percentages and release triggers should mirror; where the subcontract releases retention earlier, the main contractor funds the difference. [BPG-13 – Cash flow forecasting](#) covers the consequence.

Payment timing. Paying subcontractors on shorter terms than you are paid on is a working-capital leak whose size is work in progress multiplied by the difference in days — sometimes a deliberate commercial decision, but one that should carry a number rather than be an accident of two separately negotiated documents. Statutory payment regimes constrain what may be agreed in many jurisdictions and differ between them; the principle of pricing the gap is general.

Change and variation mechanics. Both processes must be capable of carrying the same instruction, or changes arrive downstream that cannot be passed upstream; [BPG-11 – Change orders and variations](#) owns the process itself.

The controls function's role is not to draft the contract but to hold a **back-to-back matrix** — one table per subcontract comparing head-contract and subcontract obligation on each dimension, with the gap quantified and an owner for each gap that cannot be closed.

— 5. Progress claimed versus progress verified

A subcontractor's application states a value of work done; your measurement states a different one. Both may be prepared honestly — the systematic difference is the ordinary consequence of a party valuing its own output. Three quantities are in play, and conflating any two produces a reporting error.

Quantity	What it is	What it drives
Claimed value	The subcontractor's assertion	Nothing. Never posted to the cost ledger
Certified value	What your measurement supports at contract rates, less retention and previous payments	Payment, and through the certificate, the accrual
Earned value	Budgeted value of physical work performed under your rules of credit	Performance reporting

Certified value and earned value are close relatives, not the same quantity, and their difference is usually informative: a valuation typically pays for materials delivered to site, while rules of credit credit installed work only. That gap is a timing difference reversing on installation, and reporting it as a performance variance is the mistake the worked example illustrates. [BPG-06 – Progress measurement and rules of credit](#) owns the rules that make earned value mean something.

Agree the verification method at award, because retrofitting it is a negotiation. In descending order of reliability: joint measurement of installed quantities against a bill; witnessed milestone achievement against defined criteria; independent survey on a sample with a stated rule; and last, review of the subcontractor's own records — a review of an assertion rather than a measurement, and to be labelled as such wherever the number appears. **Verify on a sample with a rule, not on whatever looks suspicious**, because a sample chosen by suspicion cannot be extrapolated. And **record the over-claim, do not just remove it**: one month's over-claim is ordinary, a trend is a forecasting signal.

— 6. When the subcontractor's own controls are weak

Sooner or later you will engage an organisation that is good at the work and poor at the data: no coded cost structure, progress claimed as a percentage judgement, a schedule in a spreadsheet. It is your problem, because their scope appears in your report. The signs show in the first two claims: percentages moving in round numbers, progress tracking the payment curve exactly, a schedule with no logic behind the bars, and no answer to what the remaining work consists of. The responses that work, roughly in order of cost:

- **Specify the data at award.** A short data schedule — format, fields, frequency, cut-off, consequence of non-delivery — costs nothing at award and is unobtainable afterwards. Specify the fields, not the tool; mandating a software package excludes capable subcontractors and does not by itself produce good data.
- **Measure and schedule it yourself.** Take the quantity survey in-house and maintain your own network for their scope, updated from physical observation. It is duplication, it should be an open resourcing decision rather than an overload absorbed by one cost engineer, and it is sometimes the only way to keep the interface dates honest.
- **Tie data to money.** Where the contract allows, make a compliant progress return a condition of a valid application — the strongest lever available, but applied as a standing requirement from the first month, because a data requirement introduced during a dispute reads as a payment tactic.
- **Escalate as a commercial matter.** Weak controls raise the probability of latent overrun, un-notified delay and disputed final accounts: a risk-register entry with an owner and an exposure.

Whatever the combination, record **the basis of the number that reaches your report**. A cost line from your own measurement and one from an assertion are different qualities of information.

— 7. How this goes wrong

The claim is posted as the cost. No verification, no accrual for the cadence gap, no distinction between claimed and certified. Everything downstream inherits the subcontractor's commercial position. The refined version of the same error removes the over-claim from the certificate and reintroduces it through an accrual calculated on the claimed run rate.

Interfaces are recorded as milestones. One owner, one date, no float column, no consequence. When the date slips, both parties can demonstrate they were waiting for the other.

Third-party interfaces are missing, because nobody in the delivery team has a contract with the utility, the authority or the neighbouring project — usually the longest-lead dependencies on the job.

Back-to-back is assumed, not checked. The two documents were reviewed by different people at different times, and the notice-period and retention differences surface when a claim arrives.

Materials on site are read as a performance variance. The index dip is investigated as a productivity problem, and next month it reverses without explanation (\$8.6).

Their schedule is filed rather than integrated. A bar chart never linked into the project network cannot tell you what an interface slip costs, and float comfortable in their network may not exist in yours.

Weak controls are managed as a relationship issue. Raised at progress meetings, never entered as a risk, never priced, never escalated — until the final account arrives with no supporting records on either side.

— 8. Worked example

Illustrative figures. Currency USD, whole dollars. One subcontract package, viewed by the main contractor. Assumptions stated as they arise; percentages to one decimal place.

8.1 The setup

Item	Value
Subcontract value	4,800,000
Programme	12 months from 1 January, planned value spread evenly
Subcontractor's period end	20th of the month
Main contractor's cut-off	Last day of the month
Retention	5 %, back-to-back with the head contract
Previously certified, net of retention	1,600,000

At 20 June the subcontractor applies for a cumulative gross value of **2,150,000**; your quantity surveyor's measure supports **2,020,000**. Measured cumulative value at 20 March was **820,000**. Earned value at 30 June under the project's rules of credit is **2,100,000**.

8.2 The over-claim

```
over-claim = claimed - measured
            = 2,150,000 - 2,020,000
            = 130,000

as a proportion of the claim = 130,000 ÷ 2,150,000 = 6.0 %
```

8.3 The payment decision — the certificate

```
net certified to date = measured × (1 - retention)
                     = 2,020,000 × 0.95
                     = 1,919,000

this certificate      = net certified to date - previously certified net
                     = 1,919,000 - 1,600,000
                     = 319,000
```

Had the application been certified as submitted:

```
2,150,000 × 0.95 = 2,042,500
2,042,500 - 1,600,000 = 442,500
difference = 442,500 - 319,000 = 123,500
```

which reconciles to the over-claim net of retention: $130,000 \times 0.95 = 123,500$. Certifying the application rather than the measure would have paid **123,500** more than the work supports.

8.4 The cost decision — closing the cadence gap

The certificate covers work to 20 June; the cost report closes on 30 June. The gap is **10 days**. The run rate comes from verified measure over the three preceding periods, not from the claims:

```
measured growth = 2,020,000 - 820,000 = 1,200,000 (20 March to 20 June)
per month       = 1,200,000 ÷ 3 = 400,000
per day (30-day) = 400,000 ÷ 30 = 13,333.33

gap accrual = 13,333.33 × 10 = 133,333
```

Cost to be recognised at 30 June for this package:

```
2,020,000 (certified measure) + 133,333 (accrued 21–30 June) = 2,153,333
```

A coincidence worth noticing, because it is a trap. The 2,153,333 is within 3,333 of the 2,150,000 claimed, so posting the claim would have been almost exactly right this month — which is why the shortcut survives. But the two figures are right for different reasons and composed differently: 2,153,333 is 2,020,000 of verified work plus 133,333 of estimated work, and that composition supports three separate actions — certify 319,000, accrue 133,333 with no payment attached,

pursue the 130,000 over-claim. The single figure supports none of them, and the agreement between the two will not survive a change in either the over-claim or the run rate.

8.5 What it does to the reported indices

Assumption: planned value is spread evenly across twelve months from 1 January, so planned value at 30 June is six months of 400,000. SPI is the schedule performance index, earned value divided by planned value.

```
PV (planned value) = 6 × 400,000 = 2,400,000
EV (earned value)  = 2,100,000
AC (actual cost)   = 2,153,333

SPI = EV ÷ PV = 2,100,000 ÷ 2,400,000 = 0.875
CPI = EV ÷ AC = 2,100,000 ÷ 2,153,333 = 0.975
```

The same month with the cadence gap ignored — certificate posted, no accrual:

```
AC = 2,020,000
CPI = 2,100,000 ÷ 2,020,000 = 1.040
swing = 1.040 - 0.975 = 0.065
```

A swing of 0.065 on the cost performance index, produced entirely by a ten-day mismatch between two calendars. On a tolerance band a few hundredths wide that is the difference between green and an exception report — and next month, when the missing ten days land on top of a full period, the index falls for a reason unconnected to how the work was performed.

8.6 Reading both indices correctly

The 0.975 is not a productivity problem. The recognised cost of 2,153,333 is a valuation basis; the earned value of 2,100,000 is a rules-of-credit basis, and the difference:

```
2,153,333 - 2,100,000 = 53,333
```

is material delivered to site and paid for, which the valuation includes and the rules of credit do not credit until installed. It is a timing difference that reverses on installation, and the narrative should say so.

The schedule reading is unaffected by any of the cost treatment above:

```
measured completion          = 2,020,000 ÷ 4,800,000 = 42.1 %
planned completion at 30 June = 2,400,000 ÷ 4,800,000 = 50.0 %
SPI = 0.875
```

A schedule performance index of 0.875 with a 6.0 per cent over-claim on the latest application is a specific management situation — behind, and valuing optimistically, which is the ordinary sequence — and the moment to re-confirm the interface dates downstream with the parties that depend on them.

— 9. Checklist

At award

- [] Valuation date aligned to the project cut-off, or a progress return required on the cut-off.
- [] Data schedule in the subcontract: fields, format, frequency, cut-off, consequence of non-delivery.
- [] Verification method stated — joint measurement, witnessed milestone, sampled survey or record review.
- [] Back-to-back matrix completed: notice periods, retention, payment timing, change mechanics — each gap quantified, each with an owner.
- [] Interface points identified, including third parties with no contract.
- [] Rules of credit agreed, distinct from the valuation basis.

Every period

- [] Claim measured before it is certified; over-claim recorded, not just removed.
- [] Certificate computed from measure, retention and previous payments, with the arithmetic shown.
- [] Cadence gap accrued, using the verified run rate.
- [] Cost basis labelled: measured, certified, accrued or asserted.
- [] Materials on site identified and excluded from the performance narrative.
- [] Subcontractor's progress update integrated into the project network, not filed.
- [] Interface register updated with dates confirmed by both parties, and float recomputed.
- [] Over-claim trend reviewed across periods, not only this month's figure.

Every quarter, or at each gate

- [] Interfaces with no bilateral confirmation in the agreed period flagged regardless of float.
- [] Subcontractors with weak controls carried as named risk entries with an exposure.
- [] Unclosed back-to-back gaps reviewed for whether they have been triggered.

Projects rarely lose control of subcontracted scope through one large failure. They do it through a claim posted as a cost, an interface with one owner and a notice period seven days out of step — each small, each invisible in the report, and all of them arriving together in the final account.

— Related

- [BPG-06 – Progress measurement and rules of credit](#) — what makes earned value on subcontracted scope mean something
- [BPG-07 – Accruals and cut-off discipline](#) — the accounting treatment behind the gap accrual in §8.4
- [BPG-11 – Change orders and variations](#) — the change process that must carry an instruction both ways
- [BPG-12 – Claims and extension of time](#) — where notice-period gaps and unpassed subcontractor claims end up

- [BPG-13 – Cash flow forecasting](#) — the working-capital effect of terms that are not back-to-back
- [BPG-05 – Schedule quality – a practical review](#) — the checks to apply before integrating their schedule

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 7 — Contracts, Commercial Management, BoQ, Invoicing and Revenue, first authored draft, August 2026: interim valuation and certification, back-to-back subcontract discipline, and the reconciliation of earned value, valuation and revenue.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 5 — Cost Management and Cost Control, first authored draft, August 2026: commitment and accrual capture, and the cost ledger's relationship to certification.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 10 — Project Scheduling, first authored draft, August 2026: integration of contractor schedules and float across networks.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

Notice periods, time bars, retention mechanics, payment terms and statutory payment regimes vary between contract forms and jurisdictions. This guide names no contract form and reproduces no clause; the arithmetic is to be applied to the documents in force on the reader's project.

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BPG-19 · BEST PRACTICE GUIDES

Project controls assurance and health checks

Testing whether the numbers can be relied on, early enough for the answer to matter



VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

A project controls health check tests whether a project's control system is capable of producing a number a decision can rest on. This guide sets out what such a review examines, how to score it on evidence rather than impression, how much independence the reviewer needs and what to do when full independence is unavailable, the difference between assurance and audit, why timing determines whether findings can be acted on at all, and how to write findings that are accepted rather than resented. The worked example scores a review and then demonstrates, arithmetically, why a composite score cannot express a blocking weakness.

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Project controls assurance and health checks

In one paragraph. A project controls health check tests whether a project's control system is capable of producing a number a decision can rest on. This guide sets out what such a review examines, how to score it on evidence rather than impression, how much independence the reviewer needs and what to do when full independence is unavailable, the difference between assurance and audit, why timing determines whether findings can be acted on at all, and how to write findings that are accepted rather than resented. The worked example scores a review and then demonstrates, arithmetically, why a composite score cannot express a blocking weakness.

Who this is for. PMO leads and heads of project controls who commission or run reviews; controls managers who are reviewed; and sponsors and assurance committees who have to decide what a review's output entitles them to conclude.

— 1. The question a health check is actually asking

A project controls health check is not an inspection of whether procedures are being followed. It asks one question: **can this project's control system produce a number that a decision can safely rest on, and if not, exactly where does the chain break?**

That framing does most of the work. It rules out the compliance sweep — a checklist of documents that exist, signed by people who exist — because a project can have every artefact the framework requires and still be incapable of telling you what it will cost. It also rules out the opinion piece, in which an experienced reviewer walks the project for two days and reports what they felt. Both are common; neither can be challenged, defended or acted on with any precision.

The chain being tested runs in one direction, and each link depends on the ones before it:

Scope is defined and structured — a breakdown that the budget, the schedule and the cost ledger all share. Without a common structure, nothing downstream reconciles and every comparison is manual.

A baseline exists and is controlled — approved, dated, with a change process that is actually used. Without it there is nothing to measure against, and variance analysis is comparison with the most recent opinion.

Cost is captured completely and on time — commitments, actuals and accruals, at the cut-off, coded to the structure. Without this, every efficiency measure has an unreliable denominator.

Physical progress is measured against rules — documented rules of credit, applied consistently, verified independently on a sample. Without this, earned value is a claim.

Forecasting is method-based and defensible — a stated method, chosen for a reason connected to the cause of the variance, with the assumption written next to the answer.

Risk and change are live — a register that changes decisions and a change process that captures scope movement before it is executed rather than after.

Reporting reaches the decision — the right measure, at the right cadence, to the person who acts, with exceptions visible.

A review that walks this chain in order produces findings that are self-prioritising, because a break early in the chain invalidates everything after it. That property is worth more than any scoring scheme, and §8 shows why the scoring scheme cannot substitute for it.

— 2. Assurance and audit are different instruments

The two words are used interchangeably and should not be. The distinction is not organisational politics; it changes what the reviewer may conclude and what the project is obliged to do.

	Audit	Assurance / health check
Question	Was the defined requirement complied with?	Will this control system produce a reliable number?
Criterion	An existing rule — procedure, contract, standard, policy	Fitness for the decisions the project must support
Output	Exceptions against the rule	Capability, with the specific weaknesses that limit it
A project can fail by	Not following the process	Following the process, where the process is inadequate
Relationship to the team	Independent testing; findings are not negotiable in substance	Advisory; findings are negotiable in how, not in whether
Typical timing	After the fact	Early enough to change the outcome

The consequential difference is the row about failing while compliant. An audit against a controls procedure can find full compliance on a project whose forecast is meaningless, because the procedure never required the forecast method to be justified. Assurance is the instrument that catches that, and it can only do so if its criterion is fitness rather than conformance.

Both are legitimate and projects need both. What causes damage is running one and reporting it as the other: an audit dressed as assurance produces a clean report on an uncontrolled project, and assurance dressed as an audit produces findings the team treats as non-negotiable when they are in fact recommendations, which is where the resentment in §6 begins.

— 3. Evidence-based scoring

A score that cannot be traced to something the reviewer inspected is an opinion with a number attached. It will be argued with, and it will be argued with successfully, because there is nothing behind it.

Three requirements make a score defensible.

A published rubric, written before the review. Each element has descriptors for each score, expressed as observable conditions rather than adjectives. Compare these two rubrics for progress measurement:

Not defensible: 0 poor, 1 weak, 2 adequate, 3 good, 4 excellent.

Defensible:

Score	Condition
0	No documented rules of credit. Progress is a judgement, recorded as a percentage.
1	Rules of credit exist as a document but progress is claimed by percentage judgement in practice.
2	Rules of credit documented and applied on major packages; no independent verification.
3	Applied across the measured scope, with independent verification on a defined sample.
4	As 3, and physical progress, valuation and earned value are reconciled each period with differences explained.

The second version can be evidenced, disputed on facts, and re-tested at the next review by a different person who will reach the same answer.

Evidence recorded per score: the artefact, its date, the test applied. "Progress measurement scored 2" is an assertion. "Progress measurement scored 2: rules of credit document rev C dated 14 March covers packages 100–400; the April progress return for package 500 records 62 per cent with no supporting quantity record; no evidence of independent verification in the last three periods" is a finding. It also survives the reviewer leaving the organisation, which the assertion does not.

A stated sample, with its coverage expressed two ways. A review that inspects a handful of control accounts should say which, why those, how many out of how many, and what proportion of value they represent. Coverage by count and coverage by value are different numbers and both are needed — §8 shows a case where they are 8.6 per cent and 65.3 per cent respectively, and reporting only the first would understate the review while reporting only the second would overstate it.

Two disciplines protect the scoring from itself. **Score the element, not the person** — a low score for progress measurement on a project whose measurement was never resourced is a finding about resourcing, and the finding should say so. And **agree the condition before writing the conclusion**: show the project the factual observations, correct any that are wrong, and only then write cause and consequence. Almost every dispute about a review turns out to be a dispute about a fact that could have been corrected in ten minutes.

— 4. Independence, and what to do without it

The reviewer must not be assessing their own work, and must not report through the person whose project is being reviewed. Both conditions are frequently breached, usually with good intentions: the most knowledgeable person available is the one who helped build the thing.

The threats worth naming, because naming them is most of the management:

Self-review — the reviewer designed the process, the template or the baseline being examined. They cannot find it wanting without finding themselves wanting.

Reporting line — the reviewer's next assignment, appraisal or invoice depends on the person receiving the findings.

Familiarity — a long relationship with the team makes the uncomfortable finding costly to write, and makes it easy to accept an explanation that would not be accepted from a stranger.

Advocacy — the reviewer has previously argued publicly for the approach in use.

Full independence is often unavailable, particularly in smaller organisations. The professional answer is not to pretend, and not to refuse. It is to **declare the impairment on the face of the report, restrict the conclusions accordingly, and add a compensating control**. Pair the internal reviewer with someone from another project or another function. Have a second person re-perform the scoring on the same evidence. Exclude from the reviewer's scope the specific elements they built, and say in the report which elements were excluded and why. A report that states "the reviewer designed the reporting template; element 6 was scored by a reviewer from the adjacent programme" is more credible than one that claims an independence nobody in the room believes.

One more condition is often forgotten: **the reviewer needs access, and access is granted, not assumed**. Access to the cost system rather than to extracts prepared for the review, to the schedule file rather than a printed bar chart, to the change log, to the register, and to people without their manager in the room. Access limitations are themselves a finding and should be reported as one.

— 5. Timing: early enough to matter

The value of a health check falls sharply with elapsed project time, and the reason is structural rather than motivational.

Most controls weaknesses are foundational: a coding structure that does not support the reporting the project needs, rules of credit that were never written, a baseline that was never properly approved. The cost of fixing a foundational weakness scales with everything built on top of it. Correcting a coding structure in month two means recoding two months of transactions. Correcting it in month fourteen means recoding fourteen months of transactions, restating every report issued from them, and explaining to a board why the trend they have been watching has changed shape. The second is not fourteen times harder than the first; it is often simply not done, and the project runs to completion on a structure everyone knows is wrong.

Three timing points earn their cost:

Before baseline acceptance. The single highest-value review. Everything is still cheap to change, nothing has been reported externally, and the review's findings can become conditions of baseline approval — which is the only mechanism that reliably gets them fixed.

At the first reporting cycle after mobilisation. The first month-end is when the design meets reality. Reviewing the first cycle catches the gap between the controls plan and what the team can actually execute, while the correction is a process change rather than a restatement.

At each gate, and on trigger. Triggers worth defining in advance: a change of controls manager, a re-baseline, the first month a forecast moves by more than a defined tolerance, and the mobilisation of a major subcontract package.

A review at sixty per cent complete can still be worth doing — but its honest purpose is different. It is not improving the control system; it is establishing what the current numbers can and cannot support, so that decisions taken from here are taken with the right level of confidence. Reviewers should say which of those two jobs they are doing, because a late review reported as an improvement exercise generates recommendations nobody has the time or the appetite to implement.

— 6. Writing findings that get acted on

The difference between a finding that is implemented and one that is resented is mostly structure. Five components, in this order:

Condition — what is, factually, with the evidence and its date. No adjectives. This is the part the project must be given the chance to correct before anything else is written.

Criterion — what should be, and whose rule that is. "The project's own controls execution plan section 4.2 requires..." is far stronger than "good practice requires...", because it removes the argument about whose standard applies. Where the criterion genuinely is the reviewer's professional judgement, say so and give the reasoning.

Cause — why the gap exists, at the level of process, resourcing or design rather than person. "The progress measurement role was not filled until month four" is a cause. "The cost engineer did not prioritise this" is an accusation, and it converts a finding into a personnel dispute in which the substance is lost.

Consequence — what it costs in decisions, not in abstract risk. "The forecast presented to the September board cannot be traced to a stated method, so its sensitivity to the steelwork productivity assumption is unknown" tells a sponsor what they are exposed to. "This represents a significant risk to project delivery" tells them nothing and will be read as boilerplate.

Correction — the specific act, its owner and its date. If the reviewer cannot name the act, the finding is not ready; a finding whose correction is "improve the process" will be closed by an assertion that the process has been improved.

Four further rules that determine whether the report changes anything:

Rank by consequence, not by severity adjective. Three findings ranked by what they prevent the project from knowing beats fifteen ranked as high, medium and low.

Cap the count. A report with forty findings will have none of them fixed. If forty genuinely exist, report the five that block the chain and list the rest as observations with no owner.

Route each finding to the person who can fix it. Some findings are not the project's to fix — a corporate coding structure, an unresourced role, a template mandated by the PMO. Sending them to the project guarantees they stay open.

Track to closure with evidence. A finding is closed when the reviewer sees the changed artefact, not when the owner reports that it is done. Publishing the closure rate is what stops the next review starting from the same place.

— 7. How this goes wrong

The compliance sweep sold as assurance. Every required document exists, so the project scores well, and the forecast is still unusable. The tell is a report whose findings are all about documents and none about numbers.

The score without the rubric. Numbers assigned on impression, disputed by the project, defended by seniority. The next review reaches different scores on an unchanged project, and the trend is meaningless.

The composite score as the deliverable. A single percentage travels well and hides everything the review found. §8 demonstrates arithmetically why it cannot represent a blocking weakness.

No sample stated. "We reviewed the cost reports" — how many, which, chosen how, covering what value. Without those, the review cannot be repeated or extended.

Independence asserted rather than analysed. The report claims independence; the reviewer built the template. Declaring the impairment costs a paragraph; being found to have concealed it costs the review.

Findings written as accusations. Cause stated at the level of a person. The project's response becomes a defence of individuals, the substance is never discussed, and the next review is obstructed before it starts.

Corrections that cannot be verified. "Strengthen the change control process." Closed at the next review on the strength of an assertion, because there is no artefact to inspect.

Too late to matter, and not said. A review at sixty per cent complete reported as though its findings could still be implemented. The project agrees to everything and does none of it, which teaches everyone that assurance findings are optional.

Findings routed to the wrong owner. Corporate-level causes sent to a project team with no authority to change them, then reported as unresolved project failures at the next cycle.

The review that is never closed out. Findings raised, actions agreed, nobody tracks them, the next review raises the same findings. Reporting the closure rate from the previous review is the cheapest correction available and the most often omitted.

— 8. Worked example

Illustrative figures. One project controls health check at a single review date. Scores are on a 0–4 scale against a published rubric. Weights sum to 100 per cent. All arithmetic shown; percentages rounded to whole numbers where stated.

8.1 The sample and its coverage

The project has **140 control accounts** with a total budget of **9,500,000** (USD, illustrative). The review inspected the **12 largest by budget**, carrying **6,200,000**.

```
coverage by count = 12 ÷ 140 = 0.0857 → 8.6 %
coverage by value = 6,200,000 ÷ 9,500,000 = 0.6526 → 65.3 %
```

Both numbers belong in the report. Coverage by count alone reads as a thin review; coverage by value alone conceals that 128 accounts — including every small account, where control weaknesses concentrate — were not examined. The sampling rule ("the twelve largest by budget") should also be stated, because it is not random and its bias is knowable: it systematically excludes the small, unglamorous accounts, and a future review should sample differently for exactly that reason.

8.2 The scores

#	Element	Weight	Score (0-4)	Weighted contribution
1	Baseline integrity and change control	20 %	3	$0.20 \times 3 = 0.60$
2	Schedule quality	20 %	2	$0.20 \times 2 = 0.40$
3	Cost and commitment capture	15 %	3	$0.15 \times 3 = 0.45$
4	Progress measurement	15 %	1	$0.15 \times 1 = 0.15$
5	Forecasting	15 %	2	$0.15 \times 2 = 0.30$
6	Risk and change management	15 %	2	$0.15 \times 2 = 0.30$
Total		100 %		2.20

weight check = $20 + 20 + 15 + 15 + 15 + 15 = 100 \%$
 composite = $0.60 + 0.40 + 0.45 + 0.15 + 0.30 + 0.30 = 2.20$
 as a percentage of the 4.00 maximum = $2.20 \div 4.00 = 0.550 \rightarrow 55 \%$

8.3 Why 55 per cent is not the finding

Element 4, progress measurement, scored **1**: rules of credit exist as a document but progress is claimed by percentage judgement in practice, with no independent verification in the last three periods.

Every subsequent number on this project is computed from that element. Earned value is derived from measured progress. The cost performance index and schedule performance index are derived from earned value. The forecast is derived from those indices. A score of 1 on element 4 therefore does not mean "one element is weak"; it means the project's cost performance measurement, its schedule performance measurement and its forecast are all resting on unverified percentage judgements.

The composite cannot express this, because a weighted average treats the elements as independent and additive. They are neither.

8.4 Testing whether a dependency rule rescues the composite

A common remedy is to add a rule: **no element may be scored above the lowest score of the elements it depends on**. Forecasting (element 5) depends on progress measurement (element 4, scored 1) and cost capture (element 3, scored 3), so it is capped at 1 and falls from 2 to 1.

revised composite = $0.60 + 0.40 + 0.45 + 0.15 + (0.15 \times 1) + 0.30$
 = $0.60 + 0.40 + 0.45 + 0.15 + 0.15 + 0.30$
 = 2.05

 as a percentage = $2.05 \div 4.00 = 0.5125 \rightarrow 51 \%$

The composite moves from 55 per cent to 51 per cent — a change of four percentage points to represent a project whose entire performance measurement chain is unverified.

This is the point of the calculation, and it is not a failure of the rule. A single weighted average is structurally incapable of expressing a blocking weakness, because averaging is a com-

pensating operation: strength elsewhere always offsets weakness here. Improving the rule cannot fix a property of the arithmetic.

8.5 What the deliverable should therefore be

Three things, in this order:

The blocking finding, stated first. Element 4 scored 1. Earned value, both performance indices and the forecast presented to the board are derived from unverified percentage judgements. Until independent verification is in place on a defined sample, no performance figure from this project should be relied on for a funding decision.

The element profile, as a labelled chart or table with the rubric descriptor next to each score — never colour alone, and never a single dial.

The composite, last, and only as a trend line for this project against itself. Comparing 55 per cent here with 55 per cent on another project is meaningless unless both used the same rubric, the same weights and a comparable sample, which is rarely true and almost never checked.

8.6 Closure of the previous review

The previous review raised **14 findings**. At this review: **5** closed with evidence, **6** open with revised dates, **3** open with no movement.

```
check: 5 + 6 + 3 = 14
closure rate = 5 ÷ 14 = 0.357 → 35.7 %
```

A closure rate of 35.7 per cent with three findings showing no movement at all is itself a finding, and it belongs in the report ahead of any new observation — because a project that does not close findings will not close the ones raised today either, and that fact changes what this review can reasonably expect to achieve.

9. Checklist

Commissioning the review

- [] The question is stated as fitness for decisions, not conformance to procedure — or, if an audit is wanted, it is called one.
- [] Rubric published before the review, with observable descriptors for every score.
- [] Weights published and summing to 100 per cent.
- [] Sampling rule agreed in advance, with coverage to be reported by count and by value.
- [] Reviewer's independence analysed against self-review, reporting line, familiarity and advocacy; any impairment declared in the report.
- [] Access agreed: source systems, not extracts; people, not only managers.
- [] Timing justified — before baseline, first cycle, gate or trigger — and the review's purpose stated accordingly.
- [] Previous review's findings retrieved and their closure status tested first.

Conducting it

- [] The chain walked in order: structure, baseline, cost capture, progress, forecast, risk and change, reporting.
- [] Each score evidenced by artefact, date and test applied.
- [] Factual conditions agreed with the project before cause and consequence are written.
- [] Access limitations recorded as findings.
- [] Findings capped, ranked by consequence, and each routed to the person able to fix it.

Reporting it

- [] Blocking finding stated first, before any score.
- [] Element profile published with descriptors; no meaning carried by colour alone.
- [] Composite reported last, and only as a trend against this project's own history.
- [] Every finding carries condition, criterion, cause, consequence and correction with an owner and date.
- [] No correction is written that cannot be verified by inspecting a changed artefact.
- [] Closure rate from the previous review published.

A review run this way produces fewer findings, and the project argues with almost none of them — because the facts were agreed before the conclusions were written and every correction names something specific. The measure of whether it worked is not the score. It is whether the next review can start somewhere new.

— Related

- [TPL-15 – Project controls health check](#) — the review instrument, rubric structure and scoring sheet
- [BPG-05 – Schedule quality – a practical review](#) — the schedule element of the chain, examined in depth
- [TPL-14 – Schedule quality review checklist](#) — the schedule checks in usable form
- [BPG-04 – Baselining and baseline change control](#) — the baseline integrity element, and why pre-baseline review is the highest-value timing
- [BPG-14 – Monthly reporting that gets read](#) — the reporting element of the chain and what "reaches the decision" means
- [BPG-16 – Risk registers that work](#) — the register health metrics a reviewer can compute in minutes

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 10 — Project Scheduling, first authored draft, August 2026: schedule health-check metrics and the schedule basis document.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 8 — Project Management Lifecycle, first authored draft, August 2026: stage-gate design, gate conditions tracked to closure, and the PMO's assurance role.

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 4 — Performance Management, Variance Analysis and Management Reporting, first authored draft, August 2026: measure design, gaming patterns and the countermeasure of auditing the measure rather than only the result.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

No maturity model, assurance framework or published review methodology is reproduced or cited here, and no scoring threshold in this guide is attributed to any external body. The weights, scale and rubric in §8 are illustrative and were constructed for teaching; an organisation adopting them should set its own and publish them before its first review.

— Status and version

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BPG-20 · BEST PRACTICE GUIDES

Closeout, lessons learned and benchmarking

Collecting what the project is owed, and what it knows, before both disperse

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Closeout is where a project's remaining money and its entire accumulated knowledge are either collected or lost, and both are usually lost for the same reason: the people who hold them have already moved on. This guide covers commercial and financial closeout and the final account, the open-items position that keeps a project's reported margin provisional, why lessons-learned registers are almost universally useless and the four properties that fix them, and how to capture benchmark data that the next estimate can actually use. The worked example reconciles a final account and then shows a productivity rate that reads as 11.1 per cent over estimate or 2.2 per cent under it, depending only on which hours were counted.

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Closeout, lessons learned and benchmarking

In one paragraph. Closeout is where a project's remaining money and its entire accumulated knowledge are either collected or lost, and both are usually lost for the same reason: the people who hold them have already moved on. This guide covers commercial and financial closeout and the final account, the open-items position that keeps a project's reported margin provisional, why lessons-learned registers are almost universally useless and the four properties that fix them, and how to capture benchmark data that the next estimate can actually use. The worked example reconciles a final account and then shows a productivity rate that reads as 11.1 per cent over estimate or 2.2 per cent under it, depending only on which hours were counted.

Who this is for. Project controls managers and cost engineers running a project to closure; commercial managers settling final accounts; and estimating and PMO functions who depend on what closeout hands them.

— 1. Closeout starts before the work does

The characteristic failure of closeout is that it is scheduled as a phase and staffed as an afterthought. The team demobilises on completion, the strongest people leave first because they are wanted elsewhere, and the residual work — final account, retention recovery, record assembly, knowledge capture — falls to whoever is left, at exactly the moment the project's budget for staff has closed.

Two structural corrections do most of the work, and both are planning decisions rather than closeout decisions.

Fund and resource closeout in the baseline. A named budget line and named people, with an end date that is later than the completion date. If closeout has no budget, it will be done by people who are also doing something else, which is a reliable way to lose money on the last two per cent of a project.

Schedule the closure deliverables backwards from the last commercial event, not from practical completion. The final retention release, the last defect certificate and the last subcontract final account often sit many months after the asset is handed over. The project is not closed when the client takes the asset; it is closed when the last receivable is collected and the last liability is extinguished.

A third correction is cultural and harder: **collect while the evidence is warm.** Records, quantities, photographs, correspondence and the reasoning behind decisions are all cheap to capture during execution and expensive to reconstruct afterwards. The reconstruct-later approach fails specifically at closeout, because the person who knew has gone.

— 2. Commercial and financial closeout

Commercial closeout settles the position with every party. Financial closeout settles the position in the books. They are related and not identical, and running them as one exercise causes a project to be reported as closed while money is still outstanding.

Commercial closeout covers, at minimum: agreement of the final account with the client; agreement of every subcontract final account; resolution or provision of every claim in both directions; release of retention on both sides; discharge or expiry of bonds, guarantees and warranties; settlement of contra-charges and back-charges; and closure of insurances at the appropriate date rather than at handover.

Financial closeout covers: reversal of accruals that will never be invoiced; release of provisions that are no longer needed and recognition of those that are; final cost allocation; closing the cost collection structure so nothing further can be posted; and archiving the records that support every number.

Two disciplines are worth stating explicitly because they are where reported margin most often moves after a project is declared complete.

No provision is released without a decision and a date. Provisions carried against unsettled subcontractor accounts, latent defects or disputed variations are judgements. Releasing them quietly to improve a final result is a reporting failure and, depending on materiality and jurisdiction, potentially more than that. The correct treatment is a documented decision with the basis stated.

The cost structure is closed, not just abandoned. An open cost code on a closed project will receive postings — recharges, late invoices, corrections — and they will land on a project nobody is monitoring. Close the codes and route genuine late items through a named person.

Accounting treatment, tax treatment, retention mechanics, defects-liability periods and statutory payment regimes differ substantially between jurisdictions and contract forms. What is general is the discipline: a closeout position is not complete until every open item has a value, an owner and an expected settlement date.

— 3. The final account and the open-items position

The final account is the last definitive reconciliation of scope, cost, billing and entitlement for a contract. Its components are owned by other guides — [BPG-11 – Change orders and variations](#) for the variation account, [BPG-12 – Claims and extension of time](#) for the claim settlement — and what belongs here is the assembly and what it leaves outstanding.

The assembly is straightforward arithmetic and is worked in §7: original value, plus approved variations, adjusted for remeasurement of provisional or approximate quantities, plus agreed claim settlements, less contra-charges and back-charges. What follows the assembly is where projects go wrong.

Retention is a receivable, not a rounding. It is typically released in stages, and the last stage usually falls after a defects-liability period that has no reason to align with anyone's financial year. Track each stage with its own trigger, its own date and its own owner. The project is not closed while it is outstanding.

Every open item carries a value and an exposure. Subcontractor final accounts still in negotiation, disputed variations, unresolved back-charges and outstanding defects each have a claimed value and a provided value. The difference is the exposure, and the project's reported margin is provisional by exactly that amount. Publishing the open-items table with the claimed, provided and exposure columns is the single most useful thing closeout reporting does, because it converts "final result subject to closeout" into a number.

The final account is the reconciliation point for everything upstream. It should tie to the change log, to the cost ledger, to the billing record and to whatever was recognised as revenue — and where it does not tie, the difference should be explained rather than adjusted. Differences between measured value, earned value and recognised revenue are usually legitimate and informative; making them disappear destroys the information.

— 4. Why lessons-learned registers fail

Almost every organisation has lessons-learned registers. Almost none of them changes what the next project does. The failure is systematic enough to be diagnosed precisely, and every element of the diagnosis points at the same remedy.

They are written at the end, by the people who are left. The lesson belongs to whoever was there when it happened, and by closeout that person has moved. What gets written is what the remaining team remembers, which is systematically the recent, the visible and the uncontroversial.

They are written as feelings about the past. "Communication with the client could have been better." "Earlier engagement with the supply chain would have helped." These are true, unactionable, and identical across every project in the world. Nothing in them tells anyone what to do differently.

They have no addressee. A lesson about design co-ordination belongs to the person who owns the design management process. A lesson about schedule templates belongs to the head of planning. A register that addresses nobody is filed, and filing is where lessons go to die.

They have no route into a process. A lesson that does not change a document — a template, a checklist, a procedure, a standard scope of work, an estimating norm — changes nothing at all. Knowledge that lives only in a register is knowledge that must be searched for by someone who does not know it exists.

They are stored where nobody looks and tested at no gate. If the next project's gate review does not ask which lessons apply and what was done with them, the register has no consumer, and an artefact with no consumer decays.

They are anonymised into uselessness. Removing the specifics to avoid blame removes the content. A lesson without the quantity, the date, the package and the mechanism is a proverb.

— 5. What a usable lesson looks like

Four properties fix the failures above: a lesson must be **specific, attributable, actionable** and **routed**.

Specific — it names the package, the date, the quantity and the mechanism. A lesson that could apply to any project applies to none.

Attributable — it names who observed it and who can confirm it, so the next reader can ask a question. This is not blame; it is provenance, and it is the difference between a lesson and a rumour.

Actionable — it states a change to a named artefact. Not "we should plan commissioning earlier" but "the standard schedule template adds an activity for each witness test, showing client-supplied resource as a dependency".

Routed — it names the process owner who receives it, and the register records their disposition: accepted with a date, rejected with a reason, or deferred with a review date. A rejected lesson with a recorded reason is a healthy outcome; an unanswered lesson is not.

The structure that produces all four is a fixed statement form:

Situation (what the circumstances were) → **action** (what we did) → **outcome** (what happened, with the number) → **change** (what should be different, in which document) → **owner** (which process owner receives it) → **test** (at which gate the next project is asked about it).

Compare, on the same event:

Unusable: "Better co-ordination with the client's operations team was needed during commissioning."

Usable: "**Situation:** the water treatment package required client operations staff to witness performance tests, an obligation in the contract but not represented in the schedule. **Action:** witness tests were scheduled assuming staff availability on request. **Outcome:** commissioning of the package was delayed 23 days waiting for witness attendance; time-related site costs continued throughout. **Change:** the standard schedule template adds a client-resource activity for every witness test, and the contract review checklist adds a check for client-supplied resource obligations. **Owner:** head of planning for the template; head of commercial for the checklist. **Test:** gate 3 schedule acceptance asks whether client-supplied resource obligations are represented in the network."

The second is longer, and length is not the point — the point is that someone can act on it without asking a question, and that a gate reviewer can check whether they did.

Capture while the pain is fresh. The strongest single change an organisation can make in this area is to stop treating lessons capture as a closeout activity. Capture at each phase gate, at each significant event, and at the resolution of each major risk or claim — while the people involved are still present and still remember why the decision looked reasonable at the time. Closeout then assembles and routes what has already been captured, which is a manageable task, instead of attempting recall, which is not.

TPL-16 – [Lessons learned and closeout register](#) carries the field structure and disposition columns this implies.

— 6. Benchmarking: capturing data the next estimate can use

The controls function is the natural custodian of a project's quantitative legacy, because it holds the performance data. What it hands over determines the quality of the next estimate — and, increasingly, the quality of whatever is fitted to the organisation's historical data.

What is worth capturing at closeout:

Outturn performance indices — final cost and schedule performance, and the shape of their history rather than only the end point.

Unit rates and productivity — hours per unit and cost per unit for the significant work types, at a level of detail the next estimator will actually use.

Change volume — the number and value of variations against original value, which is the most transferable single indicator of how well the scope was defined.

Contingency behaviour — how much was funded, at what confidence level, how much was drawn, and against which risks. This is what calibrates the next project's contingency, and it is almost never recorded.

Estimate accuracy by gate — the estimate at each gate against the outturn, which tells the organisation how much its own early numbers are worth.

All of that is worthless without the property that makes it comparable: **normalisation metadata**. A rate without its definition is not data; it is a number that will be misapplied. Five fields are the minimum, and §7.3 shows what happens when they are missing:

- **What is in the numerator** — which hours, which costs; direct only, or with supervision, scaffold, preliminaries, overtime premium?
- **What is in the denominator** — which quantity, measured how; net installed, gross delivered, contract measure?
- **Conditions** — site access, working hours, weather exposure, congestion, height, shift pattern.
- **Basis of currency and time** — currency, price base date, and whether escalation has been removed.
- **Scope boundary** — what the rate includes and specifically excludes.

Two further rules. **Capture the metadata at the same time as the number**, because it cannot be reconstructed later — nobody will remember in two years whether the supervision hours were in. And **publish the sample size and the variability**, not only the average: a rate derived from one package on one project is an observation, and presenting it as a norm is how one project's bad month becomes an organisation's estimating standard.

— 7. Worked example

Illustrative figures. Currency is USD, whole dollars. One construction contract within a larger project, viewed by the party that engaged the contractor. Retention and defects mechanics are those assumed for this example and vary by contract form and jurisdiction. No escalation or discounting is applied.

7.1 The final account

Component	Value
Original contract value	8,400,000
Approved variations	+ 610,000
Remeasurement of provisional quantities	– 85,000
Agreed claim settlement	+ 240,000
Contra-charges and back-charges	– 55,000
Final account	9,110,000

$$\begin{aligned}
 8,400,000 + 610,000 &= 9,010,000 \\
 9,010,000 - 85,000 &= 8,925,000 \\
 8,925,000 + 240,000 &= 9,165,000 \\
 9,165,000 - 55,000 &= 9,110,000
 \end{aligned}$$

7.2 Retention and what is still to collect

Assumption for this example: retention is 5 per cent of the final account, half released at practical completion and the balance at the end of the defects-liability period.

$$\begin{aligned}
 \text{total retention} &= 9,110,000 \times 0.05 = 455,500 \\
 \text{released at practical completion} &= 455,500 \div 2 = 227,750 \\
 \text{still held} &= 455,500 - 227,750 = 227,750
 \end{aligned}$$

With **8,760,000** certified and paid to date, net of retention still held:

$$\begin{aligned}
 \text{amount outstanding excluding retention} &= \text{final account} - \text{retention still held} - \text{paid to date} \\
 &= 9,110,000 - 227,750 - 8,760,000 \\
 &= 122,250 \\
 \text{total still to collect} &= 122,250 + 227,750 = 350,000
 \end{aligned}$$

The contract is not closed while **350,000** — nearly four per cent of the final account ($350,000 \div 9,110,000 = 3.8\%$) — remains outstanding, and roughly two thirds of it depends on a defects-liability period whose end date has nothing to do with the reporting calendar.

7.3 The open-items position

Open item	Claimed against us	Provided	Exposure
Subcontractor A — final account in negotiation	210,000	140,000	70,000
Subcontractor B — disputed variation	165,000	110,000	55,000
Subcontractor C — back-charge counter-claim	105,000	60,000	45,000
Total	480,000	310,000	170,000

column check: $210,000 + 165,000 + 105,000 = 480,000$
 $140,000 + 110,000 + 60,000 = 310,000$
 exposure = $480,000 - 310,000 = 170,000$

The project's reported result is provisional by up to **170,000**, and saying so with the number attached is the difference between a closeout report and a closeout announcement.

7.4 The benchmark that means two opposite things

The project erected **1,240 tonnes** of structural steel using **18,600 recorded labour hours**, of which **12 per cent** were supervision. The estimate for this work assumed **13.5 direct hours per tonne**.

All-in rate, as most closeout records capture it:

$18,600 \div 1,240 = 15.0$ hours per tonne
 variance against the 13.5 estimate = $15.0 - 13.5 = 1.5$ hours per tonne
 $= 1.5 \div 13.5 = 11.1$ % above estimate

Direct-only rate, on the same basis as the estimate:

supervision hours = $18,600 \times 0.12 = 2,232$
 direct hours = $18,600 - 2,232 = 16,368$
 direct rate = $16,368 \div 1,240 = 13.2$ hours per tonne
 variance against the 13.5 estimate = $13.2 - 13.5 = -0.3$ hours per tonne
 $= 0.3 \div 13.5 = 2.2$ % below estimate

Same job. Read as 11.1 per cent over estimate, or 2.2 per cent under it, entirely according to which hours were counted. No performance question separates them; only a definition does.

What the definitional error costs the next estimate. Suppose the 15.0 figure is filed without its definition and the next estimator applies it to a 1,500-tonne job, believing it comparable with the 13.5 norm:

estimated on the all-in rate = $1,500 \times 15.0 = 22,500$ hours
 like-for-like on the direct rate = $1,500 \times 13.2 = 19,800$ hours
 difference = $22,500 - 19,800 = 2,700$ hours
 overstatement = $2,700 \div 19,800 = 13.6$ %

A **2,700-hour** error on one line, produced by a missing metadata field, not by any disagreement about how long steel takes to erect. And the error is invisible: both numbers are correct, both are traceable to real records, and nothing in the estimate signals that the two rates are measured differently.

7.5 The rest of the benchmark set from this contract

```
change volume = approved variations ÷ original value
               = 610,000 ÷ 8,400,000 = 7.3 %

estimate accuracy at the sanction gate:
gate estimate for this contract = 8,000,000
outturn (final account)         = 9,110,000
variance = (9,110,000 - 8,000,000) ÷ 8,000,000
          = 1,110,000 ÷ 8,000,000 = 13.9 % above the gate estimate
```

Both figures need the same treatment as the steel rate. The 7.3 per cent change volume is comparable only with projects whose variation definition matches — some organisations count client changes only, others include design development and remeasurement. The 13.9 per cent estimate variance is measured against the sanction-gate estimate for this contract alone, and would be a different number measured against the feasibility estimate, the tender sum or the total project outturn. Record which, or the number will be compared with something it does not describe.

— 8. How this goes wrong

Closeout is unfunded and unstaffed. The last two per cent of the project is delivered by whoever has not yet been reassigned. Retention goes uncollected, final accounts are settled on the counterparty's terms because nobody has time to argue, and the lessons are never captured.

The project is declared closed at handover. The asset is transferred, the report says complete, and 350,000 of receivables and 170,000 of exposure are still live.

Provisions are released to improve the result. Quietly, without a documented decision. The number improves and the basis for it is gone.

Lessons are captured at the end, from memory. Covered in §4. The tell is a register whose entries are all about communication and stakeholder engagement, because those are the things people remember feeling.

Lessons are written without an addressee. No process owner, no artefact to change, no disposition. The register is complete and inert.

Specifics are removed to avoid blame. The quantity, the date and the package go, the mechanism becomes a generality, and the entry joins the proverbs.

The gate does not ask. No project is required to demonstrate what it took from the register, so nobody reads it, so nobody writes it well.

Benchmarks are captured without definitions. §7.4 prices this. The organisation ends up with a respectable-looking rate library that cannot be safely used and, worse, that people use anyway.

One project's rate becomes a norm. A single observation, published as an organisational standard, with no sample size and no indication of variability.

Contingency behaviour is never recorded. How much was funded, at what confidence level, how much was drawn and against what. Without it, the next contingency argument is conducted entirely on assertion, and [BPG-10 – Contingency and management reserve](#) has nothing to calibrate against.

The archive is assembled but not indexed. Everything is kept and nothing can be found, which is operationally identical to keeping nothing while costing considerably more.

— 9. Checklist

Planned at the start, not at the end

- [] Closeout has a budget line, named people and an end date later than completion.
- [] Closure deliverables scheduled backwards from the last commercial event, not from handover.
- [] Lessons capture scheduled at each gate and at each significant event, not only at closeout.
- [] Benchmark metadata fields defined at the same time as the cost and quantity structures.

Commercial and financial

- [] Final account assembled and tied to the change log, cost ledger, billing record and revenue.
- [] Every subcontract final account settled or provided for, with the exposure stated.
- [] Retention tracked by stage, each with its trigger, date and owner, to zero.
- [] Bonds, guarantees, warranties and insurances discharged or diarised to their true end dates.
- [] Accruals reversed, provisions released only on a documented decision with a stated basis.
- [] Cost collection codes closed; a named person routes genuine late items.
- [] Open-items table published with claimed, provided and exposure columns.

Lessons

- [] Every entry is specific, attributable, actionable and routed.
- [] Every entry names a document to change and a process owner to receive it.
- [] Every entry has a recorded disposition: accepted with a date, rejected with a reason, or deferred.
- [] Entries carry the quantity, the date and the package — specifics not stripped.
- [] The next project's gate review asks which entries apply and what was done with them.

Benchmarking

- [] Unit rates and productivity captured with all five normalisation fields.
- [] Numerator and denominator definitions recorded at the moment of capture.
- [] Currency, price base date and escalation treatment stated.
- [] Change volume recorded with its variation definition.
- [] Estimate accuracy recorded against each gate estimate, with the gate named.
- [] Contingency funded, confidence level, drawn amount and the risks it was drawn against, all recorded.
- [] Sample size and variability published alongside any rate offered as a norm.
- [] Archive indexed to the structure the next project will search by, not the one this project used.

A project that closes this way hands its successor two things that are otherwise unobtainable: a rate library whose numbers mean what they say, and a small set of lessons that have already changed a document. Everything else in a closeout file is a record. Those two are an inheritance, and they are the only part of a finished project that goes on earning.

— Related

- [TPL-16 – Lessons learned and closeout register](#) — the field structure, disposition columns and routing this guide assumes
- [BPG-19 – Project controls assurance and health checks](#) — the review that tests whether closeout data is being captured while it is still cheap
- [BPG-11 – Change orders and variations](#) — the variation account that feeds the final account
- [BPG-12 – Claims and extension of time](#) — the claim settlements that feed it, and the records that support them
- [BPG-10 – Contingency and management reserve](#) — what closeout's contingency-behaviour data is used to calibrate
- [BPG-09 – Estimate at completion – choosing and defending a method](#) — the forecasting method whose accuracy closeout finally measures

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 8 — Project Management Lifecycle, first authored draft, August 2026: contract and project closure, handover and the final account, completions and turnover, and the controls function's custody of quantitative lessons.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 7 — Contracts, Commercial Management, BoQ, Invoicing and Revenue, first authored draft, August 2026: remeasurement, retention, certification and the reconciliation of valuation with earned value and recognised revenue.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 3 — Budgeting and Forecasting, first authored draft, August 2026: the use of historical performance data in estimating, and estimating bias.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

Retention percentages, release triggers, defects-liability periods, remeasurement rules and the accounting treatment of provisions differ between contract forms and between jurisdictions; the mechanics in §7 are stated as assumptions of the example and are not presented as general rules. No industry benchmark, productivity norm or estimate-accuracy range is cited in this guide, because none was verified for it; every figure in §7 is illustrative and was constructed for teaching.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.